

PHYSICAL CHEMISTRY

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SURFACE CHEMISTRY

EXERCISE-I

- The size of the colloidal particles is in between:
 - $10^{-7} - 10^{-9}$ cm.
 - $10^{-9} - 10^{-11}$ cm.
 - $10^{-5} - 10^{-7}$ cm.
 - $10^{-2} - 10^{-3}$ cm.
- Peptization is a process of :
 - Precipitating the colloidal particles
 - Purifying the colloidal sol
 - Dispersing the precipitate into colloidal sol
 - Movement of colloidal particles towards the oppositely charged electrodes
- Colloids are purified by :
 - Brownian motion
 - Precipitation
 - Dialysis
 - Filtration
- Flocculation value is expressed in terms of:
 - Millimole per litre
 - Mol per litre
 - Gram per litre
 - Mol per millilitre
- Which is kinetic phenomenon?
 - Brownian motion
 - Tyndall effect
 - Both (1) and (2)
 - None of these
- The heats of adsorption in physisorption lie in the range (in KJ mol^{-1}) :
 - 40-400
 - 40-100
 - 20-40
 - 1-10
- According to Langmuir adsorption isotherm, the amount of gas adsorbed at very high pressure :
 - Reaches a constant limiting value
 - Goes on increasing with pressure
 - Goes on decreasing with pressure
 - Increases first and decreases later with pressure
- Which one of the following is an incorrect statement for physisorption :
 - It is a reversible process
 - It requires less heat of adsorption
 - It requires activation energy
 - It takes place at low temperature
- Lyophobic colloids are :-
 - Reversible
 - Irreversible
 - Water loving
 - Solvent loving
- When freshly precipitated $\text{Fe}(\text{OH})_3$ is boiled with water in the presence of few drops of dil HCl, a hydrated ferric hydroxide sol is obtained. This method is termed as :-
 - Dialysis
 - Peptization
 - Ultrafiltration
 - Electrodispersion
- To coagulate Pt sol, which of the following is most effective :-
 - NaCl
 - Na_3PO_4
 - AlCl_3
 - Alcohol
- A colloidal mixture of $\text{Fe}(\text{OH})_3$ in water is :-
 - A hydrophilic colloid
 - A hydrophobic colloid
 - An emulsion
 - None
- A catalyst is a substance which :
 - Increases the equilibrium concentration of the product
 - Change the equilibrium constant of the reaction
 - Shortens the time to reach equilibrium
 - Supplies energy to the reaction
- Which of the following statement is more correct:
 - Catalyst only accelerates the rate of a chemical reaction
 - A catalyst can retard the rate of a chemical reaction
 - A catalyst does not affect the speed of a reaction
 - A catalyst alters the speed of a reaction
- Efficiency of the catalyst depends on its :-
 - Molecular weight
 - Number of free valencies
 - Physical state
 - Amount used
- In a reversible reaction, a catalyst :-
 - Increases the rate of forward reaction only
 - Increases the rate of forward reaction to a greater extent than that of the backward reaction
 - Increases the rate of forward reaction and decreases that of the backward reaction
 - alters the rate of forward and backward reaction equally
- Which is false for catalyst :-
 - A catalyst can initiate a reaction
 - It does not alter the position of equilibrium in a reversible reaction
 - A catalyst remains unchanged in quality and composition at the end of reaction
 - Catalysts are sometimes very specific in respect of a reaction

- 18.** Shape selective catalysts are so called because of :
 (1) The shape of the catalysts
 (2) The specificity of the catalysts
 (3) The size of the pores of the catalysts which can trap only selective molecules
 (4) Their use for only some selected reactions
- 19.** Which gas will be adsorbed on a solid to greater extent.
 (1) A gas having non polar molecule
 (2) A gas having highest critical temperature (T_c)
 (3) A gas having lowest critical temperature.
 (4) A gas having highest critical pressure.
- 20.** The nature of bonding forces in chemisorption
 (1) purely physical such as Van Der Waal's forces
 (2) purely chemical
 (3) both chemical and physical simultaneously.
 (4) none of these
- 21.** The Tyndall effect associated with colloidal particles is due to
 (1) presence of electrical charges
 (2) scattering of light
 (3) absorption of light
 (4) reflection of light
- 22.** Which one of the following is not applicable to chemisorption?
 (1) Its heat of adsorption is high
 (2) It takes place at high temperature
 (3) It is reversible
 (4) It forms mono-molecular layers
- 23.** Milk is an example of
 (1) emulsion (2) suspension
 (3) foam (4) sol.
- 24.** Colloidal particles in a sol. can be coagulated by
 (1) heating
 (2) adding an electrolyte
 (3) adding oppositely charged sol
 (4) any of the above methods
- 25.** Given below are a few electrolytes, indicate which one among them will bring about the coagulation of a gold sol. quickest and in the least of molar concentration?
 (1) NaCl (2) $MgSO_4$
 (3) $Al_2(SO_4)_3$ (4) $K_4[Fe(CN)_6]$
- 26.** The minimum concentration of an electrolyte required to cause coagulation of a sol is called
 (1) flocculation value (2) gold number
 (3) protective value (4) none of these
- 27.** Which one of following statements is not correct in respect of lyophilic sols?
 (1) There is a considerable interaction between the dispersed phase and dispersion medium
 (2) These are quite stable and are not easily coagulated
 (3) They need stabilizing agent
 (4) The particle are hydrated
- 28.** At the critical micelle concentration (CMC) the surfactant molecules
 (1) decompose
 (2) dissociate
 (3) associate
 (4) become completely soluble
- 29.** Although nitrogen does not adsorb on surface at room temperature, it adsorbs on surface at 83K. Which one of the following statements is correct -
 (1) At 83K, there is formation of monomolecular layer
 (2) At 83K, there is formation of multimolecular layer
 (3) At 83K, nitrogen molecules are held by chemical bonds
 (4) At 83K, nitrogen is adsorbed as atoms.
- 30.** The volume of a colloidal particle V_C , volume of a solute particle in a true solution V_t , the volume of suspension particle is V_s can be arranged
 (1) $V_C = V_t = V_s$ (2) $V_s < V_C < V_t$
 (3) $V_s > V_C > V_t$ (4) $V_C > V_s > V_t$

				ANSWER KEY			Exercise-I			
Que.	1	2	3	4	5	6	7	8	9	10
Ans.	3	3	3	1	1	3	1	3	2	2
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	3	2	3	4	2	4	1	1	2	2
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	2	3	1	4	3	1	3	3	2	3

PREVIOUS YEARS' QUESTIONS

EXERCISE-II

1. Gold numbers of protective colloids A, B, C and D are 0.50, 0.01, 0.10 and 0.005 respectively. The correct order of their protective power is :-
[AIEEE-2008]
(1) $D < A < C < B$ (2) $C < B < D < A$
(3) $A < C < B < D$ (4) $B < D < A < C$
 2. Which of the following statements is incorrect regarding physisorptions ? [AIEEE-2009]
(1) Under high pressure it results into multi molecular layer on adsorbent surface
(2) Enthalpy of adsorption ($\Delta H_{\text{adsorption}}$) is low and positive
(3) It occurs because of Van der Waal's forces
(4) More easily liquefiable gases are adsorbed readily
 3. Among the electrolytes Na_2SO_4 , CaCl_2 , $\text{Al}(\text{SO}_4)_3$ and NH_4Cl , the most effective coagulating agent for Sb_2S_3 sol is : [IIT-2009]
(1) Na_2SO_4
(2) CaCl_2
(3) $\text{Al}_2(\text{SO}_4)_3$
(4) NH_4Cl
 4. The correct statement(s) pertaining to the adsorption of a gas on a solid surface is (are) - [IIT-2011]
(1) Adsorption is always exothermic
(2) Physisorption may transform into chemisorption at high temperature
(3) Physisorption increases with increasing temperature but chemisorption decreases with increasing temperature
(4) Chemisorption is more exothermic than physisorption, however it is very slow due to higher energy of activation
 5. According to Freundlich adsorption isotherm, which of the following is correct ? [AIEEE-2012]
(1) $\frac{x}{m} \propto p^0$
(2) $\frac{x}{m} \propto p^1$
(3) $\frac{x}{m} \propto p^{1/n}$
(4) All the above are correct for different ranges of pressure
 6. If x is the mass of the gas adsorbed on mass m of the absorbent at pressure p , Freundlich adsorption isotherm gives a straight line on plotting :- [AIEEE-2012 (Online)]
(1) $\frac{x}{m}$ vs p (2) $\log \frac{x}{m}$ vs $\log p$
(3) $\log \frac{x}{m}$ vs p (4) $\frac{x}{m}$ vs $\frac{1}{p}$
 7. Fog is a colloidal solution of :- [AIEEE-2012 (Online)]
(1) Gaseous particles dispersed in a liquid
(2) Solid particles dispersed in a liquid
(3) Liquid particles dispersed in gas
(4) Solid particle dispersed in gas
 8. The coagulating power of electrolytes having ions Na^+ , Al^{3+} and Ba^{2+} for arsenic sulphide sol increases in the order :- [JEE (MAIN) 2013]
(1) $\text{Al}^{3+} < \text{Ba}^{2+} < \text{Na}^+$ (2) $\text{Na}^+ < \text{Ba}^{2+} < \text{Al}^{3+}$
(3) $\text{Ba}^{2+} < \text{Na}^+ < \text{Al}^{3+}$ (4) $\text{Al}^{3+} < \text{Na}^+ < \text{Ba}^{2+}$
 9. The migration of dispersion medium under the influence of an electric potential is called : [JEE (MAIN) 2013 (Online)]
(1) Electrophoresis (2) Cataphoresis
(3) Electroosmosis (4) Sedimentation
 10. Smoke is an example of : [JEE (MAIN) 2013 (Online)]
(1) Solid dispersed in solid
(2) Solid dispersed in gas
(3) Gas dispersed in solid
(4) Gas dispersed in liquid
 11. For a linear plot of $\log(x/m)$ versus $\log p$ in a Freundlich adsorption isotherm, which of the following statements is correct ? (k and n are constants) [JEE (MAIN) 2016]
(1) $\log (1/n)$ appears as the intercept
(2) Both k and $1/n$ appear in the slope term
(3) $1/n$ appears as the intercept
(4) Only $1/n$ appears as the slope
 12. Gold numbers of some colloids are : Gelatin : 0.005 – 0.01, Gum Arabic : 0.15 – 0.25 ; Oleate : 0.04 – 1.0, Starch : 15 – 25. Which among these is a better protective colloid ? [JEE (MAIN) 2016 (Online)]
(1) Oleate (2) Gelatin
(3) Gum-Arabic (4) Starch

- 13.** A particular adsorption process has the following characteristics: (i) It arises due to van der Waals forces and (ii) it is reversible. Identify the correct statement that describes the above adsorption process : **[JEE (MAIN) 2016 (Online)]**
- (1) Enthalpy of adsorption is greater than 100 kJ mol^{-1}
 - (2) Energy of activation is low.
 - (3) Adsorption is monolayer
 - (4) Adsorption increases with increase in temperature.
- 14.** For a linear plot of $\log(x/m)$ versus $\log p$ in a Freundlich adsorption isotherm, which of the following statements is correct ? (k and n are constants) **[JEE (MAIN) 2016 (Offline)]**
- (1) $\log (1/n)$ appears as the intercept
 - (2) Both k and $1/n$ appear in the slope term
 - (3) $1/n$ appears as the intercept
 - (4) Only $1/n$ appears as the slope
- 15.** The Tyndall effect is observed only when following conditions are satisfied :- **[JEE (MAIN) 2017]**
- (a) The diameter of the dispersed particles is much smaller than the wavelength of the light used.
 - (b) The diameter of the dispersed particle is not much smaller than the wavelength of the light used.
 - (c) The refractive indices of the dispersed phase and dispersion medium are almost similar in magnitude.
 - (d) The refractive indices of the dispersed phase and dispersion medium differ greatly in magnitude.
- (1) (a) and (d)
 - (2) (b) and (d)
 - (3) (a) and (c)
 - (4) (b) and (c)
- 16.** Among the following, correct statement is : **[JEE-Main (online)2017]**
- (1) One would expect charcoal to adsorb chlorine more than hydrogen sulphide.
 - (2) Brownian movement is more pronounced for smaller particles than for bigger-particles.
 - (3) Hardy Schulze law states that bigger the size of the ions, the greater is its coagulating power
 - (4) Sols of metal sulphides are lyophilic
- 17.** Adsorption of a gas on a surface follows Freundlich adsorption isotherm. Plot of $\log \frac{x}{m}$ versus $\log p$ gives a straight line with slope equal to 0.5, then : **[JEE-Main (online)2017]**
- $\left(\frac{x}{m}\right)$ is the mass of the gas adsorbed per gram of adsorbent)
- (1) Adsorption is proportional to the square of pressure.
 - (2) Adsorption is independent of pressure.
 - (3) Adsorption is proportional to the pressure.
 - (4) Adsorption is proportional to the square root of pressure.
- 18.** Which one of the following is not a property of physical adsorption **[JEE-Main (online)2018]**
- (1) Unilayer adsorption occurs
 - (2) Greater the surface area, more the adsorption
 - (3) Lower the temperature, more the adsorption
 - (4) Higher the pressure, more the adsorption

PREVIOUS YEARS QUESTIONS				ANSWER KEY			Exercise-II			
Que.	1	2	3	4	5	6	7	8	9	10
Ans.	3	2	3	1,2,4	4	2	3	2	3	2
Que.	11	12	13	14	15	16	17	18		
Ans.	4	2	2	4	2	2	4	1		

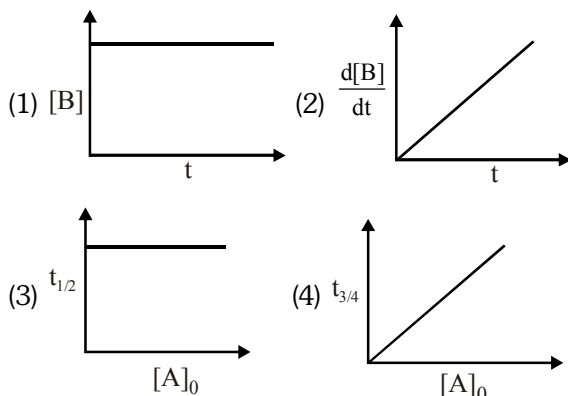
CHEMICAL KINETICS

EXERCISE-I

1. In a reaction $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \longrightarrow 2\text{NH}_3(\text{g})$, the rate of appearance of NH_3 is $2.5 \times 10^{-4} \text{ mol L}^{-1} \text{ sec}^{-1}$. The rate of reaction & rate of disappearance of H_2 will be (In $\text{molL}^{-1}\text{sec}^{-1}$)
- (1) 3.75×10^{-4} , 1.25×10^{-4}
(2) 1.25×10^{-4} , 2.5×10^{-4}
(3) 1.25×10^{-4} , 3.75×10^{-4}
(4) 5.0×10^{-4} , 3.75×10^{-4}

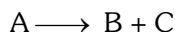
- 2.** The rate constant of a zero order reaction is $0.2 \text{ mol dm}^{-3}\text{h}^{-1}$. If the concentration of the reactant after 30 minutes is 0.05 mol dm^{-3} . Then its initial concentration would be :
- (1) 0.15 mol dm^{-3}
(2) 1.05 mol dm^{-3}
(3) 0.25 mol dm^{-3}
(4) 4.00 mol dm^{-3}

3. Which graph represents zero order reaction $[A(g) \rightarrow B(g)]$:



4. A reaction is of first order. After 100 minutes, 75 g of the reactant A are decomposed when 100g are taken initially. Calculate the time required when 150g of the reactant A are decomposed, the initial weight taken is 200g.
- (1) 100 minutes
 - (2) 200 minutes
 - (3) 150 minutes
 - (4) 175 minutes

- 5.** Consider the reaction :



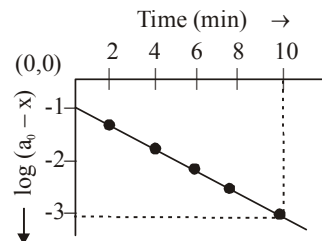
Initial concentration of A is 1 M. 20 minutes time is required for completion of 80 % reaction.

If $\frac{d[B]}{dt} = k[A]$, then half life ($t_{1/2}$) is

(Use : $\ln 5 = 1.6$, $\ln 2 = 0.7$)

- (1) 55.44 min. (2) 50 min
(3) 8.75 min (4) 12.5 min

6. For the first order decomposition of $\text{SO}_2\text{Cl}_2(\text{g})$, $\text{SO}_2\text{Cl}_2(\text{g}) \rightarrow \text{SO}_2(\text{g}) + \text{Cl}_2(\text{g})$ a graph of $\log(a_0 - x)$ vs t is shown in figure. What is the rate constant (sec^{-1})?



- (1) 0.2
(2) 4.6×10^{-1}
(3) 7.7×10^{-3}
(4) 1.15×10^{-2}

7. At certain temperature, the half life period for the thermal decomposition of a gaseous substance depends on the initial partial pressure of the substance as follows

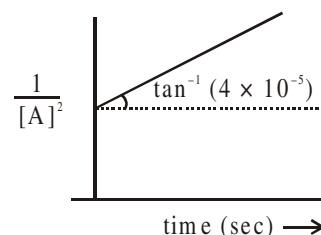
P(mmHg)	500	250
t _{1/2} (in min.)	235	950

Find the order of reaction

[Given $\log (23.5) = 1.37$; $\log (95) = 1.97$;
 $\log 2 = 0.30$]

- (1) 1 (2) 2 (3) 2.5 (4) 3

- 8.** For a reaction $A \longrightarrow \text{Product}$



What is the value of k for the given reaction-

- (1) $4 \times 10^{-5} \text{ M}^{-1}\text{s}^{-1}$

(3) $2 \times 10^{-5} \text{ M}^{-2}\text{s}^{-1}$

(2) $\frac{4}{3} \times 10^{-5} \text{ M}^{-2}\text{s}^{-1}$

(4) $\frac{2}{3} \times 10^{-5} \text{ M}^{-2}\text{s}^{-1}$

9. For a 1st order homogeneous gaseous reaction $A \rightarrow 2B + C$, if the pressure after time t was P_t and after long time was P_∞ , then rate constant (k) in terms of P_t & P_∞ and t is -

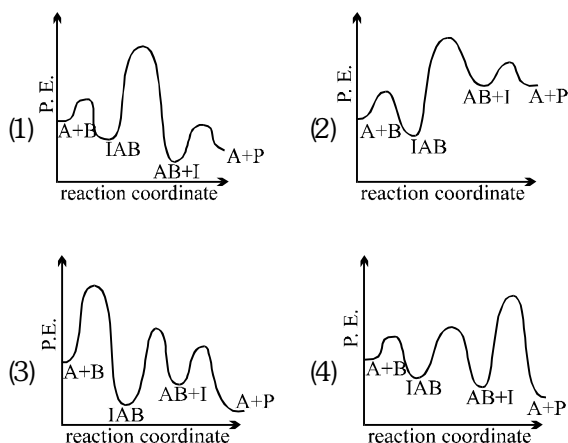
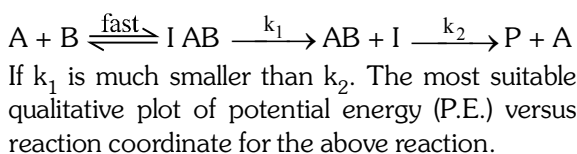
$$(1) k = \frac{2.303}{t} \log \left(\frac{P_{\infty}}{P_{\infty} - P_t} \right)$$

$$(2) k = \frac{2.303}{t} \log \left(\frac{2P_{\infty}}{P_{\infty} - P_t} \right)$$

$$(3) \quad k = \frac{2.303}{t} \log \left(\frac{2P_{\infty}}{3(P_{\infty} - P_t)} \right)$$

- (4) None of these

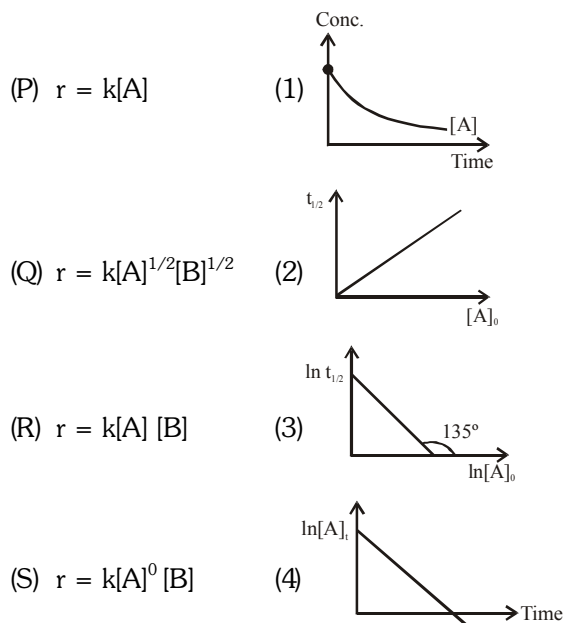
10. Activation energy of a reaction is –
 (1) The energy released during the reaction
 (2) The energy evolved when activated complex is formed
 (3) Minimum amount of energy needed to overcome the potential barrier of reaction
 (4) The energy needed to form one mole of the product
11. For producing the effective collisions, the colliding molecules must possess:-
 (1) A certain minimum amount of energy
 (2) Energy equal to or greater than threshold energy
 (3) Proper orientation
 (4) Threshold energy as well as proper orientation of collision
12. For the first order reaction $A \longrightarrow B + C$, carried out at 27°C if $3.8 \times 10^{-16}\%$ of the reactant molecules can overcome energy barrier, the E_a (activation energy) of the reaction is $[\log 3.8 = 0.58, 2.303 \times 8.314 \times 17.42 = 333.33]$
 (1) 12 kJ/mole (2) 831.4 kJ/mole
 (3) 100 kJ/mole (4) 111.11 J/mole
13. The following mechanism has been proposed for the exothermic catalyzed complex reaction.



14. Which of the following is **INCORRECT** for first order reaction ?
 (1) On introducing catalyst, both rate constant and rate of reaction increases.
 (2) On increasing temperature both rate constant & rate of reaction increases.
 (3) On decreasing volume both rate constant & rate of gaseous reaction increases.
 (4) On increasing concentration of gaseous reactant at constant volume & constant temperature both total pressure and rate of the reaction increases.

15. Thermal decomposition of compound X is a first order reaction. If 75% of X is decomposed in 100 min. How long will it take for 90% of the compound to decompose? **Given : $\log 2 = 0.30$**
 (1) 190 min (2) 176.66 min
 (3) 166.66 min (4) 156.66 min
16. For the reaction $A + B \rightarrow \text{product}$,
 Given : $[A]_0 = [B]_0$

List-I (Observed Rate Law) is- **List-II (Graph)**



Code :

P	Q	R	S
(1) 4	1	3	2
(2) 2	3	1	4
(3) 1	2	3	4
(4) 4	3	2	1

Match the Columns for Reaction $A \rightarrow P$

Column - I	Column - II	Column - III
(I) First Order	(i) Reaction complete in finite time	(P) Rate depends on concentration
(II) Second Order	(ii) Reaction complete in infinite time	(Q) After equal interval of time concentration of reactant left are in G.P.
(III) Third Order	(iii) Half life is independent of concentration of reactant	(R) After equal interval of time concentration of reactant left are in A.P.
(IV) Zero Order	(iv) Half life decreases when concentration of reactant increases	(S) Half life depends on temperature

17. For the reaction $H_2 + I_2 \rightleftharpoons 2HI$, the true relationship is -

$$\begin{aligned}
 (1) - \frac{d[H_2]}{dt} &= \frac{d[I_2]}{dt} = \frac{d[HI]}{dt} \\
 (2) \frac{d[H_2]}{dt} &= - \frac{d[I_2]}{dt} = \frac{1}{2} \frac{d[HI]}{dt} \\
 (3) - \frac{2d[H_2]}{dt} &= - \frac{2d[I_2]}{dt} = \frac{d[HI]}{dt} \\
 (4) \frac{2d[H_2]}{dt} &= - \frac{2d[I_2]}{dt} = \frac{1}{2} \frac{d[HI]}{dt}
 \end{aligned}$$

18. A chemical reaction $[2A] + [2B] + [C] \longrightarrow \text{product}$ follows the rate equation : $r \propto [A][B]^2$ then order of reaction is -
 (1) 0 (2) 1 (3) 2 (4) 3
19. The unit of rate constant of first & second order reaction is respectively -
 (1) time^{-1} , $\text{mole}^{-1} \cdot \text{litre} \cdot \text{time}^{-1}$
 (2) $\text{mole}^{-1} \text{litre}^{-1}$, time^{-1}
 (3) $\text{mole}^{-1} \cdot \text{litre} \cdot \text{time}^{-1}$, time^{-1}
 (4) sec^{-1} , litre^{-1}
20. In the equation $kt = \ln C_0 - \ln C_t$, the curve between t and $\ln C_t$ is -
 (1) a straight line (2) a parabola
 (3) a hyperbola (4) none of these
21. In respect of the equation $k = Ae^{-E_a/RT}$ in chemical kinetics, which one of the following statement is correct ?
 (1) E_a is energy of activation
 (2) R is Rydberg's constant
 (3) k is equilibrium constant
 (4) A is adsorption factor
22. For the reaction system : $2\text{NO}(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2\text{NO}_2(\text{g})$ volume is suddenly reduced to half its value by increasing the pressure on it. If the reaction is of first order with respect to O_2 and second order with respect to NO , the rate of reaction will -
 (1) Increase to eight times of its initial value
 (2) Increase to four times of its initial value
 (3) Diminish to one-fourth of its initial value
 (4) Diminish to one-eighth of its initial value
23. The rate law for a reaction between substances A and B is given by, $\text{Rate} = k[A]^n[B]^m$. On doubling the concentration of A and halving the concentration of B, the ratio of the new rate to the earlier rate of the reaction will be as :
 (1) $(n - m)$ (2) $2^{(n - m)}$
 (3) $\frac{1}{2^{(m+n)}}$ (4) $(m + n)$
24. The rate equation for the reaction $2A + B \rightarrow C$ is found to be : $\text{rate} = k[A][B]$. The correct statement in relation to this reaction is that the
 (1) rate of formation of C is twice the rate of disappearance of A.
 (2) Half life is a constant
 (3) unit of k must be s^{-1}
 (4) value of k is independent of the initial concentrations of A and B
25. Consider an endothermic reaction : $X \rightarrow Y$ with the activation energies E_b and E_f for the backward and forward reactions, respectively. In general
 (1) $E_b > E_f$
 (2) $E_b < E_f$
 (3) there is no definite relation between E_b and E_f
 (4) $E_b = E_f$
26. A reaction involving two different reactant can never be -
 (1) first order reaction
 (2) unimolecular reaction
 (3) biomolecular reaction
 (4) second order reaction
27. Which of the following statement(s) is (are) correct
 (1) A plot of $\log K_p$ versus $1/T$ is linear
 (2) A plot of $\log [X]$ versus time is linear for a first order reaction, $X \longrightarrow P$
 (3) A plot of $\log P$ versus $1/T$ is linear at constant volume.
 (4) A plot of P versus $1/V$ is linear at constant temperature.
28. The rate constant for the reaction
 $2\text{N}_2\text{O}_5 \longrightarrow 4\text{NO}_2 + \text{O}_2$
 is $3.0 \times 10^{-5} \text{ sec}^{-1}$. if the rate is $2.4 \times 10^{-5} \text{ mol litre}^{-1} \text{ sec}^{-1}$, then the concentration of N_2O_5 (in mol litre^{-1}) is
 (1) 1.4 (2) 1.2 (3) 0.004 (4) 0.8
29. In a first order reaction the concentration of reactant decreases from 800 mol/dm^3 to 50 mol/dm^3 in $2 \times 10^4 \text{ sec}$. The rate constant of reaction in sec^{-1} is
 (1) 2×10^4 (2) 3.45×10^{-5}
 (3) 1.3486×10^{-4} (4) 2×10^{-4}
30. The reaction, $X \longrightarrow \text{Product}$ follows first order kinetics. In 40 minutes the concentration of X changes from 0.1 M to 0.025 M. Then the rate of reaction when concentration of X is 0.01 M
 (1) $1.73 \times 10^{-4} \text{ M min}^{-1}$
 (2) $3.47 \times 10^{-5} \text{ M min}^{-1}$
 (3) $3.47 \times 10^{-4} \text{ M min}^{-1}$
 (4) $1.73 \times 10^{-5} \text{ M min}^{-1}$

				ANSWER KEY			Exercise-I			
Que.	1	2	3	4	5	6	7	8	9	10
Ans.	3	1	4	1	3	3	4	3	3	3
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	4	3	1	3	3	1	3	4	1	1
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	1	1	2	4	2	2	1,2,4	4	3	3

PREVIOUS YEARS' QUESTIONS

EXERCISE-II

- $t_{1/4}$ can be taken as the time taken for the concentration of a reactant to drop to $3/4$ of its initial value. If the rate constant for a first order reaction is k , then $t_{1/4}$ can be written as -
[$\ln 2 = 0.7$, $\ln 3 = 1.11$] [AIEEE-2005]
(1) $0.29/k$ (2) $0.10/k$
(3) $0.75/k$ (4) $0.69/k$
- A reaction was found to be second order with respect to the concentration of carbon monoxide. If the concentration of carbon monoxide is doubled, with everything else kept the same, the rate of reaction will [AIEEE-2005]
(1) triple
(2) increase by a factor of 4
(3) double
(4) remain unchanged
- Rate of reaction can be expressed by Arrhenius equation as $k = Ae^{-E/RT}$. In this equation, E represents [AIEEE-2005]
(1) the energy below which colliding molecules will not react
(2) the total energy of the reacting molecule at a temperature, T
(3) The fraction of molecules with energy greater than the activation energy of the reaction
(4) None of these
- The following mechanism has been proposed for the reaction of NO with Br_2 to form NOBr

$$\text{NO(g)} + \text{Br}_2\text{(g)} \rightleftharpoons \text{NOBr}_2\text{(g)}$$

$$\text{NOBr}_2\text{(g)} + \text{NO(g)} \rightarrow 2\text{NOBr(g)}$$
 If the second step is the rate determining step, the order of the reaction with respect to NO(g) is : [AIEEE 2006]
(1) 0 (2) 1 (3) 2 (4) 3
- The energies of activation for forward and reverse reactions for $\text{A}_2 + \text{B}_2 \rightleftharpoons 2\text{AB}$ are 180 kJ mol^{-1} and 200 kJ mol^{-1} respectively. The presence of a catalyst lowers the activation energy of both (forward and reverse) reactions by 100 kJ mol^{-1} . The enthalpy change of the reaction ($\text{A}_2 + \text{B}_2 \rightarrow 2\text{AB}$) in the presence of catalyst will be (in kJ mol^{-1}) - [AIEEE-2007]
(1) 300 (2) 120 (3) 280 (4) -20
- The half life period of a first order chemical reaction is 6.93 minutes. The time required for the completion of 99% of the chemical reaction will be ($\log 2 = 0.301$) :- [AIEEE-2009]
(1) 46.06 minutes (2) 460.6 minutes
(3) 230.3 minutes (4) 23.03 minutes

- Consider the reaction:

$$\text{Cl}_2\text{(aq)} + \text{H}_2\text{S(aq)} \rightarrow \text{S(s)} + 2\text{H}^+\text{(aq)} + 2\text{Cl}^-\text{(aq)}$$
 The rate equation for this reaction is

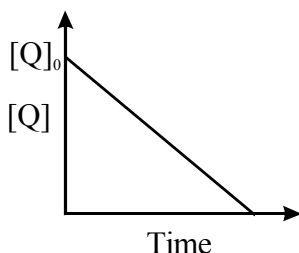
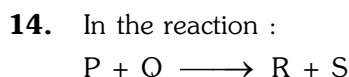
$$\text{rate} = k[\text{Cl}_2][\text{H}_2\text{S}]$$
 Which of these mechanisms is/are consistent with this rate equation ? [AIEEE-2010]
 A. $\text{Cl}_2 + \text{H}_2\text{S} \rightarrow \text{H}^+ + \text{Cl}^- + \text{Cl}^+ + \text{HS}^-$ (slow)
 $\text{Cl}^+ + \text{HS}^- \rightarrow \text{H}^+ + \text{Cl}^- + \text{S}$ (fast)
 B. $\text{H}_2\text{S} \rightleftharpoons \text{H}^+ + \text{HS}^-$ (fast equilibrium)
 $\text{Cl}_2 + \text{HS}^- \rightarrow 2\text{Cl}^- + \text{H}^+ + \text{S}$ (slow)
 (1) A only
 (2) B only
 (3) Both A and B
 (4) Neither A nor B
- The time for half life period of a certain reaction $\text{A} \rightarrow \text{Products}$ is 1 hour, when the initial concentration of the reactant 'A' is 2.0 mol L^{-1} . How much time does it take for its concentration to come from 0.50 to 0.25 mol L^{-1} if it is a zero order reaction ? [AIEEE-2010]
(1) 1 h (2) 4 h
(3) 0.5 h (4) 0.25 h
- The concentration of R in the reaction $\text{R} \rightarrow \text{P}$ was measured as a function of time and the following data is obtained :

[R] (molar)	1.0	0.75	0.40	0.10
t(min.)	0.0	0.05	0.12	0.18

- The order of the reaction is. [JEE 2010]
- The rate of a chemical reaction doubles for every 10°C rise of temperature. If the temperature is raised by 50°C , the rate of the reaction increases by about :- [AIEEE-2011]
(1) 32 times (2) 64 times
(3) 10 times (4) 24 times
 - A reactant (1) forms two products : [AIEEE-2011]

$$\text{A} \xrightarrow{k_1} \text{B, Activation Energy } E_{a1}$$

$$\text{A} \xrightarrow{k_2} \text{C, Activation Energy } E_{a2}$$
 If $E_{a2} = 2 E_{a1}$, then k_1 and k_2 are related as :-
 (1) $k_1 = 2k_2 e^{E_{a2}/RT}$ (2) $k_1 = k_2 e^{E_{a1}/RT}$
 (3) $k_2 = k_1 e^{E_{a2}/RT}$ (4) $k_1 = [A] k_2 e^{E_{a1}/RT}$



the time taken for 75% reaction of P is twice the time taken for 50% reaction of P. The concentration of Q varies with reaction time as shown in the figure. The overall order of the reaction is - **[JEE 2013]**

15. For the non-stoichiometric reaction $2A + B \rightarrow C + D$, the following kinetic data were obtained in three separate experiments, all at 298 K.

Initial Concentration (A)	Initial Concentration (B)	Initial rate of formation of C (mol L ⁻¹ S ⁻¹)
0.1M	0.1M	1.2×10 ⁻³
0.1M	0.2M	1.2×10 ⁻³
0.2M	0.1M	2.4×10 ⁻³

- (1) $\frac{dc}{dt} = k[A][B]^2$ (2) $\frac{dc}{dt} = k[A]$
 (3) $\frac{dc}{dt} = k[A][B]$ (4) $\frac{dc}{dt} = k[A]^2[B]$

- 16.** For the elementary reaction $\mathbf{M} \rightarrow \mathbf{N}$, the rate of disappearance of $\mathbf{M}$ increases by a factor of 8 upon doubling the concentration of $\mathbf{M}$. The order of the reaction with respect to $\mathbf{M}$ is [JEE 2014]

- | | |
|-------|-------|
| (1) 4 | (2) 3 |
| (3) 2 | (4) 1 |

- 17.** Higher order (>3) reactions are rare due to :-
[JEE-MAIN-(Offline)2015]

- (1) shifting of equilibrium towards reactants due to elastic collision
- (2) loss of active species on collision
- (3) low probability of simultaneous collision of all the reacting species
- (4) increase in entropy and activation energy as more molecules are involved.

- 18.** The reaction : $2\text{N}_2\text{O}_5(\text{g}) \rightarrow 4\text{NO}_2(\text{g}) + \text{O}_2(\text{g})$, follows first order kinetics. The pressure of a vessel containing only N_2O_5 was found to increase from 50 mm Hg to 87.5 mm Hg in 30 min. The pressure exerted by the gases after 60 min. will be (Assume temperature remains constant)

- [JEE-MAIN-(Online)2015]**
- (1) 106.25 nm Hg (2) 116.25 nm Hg
(3) 125 mm Hg (4) 150 mm Hg

- 19.** For the equilibrium, $A(g) \rightleftharpoons B(g)$, ΔH is -40 kJ/mol .
If the ratio of the activation energies of the forward
(E_f) and reverse (E_b) reactions is $\frac{2}{3}$ then :-

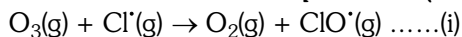
- [JEE-MAIN-(Online)2015]**
- (1) $E_f = 60 \text{ kJ/mol}$; $E_b = 100 \text{ kJ/mol}$
(2) $E_f = 30 \text{ kJ/mol}$; $E_b = 70 \text{ kJ/mol}$
(3) $E_f = 80 \text{ kJ/mol}$; $E_b = 120 \text{ kJ/mol}$
(4) $E_f = 70 \text{ kJ/mol}$; $E_b = 30 \text{ kJ/mol}$

- 20.** Decomposition of H_2O_2 follows a first order reaction. In fifty minutes the concentration of H_2O_2 decreases from 0.5 to 0.125 M in one such decomposition. When the concentration of H_2O_2 reaches 0.05 M, the rate of formation of O_2 will be :-

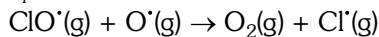
- [JEE-MAIN-(Offline)2016]**
- (1) $1.34 \times 10^{-2} \text{ mol min}^{-1}$
(2) $6.93 \times 10^{-2} \text{ mol min}^{-1}$
(3) $6.93 \times 10^{-4} \text{ mol min}^{-1}$
(4) 2.66 L min^{-1} at STP

21. The reaction of ozone with oxygen atoms in the presence of chlorine atoms can occur by a two step process shown below :

[JEE-MAIN-(Online)2016]

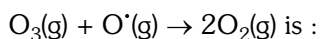


$$k_i = 5.2 \times 10^9 \text{ L mol}^{-1}\text{s}^{-1}$$



$$k_{ii} = 2.6 \times 10^{10} \text{ L mol}^{-1}\text{s}^{-1} \dots\dots(\text{ii})$$

The closest rate constant for the overall reaction



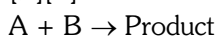
$$(1) 3.1 \times 10^{10} \text{ L mol}^{-1}\text{s}^{-1}$$

$$(2) 2.6 \times 10^{10} \text{ L mol}^{-1}\text{s}^{-1}$$

$$(3) 5.2 \times 10^9 \text{ L mol}^{-1}\text{s}^{-1}$$

$$(4) 1.4 \times 10^{20} \text{ L mol}^{-1}\text{s}^{-1}$$

22. the rate law for the reaction below is given by the expression $k[A][B]$



If the concentration of B is increased from 0.1 to 0.3 mole, keeping the value of A at 0.1 mole, the rate constant will be : [JEE-MAIN-(Online)2016]

$$(1) 3k \quad (2) 9k \quad (3) k \quad (4) k/3$$

23. According to the Arrhenius equation,

[JEE 2016]

- (1) A high activation energy usually implies a fast reaction
- (2) Rate constant increase with increase in temperature. This is due to a greater number of collisions whose energy exceeds the activation energy
- (3) Higher the magnitude of activation energy, stronger is the temperature dependence of the rate constant
- (4) The pre-exponential factor is a measure of the rate at which collisions occur, irrespective of their energy.

24. Two reactions R_1 and R_2 have identical pre-exponential factors. Activation energy of R_1 exceeds that of R_2 by 10 kJ mol^{-1} . If k_1 and k_2 are rate constants for reactions R_1 and R_2 respectively at 300 K, then $\ln(k_2/k_1)$ is equal to :

$$(R = 8.314 \text{ J mol}^{-1}\text{K}^{-1})$$

[JEE-MAINS-2017]

$$(1) 8 \quad (2) 12 \quad (3) 6 \quad (4) 4$$

25. The rate of a reaction quadruples when the temperature changes from 300 to 310 K. The activation energy of this reaction is (Assume activation energy and pre-exponential factor are independent of temperature ; $\ln 2 = 0.693$, $R = 8.314 \text{ J mol}^{-1}\text{K}^{-1}$) : [MAINS-2017(online)]

$$(1) 107.2 \text{ kJ mol}^{-1}\text{K}^{-1} \quad (2) 53.6 \text{ kJ mol}^{-1}\text{K}^{-1}$$

$$(3) 214.4 \text{ kJ mol}^{-1}\text{K}^{-1} \quad (4) 26.8 \text{ kJ mol}^{-1}\text{K}^{-1}$$

26. The rate of a reaction A doubles on increasing the temperature from 300 to 310 K. By how much, the temperature of reaction B should be increased from 300 K so that rate doubles if activation energy of the reaction B is twice to that of reaction A :

[MAINS-2017(online)]

$$(1) 2.45 \text{ K}$$

$$(2) 4.92 \text{ K}$$

$$(3) 9.84 \text{ K}$$

$$(4) 19.67 \text{ K}$$

27. At 518°C , the rate of decomposition of a sample of gaseous acetaldehyde, initially at a pressure of 363 Torr, was 1.00 Torr s^{-1} when 5% had reacted and 0.5 Torr s^{-1} when 33% had reacted. The order of the reaction is : [JEE-MAINS-2018(offline)]

$$(1) 3$$

$$(2) 1$$

$$(3) 0$$

$$(4) 2$$

28. For a first order reaction, $\text{A} \rightarrow \text{P}$, $t_{1/2}$ (half life) is

10 days. The time required for $\frac{1}{4}$ conversion of

A (in days) is :- ($\ln 2 = 0.693$, $\ln 3 = 1.1$)

[MAINS-2018(online)]

$$(1) 5$$

$$(2) 4.1$$

$$(3) 3.2$$

$$(4) 2.5$$

29. N_2O_5 decomposes to NO_2 and O_2 and follows first order kinetics. After 50 minutes, the pressure inside the vessel increases from 50 mmHg to 87.5 mmHg. The pressure of the gaseous mixture after 100 minute at constant temperature will be:

[MAINS-2018(online)]

$$(1) 116.25 \text{ mmHg}$$

$$(2) 175.0 \text{ mmHg}$$

$$(3) 106.25 \text{ mmHg}$$

$$(4) 136.25 \text{ mmHg}$$

PREVIOUS YEARS QUESTIONS				ANSWER KEY			Exercise-II			
Que.	1	2	3	4	5	6	7	8	9	10
Ans.	1	2	1	3	4	1	1	4	0	1
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	2	1,2,4	9	4	2	2	3	1	3	3
Que.	21	22	23	24	25	26	27	28	29	
Ans.	3	3	2,3,4	4	1	2	4	2	3	

SOLUTION & COLLIGATIVE PROPERTIES

EXERCISE-I

- The boiling point of C_6H_6 , CH_3OH , $C_6H_5NH_2$ and $C_6H_5NO_2$ are $80^\circ C$, $65^\circ C$, $184^\circ C$ and $212^\circ C$ respectively which will show highest vapour pressure at room temperature :
 (1) C_6H_6 (2) CH_3OH
 (3) $C_6H_5NH_2$ (4) $C_6H_5NO_2$
- Mole fraction of A vapours above the solution in mixture of A and B ($X_A = 0.4$) will be
 [Given : $P_A^\circ = 100$ mm Hg and $P_B^\circ = 200$ mm Hg]
 (1) 0.4 (2) 0.8
 (3) 0.25 (4) none of these
- At a given temperature, total vapour pressure in Torr of a mixture of volatile components A and B is given by

$$P_{Total} = 120 - 75 X_B$$
 hence, vapour pressure of pure A and B respectively (in Torr) are
 (1) 120, 75 (2) 120, 195
 (3) 120, 45 (4) 75, 45
- Two liquids A & B form an ideal solution. What is the vapour pressure of solution containing 2 moles of A and 3 moles of B at 300 K? [Given : At 300 K, Vapour pr. of pure liquid A (P_A°) = 100 torr, Vapour pr. of pure liquid B (P_B°) = 300 torr]
 (1) 200 torr (2) 140 torr
 (3) 180 torr (4) None of these
- If Raoult's law is obeyed, the vapour pressure of the solvent in a solution is directly proportional to
 (1) Mole fraction of the solvent
 (2) Mole fraction of the solute
 (3) Mole fraction of the solvent and solute
 (4) The volume of the solution
- 1 mole of heptane (V. P. = 92 mm of Hg) was mixed with 4 moles of octane (V. P. = 31 mm of Hg). The vapour pressure of resulting ideal solution is :
 (1) 46.2 mm of Hg (2) 40.0 mm of Hg
 (3) 43.2 mm of Hg (4) 38.4 mm of Hg
- Mole fraction of A vapours above solution in mixture of A and B ($X_A = 0.4$) will be :
 ($P_A^\circ = 100$ mm, $P_B^\circ = 200$ mm)
 (1) 0.4 (2) 0.8
 (3) 0.25 (4) None
- The vapour pressure of a pure liquid 'A' is 70 torr at $27^\circ C$. It forms an ideal solution with another liquid B. The mole fraction of B is 0.2 and total vapour pressure of the solution is 84 torr at $27^\circ C$. The vapour pressure of pure liquid B at $27^\circ C$ is
 (1) 14 (2) 56 (3) 140 (4) 70
- At $88^\circ C$ benzene has a vapour pressure of 900 torr and toluene has a vapour pressure of 360 torr. What is the mole fraction of benzene in the mixture with toluene that will boil at $88^\circ C$ at 1 atm. pressure, benzene - toluene form an ideal solution:
 (1) 0.416 (2) 0.588 (3) 0.688 (4) 0.740
- The exact mathematical expression of Raoult's law is (n = moles of solute ; N = moles of solvent)
 (1) $\frac{P^0 - P_s}{P^0} = \frac{n}{N}$ (2) $\frac{P^0 - P_s}{P^0} = \frac{N}{n}$
 (3) $\frac{P^0 - P_s}{P_s} = \frac{n}{N}$ (4) $\frac{P^0 - P_s}{P^0} = n \times N$
- The vapour pressure of a solvent decreased by 10 mm of Hg when a non-volatile solute was added to the solvent. The mole fraction of solute in solution is 0.2, what would be mole fraction of the solvent if decrease in vapour pressure is 20 mm of Hg
 (1) 0.2 (2) 0.4 (3) 0.6 (4) 0.8
- The vapour pressure of a solution having solid as solute and liquid as solvent is :
 (1) Directly proportional to mole fraction of the solvent
 (2) Inversely proportional to mole fraction of the solvent
 (3) Directly proportional to mole fraction of the solute
 (4) Inversely proportional to mole fraction of the solute
- One mole of non volatile solute is dissolved in two moles of water. The vapour pressure of the solution relative to that of water is
 (1) $\frac{2}{3}$ (2) $\frac{1}{3}$ (3) $\frac{1}{2}$ (4) $\frac{3}{2}$
- The vapour pressure of pure A is 10 torr and at the same temperature when 1 g of B is dissolved in 20 gm of A, its vapour pressure is reduced to 9.0 torr. If the molecular mass of A is 200 amu, then the molecular mass of B is :
 (1) 100 amu (2) 90 amu
 (3) 75 amu (4) 120 amu

15. The vapour pressure of a pure liquid solvent (X) is decreased to 0.60 atm. from 0.80 atm on addition of a non volatile substance (Y). The mole fraction of (Y) in the solution is:-
 (1) 0.20 (2) 0.25 (3) 0.5 (4) 0.75
16. Among the following, that does not form an ideal solution is :
 (1) C_6H_6 and $C_6H_5CH_3$ (2) C_2H_5Cl and C_6H_5OH
 (3) C_6H_5Cl and C_6H_5Br (4) C_2H_5Br and C_2H_5I
17. Colligative properties of the solution depend upon
 (1) Nature of the solution
 (2) Nature of the solvent
 (3) Number of solute particles
 (4) Number of moles of solvent
18. When common salt is dissolved in water
 (1) Melting point of the solution increases
 (2) Boiling point of the solution increases
 (3) Boiling point of the solution decreases
 (4) Both Melting point and Boiling point is decreases
19. What should be the freezing point of aqueous solution containing 17 gm of C_2H_5OH in 1000 gm of water (water $K_f = 1.86 \text{ deg} - \text{kg mol}^{-1}$)
 (1) -0.69°C (2) -0.34°C
 (3) 0.0°C (4) 0.34°C
20. 5% solution of sucrose is isotonic with 1% solution of a compound 'A' then the molecular weight of compound 'A' is -
 (1) 32.4 (2) 68.4 (3) 121.6 (4) 34.2
21. Osmotic pressure of a sugar solution at 24°C is 2.5 atmosphere. The concentration of the solution in mole per litre is :
 (1) 10.25 (2) 1.025
 (3) 1025 (4) 0.1025
22. A solution containing 4 g of a non volatile organic solute per 100 ml was found to have an osmotic pressure equal to 500 cm of mercury at 27°C . The molecular weight of solute is :
 (1) 14.97 (2) 149.7 (3) 1697 (4) 1.497
23. Which of the following aqueous solution will show maximum vapour pressure at 300 K?
 (1) 1 M NaCl (2) 1 M $CaCl_2$
 (3) 1 M $AlCl_3$ (4) 1 M $C_{12}H_{22}O_{11}$
24. 1.0 molal aqueous solution of an electrolyte A_2B_3 is 60% ionised. The boiling point of the solution at 1 atm is ($K_{b(H_2O)} = 0.52 \text{ K kg mol}^{-1}$)
 (1) 274.76 K (2) 377 K
 (3) 376.4 K (4) 374.76 K
25. The Vant Hoff factor (i) for a dilute solution of $K_3[Fe(CN)_6]$ is (Assuming 100% ionsation) :
 (1) 10 (2) 4 (3) 5 (4) 0.25
26. The substance A when dissolved in solvent B shows the molecular mass corresponding to A_3 . The vant Hoff's factor will be -
 (1) 1 (2) 2 (3) 3 (4) $1/3$
27. The freezing point of 1 molal NaCl solution assuming NaCl to be 100% dissociated in water is : ($K_f = 1.86 \text{ K Molality}^{-1}$)
 (1) -1.86°C (2) -3.72°C
 (3) $+1.86^\circ\text{C}$ (4) $+3.72^\circ\text{C}$
28. What is the freezing point of a solution containing 8.1 gm. of HBr in 100gm. water assuming the acid to be 90% ionised (K_f for water = $1.86 \text{ K molality}^{-1}$):-
 (1) 0.85°C (2) -3.53°C
 (3) 0°C (4) -0.35°C
29. If a ground water contains H_2S at concentration of 2 mg/l, determine the pressure of H_2S in head space of a closed tank containing the ground water at 20°C . Given that for H_2S , Henry's constant is equal to $6.8 \times 10^3 \text{ bar}$ at 20°C .
 (1) 720 Pa (2) $77 \times 10^2 \text{ Pa}$
 (3) 553 Pa (4) $55 \times 10^2 \text{ Pa}$
30. A pressure cooker reduces cooking time for food because -
 (1) The higher pressure inside the cooker crushes the food material
 (2) Cooking involves chemical changes helped by a rise in temperature
 (3) Heat is more evenly distributed in the cooking space
 (4) Boiling point of water involved in cooking is increased

ANSWER KEY						Exercise-I				
Que.	1	2	3	4	5	6	7	8	9	10
Ans.	2	3	3	4	1	3	3	3	4	3
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	3	1	1	2	2	2	3	2	1	2
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	4	2	4	4	2	4	2	2	1	4

PREVIOUS YEARS' QUESTIONS

EXERCISE-II

- A 0.004 M solution of Na_2SO_4 is isotonic with a 0.010 M solution of glucose at same temperature. The apparent degree of dissociation of Na_2SO_4 is [JEE 2004]
(1) 25% (2) 50% (3) 75% (4) 85%
- The elevation in boiling point, when 13.44 g of freshly prepared CuCl_2 are added to one kilogram of water, is [Some useful data, $K_b(\text{H}_2\text{O}) = 0.52 \text{ K kg mol}^{-1}$, mol. wt. of $\text{CuCl}_2 = 134.4 \text{ gm}$] [JEE 2005]
(1) 0.05 (2) 0.1 (3) 0.16 (4) 0.21
- 72.5 g of phenol is dissolved in 1 kg of a solvent ($k_f = 14$) which leads to dimerization of phenol and freezing point is lowered by 7 kelvin. What percent of total phenol is present in dimeric form? [JEE 2006]
(1) 20 (2) 30 (3) 35 (4) 40
- When 20 g of naphtholic acid ($\text{C}_{11}\text{H}_8\text{O}_2$) is dissolved in 50 g of benzene ($K_f = 1.72 \text{ K kg mol}^{-1}$), a freezing point depression of 2 K is observed. The van't Hoff factor (i) is [JEE 2007]
(1) 0.5 (2) 1 (3) 2 (4) 3
- The degree of dissociation (α) of a weak electrolyte, A_xB_y is related to van't Hoff factor (i) by the expression :- [AIEEE-2011]
(1) $\alpha = \frac{x+y-1}{i-1}$ (2) $\alpha = \frac{x+y+1}{i-1}$
(3) $\alpha = \frac{i-1}{(x+y-1)}$ (4) $\alpha = \frac{i-1}{x+y+1}$
- K_f for water is $1.86 \text{ K kg mol}^{-1}$. If your automobile radiator holds 1.0 kg of water, how many grams of ethylene glycol ($\text{C}_2\text{H}_6\text{O}_2$) must you add to get the freezing point of the solution lowered to -2.8°C ? [AIEEE-2012]
(1) 27 g (2) 72 g (3) 93 g (4) 39 g
- A solution containing 0.85 g of ZnCl_2 in 125.0 g of water freezes at -0.23°C . The apparent degree of dissociation of the salt is : [JEE (MAIN)-2012 ONLINE]
(k_f for water = $1.86 \text{ K kg mol}^{-1}$, atomic mass ; $\text{Zn} = 65.3$ and $\text{Cl} = 35.5$)
(1) 1.36% (2) 2.47% (3) 73.5% (4) 7.35%
- Liquids A and B form an ideal solution. At 30°C , the total vapour pressure of a solution containing 1 mol of A and 2 moles of B is 250 mm Hg. The total vapour pressure becomes 300 mm Hg when 1 more mol of A is added to the first solution. The vapour pressures of pure A and B at the same temperature are [JEE (MAIN)-2012 ONLINE]
(1) 450, 150 mm Hg (2) 250, 300 mm Hg
(3) 125, 150 mm Hg (4) 150, 450 mm Hg
- The freezing point of a 1.00 m aqueous solution of HF is found to be -1.91°C . The freezing point constant of water, K_f , is $1.86 \text{ K kg mol}^{-1}$. The percentage dissociation of HF at this concentration is [JEE (MAIN)-2012 ONLINE]
(1) 2.7% (2) 30% (3) 10% (4) 5.2%
- How many grams of methyl alcohol should be added to 10 litre tank of water to prevent its freezing at 268 K ? [JEE (MAIN)-2013 ONLINE]
(K_f for water is $1.86 \text{ K kg mol}^{-1}$)
(1) 899.04 g (2) 886.02 g
(3) 868.06 g (4) 880.07 g
- Vapour pressure of pure benzene is 119 torr and that of toluene is 37.0 torr at the same temperature. Mole fraction of toluene in vapour phase which is in equilibrium with a solution of benzene and toluene having a mole fraction of toluene 0.50, will be : [JEE (MAIN)-2013 ONLINE]
(1) 0.137 (2) 0.205 (3) 0.237 (4) 0.435
- A molecule M associates in a given solvent according to the equation $\text{M} \rightleftharpoons (\text{M})_n$. For a certain concentration of M, the van't Hoff factor was found to be 0.9 and the fraction of associated molecules was 0.2. The value of n is : [JEE (MAIN)-2013 ONLINE]
(1) 2 (2) 4 (3) 5 (4) 3
- 12g of a nonvolatile solute dissolved in 108g of water produces the relative lowering of vapour pressure of 0.1. The molecular mass of the solute is : [JEE (MAIN)-2013 ONLINE]
(1) 60 (2) 80 (3) 40 (4) 20
- The molarity of a solution obtained by mixing 750 mL of 0.5(M)HCl with 250 mL of 2(M)HCl will be :- [JEE (MAIN)-2013]
(1) 0.875 M (2) 1.00 M (3) 1.75 M (4) 0.975 M
- The observed osmotic pressure for a 0.10 M solution of $\text{Fe}(\text{NH}_4)_2(\text{SO}_4)_2$ at 25°C is 10.8 atm. The expected and experimental (observed) values of Van't Hoff factor (i) will be respectively : ($R = 0.082 \text{ L atm K}^{-1} \text{ mol}^{-1}$) [JEE (MAIN)-2014 ONLINE]
(1) 3 and 5.42 (2) 5 and 3.42
(3) 4 and 4.00 (4) 5 and 4.42
- For an ideal Solution of two components A and B, which of the following is true ? [JEE (MAIN)-2014 ONLINE]
(1) $\Delta H_{\text{mixing}} < 0$ (zero)
(2) A – A, B – B and A – B interactions are identical
(3) A – B interaction is stronger than A – A and B – B interactions
(4) $\Delta H_{\text{mixing}} > 0$ (zero)
- Consider separate solution of 0.500 M $\text{C}_2\text{H}_5\text{OH}(\text{aq})$, 0.100 M $\text{Mg}_3(\text{PO}_4)_2(\text{aq})$, 0.250 M $\text{KBr}(\text{aq})$ and 0.125 M $\text{Na}_3\text{PO}_4(\text{aq})$ at 25°C . Which statement is true about these solutions, assuming all salts to be strong electrolytes ? [JEE (MAIN)-2014]
(1) 0.125 M $\text{Na}_3\text{PO}_4(\text{aq})$ has the highest osmotic pressure.
(2) 0.500 M $\text{C}_2\text{H}_5\text{OH}(\text{aq})$ has the highest osmotic pressure.
(3) They all have the same osmotic pressure.
(4) 0.100 M $\text{Mg}_3(\text{PO}_4)_2(\text{aq})$ has the highest osmotic pressure.

18. Consider separate solution of 0.500 M $C_2H_5OH(aq)$, 0.100 M $Mg_3(PO_4)_2(aq)$, 0.250 M $KBr(aq)$ and 0.125 M $Na_3PO_4(aq)$ at 25°C. Which statement is true about these solutions, assuming all salts to be strong electrolytes ? [JEE-MAIN-2014]
 (1) 0.125 M $Na_3PO_4(aq)$ has the highest osmotic pressure.
 (2) 0.500 M $C_2H_5OH(aq)$ has the highest osmotic pressure.
 (3) They all have the same osmotic pressure.
 (4) 0.100 M $Mg_3(PO_4)_2(aq)$ has the highest osmotic pressure.
19. Determination of the molar mass of acetic acid in benzene using freezing point depression is affected by : [JEE (MAIN)-2015 ONLINE]
 (1) association (2) dissociation
 (3) complex formation (4) partial ionization
20. A solution at 20°C is composed of 1.5 mol of benzene and 3.5 mol of toluene. If the vapour pressure of pure benzene and pure toluene at this temperature are 74.7 torr and 22.3 torr, respectively, then the total vapour pressure of the solution and the benzene mole fraction in equilibrium with it will be, respectively : [JEE (MAIN)-2015 ONLINE]
 (1) 38.0 torr and 0.589
 (2) 30.5 torr and 0.389
 (3) 35.8 torr and 0.280
 (4) 35.0 torr and 0.480
21. The vapour pressure of acetone at 20°C is 185 torr. When 1.2 g of non-volatile substance was dissolved in 100 g of acetone at 20°C, its vapour pressure was 183 torr. The molar mass ($g\ mol^{-1}$) of the substance is : [JEE (MAIN)-2015]
 (1) 128 (2) 488 (3) 32 (4) 64
22. An aqueous solution of a salt MX_2 at certain temperature has a van't Hoff factor of 2. The degree of dissociation for this solution of the salt is : [JEE (MAIN)-2016-ONLINE]
 (1) 0.50 (2) 0.80 (3) 0.67 (4) 0.33
23. The solubility of N_2 in water at 300 K and 500 torr partial pressure is 0.01 $g\ L^{-1}$. The solubility (in $g\ L^{-1}$) at 750 torr partial pressure is : [JEE (MAIN)-2016-ONLINE]
 (1) 0.02 (2) 0.005
 (3) 0.015 (4) 0.0075
24. 18 g glucose ($C_6H_{12}O_6$) is added to 178.2 g water. The vapour pressure of water (in torr) for this aqueous solution is : [JEE (MAIN)-2016]
 (1) 759.0 (2) 7.6 (3) 76.0 (4) 752.4
25. The freezing point of benzene decreases by 0.45°C when 0.2 g of acetic acid is added to 20 g of benzene. If acetic acid associates to form a dimer in benzene, percentage association of acetic acid in benzene will be :- (K_f for benzene = 5.12 $K\ kg\ mol^{-1}$) [JEE (MAIN)-2017]
 (1) 64.6% (2) 80.4% (3) 74.6% (4) 94.6%
26. 5 g of Na_2SO_4 was dissolved in x g of H_2O . The change in freezing point was found to be 3.82°C. If Na_2SO_4 is 81.5% ionised, the value of x (K_f for water = 1.86°C $kg\ mol^{-1}$) is approximately. (Molar mass of S = 32 $g\ mol^{-1}$ and that of Na = 23 $g\ mol^{-1}$) [JEE (MAIN)-2017-ONLINE]
 (1) 45 g (2) 65 g (3) 15 g (4) 25 g
27. A solution is prepared by mixing 8.5 g of CH_2Cl_2 and 11.95 g of $CHCl_3$. If vapour pressure of CH_2Cl_2 and $CHCl_3$ at 298 K are 415 and 200 mmHg respectively, the mole fraction of $CHCl_3$ in vapour form is: (Molar mass of Cl = 35.5 $g\ mol^{-1}$) [JEE (MAIN)-2017-ONLINE]
 (1) 0.486 (2) 0.325 (3) 0.162 (4) 0.675
28. For 1 molal aqueous solution of the following compounds, which one will show the highest freezing point ? [JEE (MAIN)-2018]
 (1) $[Co(H_2O)_5Cl]Cl_2 \cdot H_2O$
 (2) $[Co(H_2O)_4Cl_2]Cl \cdot 2H_2O$
 (3) $[Co(H_2O)_3Cl_3] \cdot 3H_2O$
 (4) $[Co(H_2O)_6]Cl_3$
29. Two 5 molal solutions are prepared by dissolving a non-electrolyte non-volatile solute separately in the solvents X and Y. The molecular weights of the solvents are M_x and M_y , respectively where

$$M_x = \frac{3}{4} M_y$$
 The relative lowering of vapour pressure of the solution in X is "m" times that of the solution in Y. Given that the number of moles of solute is very small in comparison to that of solvent, the value of "m" is - [JEE (MAIN)-2018-ONLINE]
 (1) $\frac{3}{4}$ (2) $\frac{4}{3}$ (3) $\frac{1}{2}$ (4) $\frac{1}{4}$
30. The mass of a non-volatile, non-electrolyte solute (molar mass = 50 $g\ mol^{-1}$) needed to be dissolved in 114 g octane to reduce its vapour pressure to 75%, is :- [JEE (MAIN)-2018-ONLINE]
 (1) 50 g (2) 37.5 g (3) 75 g (4) 150 g

PREVIOUS YEARS QUESTIONS				ANSWER KEY				Exercise-II			
Que.	1	2	3	4	5	6	7	8	9	10	
Ans.	3	3	3	1	3	3	3	1	1	3	
Que.	11	12	13	14	15	16	17	18	19	20	
Ans.	3	1	4	1	4	2	3	3	1	1	
Que.	21	22	23	24	25	26	27	28	29	30	
Ans.	4	1	3	4	4	1	2	3	1	4	

ELECTROCHEMISTRY

EXERCISE-I

- Which of the following statement is wrong about galvanic cell ?
 (1) cathode is positive charged
 (2) anode is negatively charged
 (3) reduction takes place at the anode
 (4) reduction takes place at the cathode
- A standard hydrogen electrode has zero electrode potential because
 (1) hydrogen is easier to oxidise
 (2) electrode potential is assumed to be zero
 (3) hydrogen atom has only one electron
 (4) hydrogen is the lightest element.
- A standard reduction electrode potentials of four metals are
 $A = -0.250 \text{ V}$, $B = -0.140 \text{ V}$
 $C = -0.126 \text{ V}$, $D = -0.402 \text{ V}$
 The metal that displaces A from its aqueous solution is :-
 (1) B (2) C
 (3) D (4) None of the above
- The standard electrode potentials for the reactions
 $\text{Ag}^+ (\text{aq}) + \text{e}^- \longrightarrow \text{Ag}(\text{s})$
 $\text{Sn}^{2+} (\text{aq}) + 2\text{e}^- \longrightarrow \text{Sn}(\text{s})$
 at 25°C are 0.80 volt and -0.14 volt, respectively.
 The standard emf of the cell.
 $\text{Sn}_{(\text{s})} | \text{Sn}^{2+}_{(\text{aq})} (1\text{M}) || \text{Ag}^+_{(\text{aq})} (1\text{M}) | \text{Ag}_{(\text{s})}$
 is :
 (1) 0.66 volt (2) 0.80 volt
 (3) 1.08 volt (4) 0.94 volt
- The thermodynamic efficiency of cell is given by-
 (1) $\frac{\Delta H}{\Delta G}$ (2) $\frac{nFE_{\text{cell}}}{\Delta G}$ (3) $-\frac{nFE_{\text{cell}}}{\Delta H}$ (4) Zero
- The reduction potential values are given below:
 $\text{Al}^{3+} | \text{Al} = -1.67 \text{ volt}$,
 $\text{Mg}^{2+} | \text{Mg} = -2.34 \text{ volt}$
 $\text{Cu}^{2+} | \text{Cu} = +0.34 \text{ volt}$,
 $\text{I}_2 | 2\text{I}^- = +0.53 \text{ volt}$
 Which one is the best reducing agent ?
 (1) Al (2) Mg (3) Cu (4) I_2
- $E^\circ(\text{Ni}^{2+} | \text{Ni}) = -0.25 \text{ volt}$, $E^\circ(\text{Au}^{3+} | \text{Au}) = 1.50 \text{ volt}$.
 The standard emf of the voltaic cell.
 $\text{Ni}_{(\text{s})} | \text{Ni}^{2+}_{(\text{aq})} (1.0 \text{ M}) || \text{Au}^{3+}_{(\text{aq})} (1.0 \text{ M}) | \text{Au}_{(\text{s})}$ is :
 (1) 1.25 volt (2) -1.75 volt
 (3) 1.75 volt (4) 4.0 volt

- From the following E° values of half cells,
 (i) $\text{A} + \text{e}^- \rightarrow \text{A}^-$; $E^\circ = -0.24 \text{ V}$
 (ii) $\text{B}^- + \text{e}^- \rightarrow \text{B}^{2-}$; $E^\circ = +1.25 \text{ V}$
 (iii) $\text{C}^- + 2\text{e}^- \rightarrow \text{C}^{3-}$; $E^\circ = -1.25 \text{ V}$
 (iv) $\text{D} + 2\text{e}^- \rightarrow \text{D}^{2-}$; $E^\circ = +0.68 \text{ V}$
 What combination of two half cells would result in a cell with the largest potential ?
 (1) (ii) and (iii) (2) (ii) and (iv)
 (3) (i) and (iii) (4) (i) and (iv)
- Which of the following will increase the voltage of the cell with following cell reaction
 $\text{Sn}_{(\text{s})} + 2\text{Ag}^+_{(\text{aq})} \rightarrow \text{Sn}^{2+}_{(\text{aq})} + 2\text{Ag}_{(\text{s})}$
 (1) Decrease in the concentration of Ag^+ ions
 (2) Increase in the concentration of Sn^{2+} ions
 (3) Increase in the concentration of Ag^+ ions
 (4) (1) & (2) both
- At 25°C the standard emf of cell having reactions involving two electrons change is found to be 0.295V. The equilibrium constant of the reaction is -
 (1) 29.5×10^{-2} (2) 10
 (3) 10^{10} (4) 29.5×10^{10}
- The emf of the cell in which the following reaction, $\text{Zn}(\text{s}) + \text{Ni}^{2+}_{(\text{aq})} (a = 0.1) \rightarrow \text{Zn}^{2+}_{(\text{aq})} (a = 1.0) + \text{Ni}(\text{s})$ occurs, is found to be 0.5105 V at 298 K. The standard e.m.f. of the cell is :-
 (1) -0.5105 V (2) 0.5400 V
 (3) 0.4810 V (4) 0.5696 V
- What is the potential of the cell containing two hydrogen electrodes as represented below
 $\text{Pt} | \text{H}_2(\text{g}) | \text{H}^+_{(\text{aq})} (10^{-8} \text{ M}) || \text{H}^+_{(\text{aq})} (0.001 \text{ M}) | \text{H}_2(\text{g}) | \text{Pt}$
 (1) -0.295 V (2) -0.0591 V
 (3) 0.295 V (4) 0.0591 V
- Consider the cell, $\text{Cu} | \text{Cu}^{2+} || \text{Ag}^+ | \text{Ag}$. If the concentration of Cu^{2+} and Ag^+ ions becomes ten times the emf of the cell :-
 (1) Becomes 10 times
 (2) Remains same
 (3) Increase by 0.0295 V
 (4) Decrease by 0.0295 V
- Given electrode potentials :
 $\text{Fe}^{3+}_{(\text{aq})} + \text{e}^- \longrightarrow \text{Fe}^{2+}_{(\text{aq})}$; $E^\circ = 0.771 \text{ volts}$
 $\text{I}_{2(\text{g})} + 2\text{e}^- \longrightarrow 2\text{I}^-_{(\text{aq})}$; $E^\circ = 0.536 \text{ volts}$
 E°_{cell} for the cell reaction,
 $2\text{Fe}^{3+}_{(\text{aq})} + 2\text{I}^-_{(\text{aq})} \longrightarrow 2\text{Fe}^{2+}_{(\text{aq})} + \text{I}_{2(\text{g})}$ is -
 (1) $(2 \times 0.771 - 0.536) = 1.006 \text{ volts}$
 (2) $(0.771 - 0.5 \times 0.536) = 0.503 \text{ volts}$
 (3) $0.771 - 0.536 = 0.235 \text{ volts}$
 (4) $0.536 - 0.771 = -0.235 \text{ volts}$

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				ANSWER KEY			Exercise-1			
Que.	1	2	3	4	5	6	7	8	9	10
Ans.	3	2	3	4	3	2	3	1	3	3
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	2	3	3	3	1	2	3	1	2	2
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	1	1	2	3	1	1	2	4	3	3

PREVIOUS YEARS' QUESTIONS

EXERCISE-II

1. A gas X at 1 atm is bubbled through a solution containing a mixture of 1 M Y and 1 M Z at 25°C. If the reduction potential of $Z > Y > X$, then

[JEE 1999]

- (1) Y will oxidise X and not Z
 (2) Y will oxidise Z and X
 (3) Y will oxidise both X and Z
 (4) Y will reduce both X and Z.

2. For the electrochemical cell, $M | M^+ || X^- | X$, $E^\circ(M^+ | M) = 0.44 \text{ V}$ and $E^\circ(X | X^-) = 0.33 \text{ V}$. From this data, one can deduce that

[JEE 2000]

- (1) $M + X \longrightarrow M^+ + X^-$ is the spontaneous reaction
 (2) $M^+ + X^- \longrightarrow M + X$ is the spontaneous reaction
 (3) $E_{\text{cell}} = 0.77 \text{ V}$
 (4) $E_{\text{cell}} = -0.77 \text{ V}$

3. Saturated solution of KNO_3 is used to make salt bridge because

[JEE 2001]

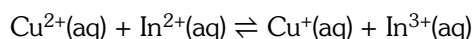
- (1) velocity of K^+ is greater than that of NO_3^-
 (2) velocity of NO_3^- is greater than that of K^+
 (3) velocities of both K^+ and NO_3^- are nearly the same
 (4) KNO_3 is highly soluble in water

4. In the electrolytic cell, flow of electrons is from:

[JEE 2003]

- (1) Cathode to anode in solution
 (2) Cathode to anode through external supply
 (3) Cathode to anode through internal supply
 (4) Anode to cathode through internal supply.

5. Find the equilibrium constant at 298 K for the reaction,



Given that

$$E^\circ_{\text{Cu}^{2+}|\text{Cu}^+} = 0.15 \text{ V}, E^\circ_{\text{In}^{3+}|\text{In}^{2+}} = -0.42 \text{ V},$$

$$E^\circ_{\text{In}^{2+}|\text{In}^+} = -0.40 \text{ V}$$

[JEE 2004]

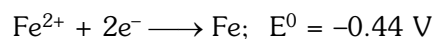
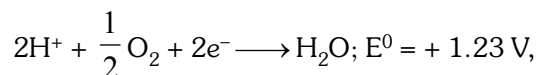
- (1) 10^4 (2) 10^6 (3) 10^8 (4) 10^{10}

6. $\text{Zn} | \text{Zn}^{2+} (a = 0.1 \text{ M}) || \text{Fe}^{2+} (a = 0.01 \text{ M}) | \text{Fe}$. The emf of the above cell is 0.2905 V. Equilibrium constant for the cell reaction is

[JEE 2004]

- (1) $10^{0.3210.0591}$ (2) $10^{0.3210.0295}$
 (3) $10^{0.2610.0295}$ (4) $e^{0.3210.295}$

7. The half cell reactions for rusting of iron are:

 ΔG^0 (in kJ) for the reaction is: [JEE 2005]

- (1) -76 (2) -322 (3) -122 (4) -176

8. The molar conductivities, $\Lambda^\circ_{\text{NaOAc}}$ and $\Lambda^\circ_{\text{HCl}}$ at infinite dilution in water at 25°C are 91.0 and 426.2 $\text{S cm}^2 \text{ mol}^{-1}$ respectively. To calculate $\Lambda^\circ_{\text{HOAc}}$

the additional value required is: [AIEEE 2006]

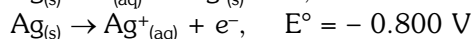
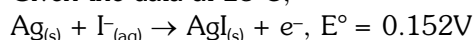
- (1) KCl (2) NaOH (3) NaCl (4) H_2O

9. Resistance of a conductivity cell filled with a solution of an electrolyte of concentration 0.1 M is 100 Ω . The conductivity of this solution is 1.29 S m^{-1} . Resistance of the same cell when filled with 0.02 M of the same solution is 520 Ω . The molar conductivity of 0.02 M solution of the electrolyte will be.

[AIEEE 2006]

- (1) $124 \times 10^{-4} \text{ S m}^2 \text{ mol}^{-1}$
 (2) $1240 \times 10^{-4} \text{ S m}^2 \text{ mol}^{-1}$
 (3) $1.24 \times 10^4 \text{ S m}^2 \text{ mol}^{-1}$
 (4) $12.4 \times 10^{-4} \text{ S m}^2 \text{ mol}^{-1}$

10. Given the data at 25°C,

What is the value of $\log K_{\text{sp}}$ for AgI?(Where K_{sp} = solubility product)

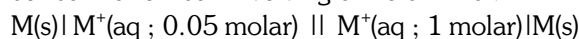
$$\left(2.303 \frac{RT}{F} = 0.059 \text{ V} \right)$$

[AIEEE 2006]

- (1) -8.12 (2) +8.612 (3) -37.83 (4) -16.13

Paragraph for Questions 11 to 12

The concentration of potassium ions inside a biological cell is at least twenty times higher than the outside. The resulting potential difference across the cell is important in several processes such as transmission of nerve impulses and maintaining the ion balance. A simple model for such a concentration cell involving a metal M is:



For the above electrolytic cell the magnitude of the cell potential $|E_{\text{cell}}| = 70 \text{ mV}$.

[JEE 2010]

11. For the above cell :-

- (1) $E_{\text{cell}} < 0$; $\Delta G > 0$ (2) $E_{\text{cell}} > 0$; $\Delta G < 0$
 (3) $E_{\text{cell}} < 0$; $\Delta G^0 > 0$ (4) $E_{\text{cell}} > 0$; $\Delta G^0 < 0$

12. If the 0.05 molar solution of M^+ is replaced by a 0.0025 molar M^+ solution, then the magnitude of the cell potential would be :-

- (1) 35 mV (2) 70 mV
 (3) 140 mV (4) 700 mV

24. Given [JEE-MAINS - 2017]

$$E^\circ_{\text{Cl}_2/\text{Cl}^-} = 1.36 \text{ V}, E^\circ_{\text{Cr}^{3+}/\text{Cr}} = -0.74 \text{ V}$$

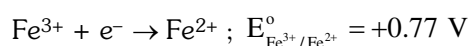
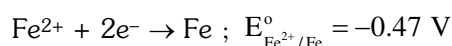
$$E^\circ_{\text{Cr}_2\text{O}_7^{2-}/\text{Cr}^{3+}} = 1.33 \text{ V}, E^\circ_{\text{MnO}_4^-/\text{Mn}^{2+}} = 1.51 \text{ V}$$

Among the following, the strongest reducing agent is

- (1) Cr (2) Mn^{2+} (3) Cr^{3+} (4) Cl^-

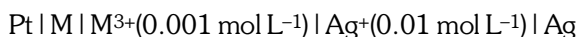
25. What is the standard reduction potential (E°) for $\text{Fe}^{3+} \rightarrow \text{Fe}$? [JEE-MAINS (ONLINE) 2017]

Given that :

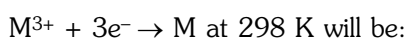


- (1) +0.30 V (2) +0.057 V
(3) -0.057 V (4) -0.30 V

26. To find the standard potential of $\text{M}^{3+} | \text{M}$ electrode, the following cell is constituted:



The emf of the cell is found to be 0.421 volt at 298 K. The standard potential of half reaction



[JEE-MAINS (ONLINE) 2017]

(Given $E^\circ_{\text{Ag}^+/\text{Ag}} = 0.80 \text{ Volt}$)

- (1) +0.30 V (2) +0.057 V
(3) -0.057 V (4) -0.30 V

27. How long (approximate) should water be electrolysed by passing through 100 amperes current so that the oxygen released can completely burn 27.66 g of diborane ?

[JEE-MAINS (OFFLINE) 2017]

(Atomic weight of B = 10.8 u)

- (1) 0.8 hours (2) 3.2 hours
(3) 1.6 hours (4) 6.4 hours

28. For the following cell : [JEE-Adv. 2017]



when the concentration of Zn^{2+} is 10 times the concentration of Cu^{2+} , the expression for ΔG (in J mol^{-1}) is

[F is Faraday constant, R is gas constant, T is temperature, $E^\circ(\text{cell}) = 1.1 \text{ V}$]

- (1) $2.303 RT + 1.1F$ (2) $2.303 RT - 2.2F$
(3) $1.1 F$ (4) $-2.2 F$

29. When an electric current is passed through acidified water, 112 mL of hydrogen gas at N.T.P. was collected at the cathode in 965 seconds. The current passed, in ampere, is : [JEE-MAINS (ONLINE) 2018]

- (1) 2.0 (2) 1.0
(3) 0.1 (4) 0.5

30. When 9.65 ampere current was passed for 1.0 hour into nitrobenzene in acidic medium, the amount of p-aminophenol produced is :-

[JEE-MAINS (ONLINE) 2018]

- (1) 10.9 g (2) 98.1 g
(3) 109.0 g (4) 9.81 g

PREVIOUS YEARS QUESTIONS				ANSWER KEY			Exercise-II			
Que.	1	2	3	4	5	6	7	8	9	10
Ans.	1	2	3	3	4	2	2	3	1	4
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	2	3	2	1	4	1	4	4	3	2
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	2	1	4	1	3	2	2	2	2	4

SOLID STATE

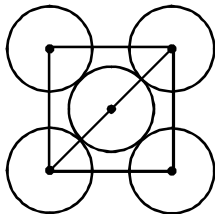
EXERCISE-I

- If the anions (A) form hexagonal closest packing and cations (C) occupy only $2/3$ octahedral voids in it, then the general formula of the compound is
(1) CA (2) CA_2 (3) C_2A_3 (4) C_3A_2
- Which one of the following schemes of ordering closest packed sheets of equal sized spheres do not generate close packed lattice.
(1) ABCABC (2) ABACABAC
(3) ABBAABBA (4) ABCBCABCBC
- Packing fraction in 2-D hexagonal arrangement of identical sphere is
(1) $\frac{\pi}{3\sqrt{2}}$ (2) $\frac{\pi}{3\sqrt{3}}$ (3) $\frac{\pi}{2\sqrt{3}}$ (4) $\pi/6$
- In fcc unit cell smallest distance between octahedral void & tetrahedral void is -
(a = edge length of unit cell)
(1) $\frac{a}{\sqrt{2}}$ (2) $\frac{\sqrt{3}a}{2}$ (3) a (4) $\frac{\sqrt{3}a}{4}$
- What is not true regarding hexagonal close packing (hcp)
(1) packing fraction is 0.74
(2) coordination number is 12
(3) ABC ABC.....type packing
(4) Containing both tetrahedral and octahedral voids
- How many unit cell are there in 1 gram cubic crystal of NaCl?
(1) $\frac{4 \times N_A}{58.5}$ (2) $\frac{N_A}{58.5}$
(3) $\frac{N_A}{58.5 \times 4}$ (4) $\frac{N_A}{58.5 \times 8}$
- The mass of a unit cell of CsCl corresponds to
(1) 1 Cs^+ and 1 Cl^- (2) 1 Cs^+ and 6 Cl^-
(3) 4 Cs^+ and 4 Cl^- (4) 8 Cs^+ and 1 Cl^-
- Edge length of M^+X^- (NaCl structure) is 7.2 Å. Assuming $M^+ - X^-$ contact along the cell edge, radius of X^- ion is ($r_{M^+} = 1.6 \text{ Å}$):
(1) 2.0 Å (2) 5.6 Å (3) 2.8 Å (4) 38 Å
- $r_{Na^+} = 95 \text{ pm}$ and $r_{Cl^-} = 181 \text{ pm}$ in NaCl (rock salt) structure. What is the shortest distance between Na^+ ions?
(1) 778.3 pm (2) 276 pm
(3) 195.7 pm (4) 390.3 pm
- The measured density of AgI is 6.94 g/cm^3 and the theoretical density is 5.67 g/cm^3 . These data indicate that solid AgI has -
(1) Schottky defect
(2) Frenkel defect
(3) Interstitial impurities defect
(4) Both (1) and (2)
- Which of the following statement is **CORRECT**?
(1) A metal can show only non- stoichiometric defects
(2) Schottky defect reduces the density of a solid due to significant increase in volume.
(3) Impurity defect always change the density.
(4) Solids having F-centres may have metal excess defect due to missing anions.
- Correct statement for ccp is :
(1) Each octahedral void is surrounded by 6 spheres and each sphere is surrounded by 4 octahedral voids
(2) Each octahedral void is surrounded by 6 spheres and each sphere is surrounded by 6 octahedral voids
(3) Each octahedral void is surrounded by 6 spheres and each sphere is surrounded by 8 octahedral voids
(4) Each octahedral void is surrounded by 6 spheres and each sphere is surrounded by 12 octahedral voids
- Which of the following statements is correct in the rock-salt structure of an ionic compounds?
(1) coordination number of cation is four whereas that of anion is six.
(2) coordination number of cation is six whereas that of anion is four.
(3) coordination number of each cation and anion is four.
(4) coordination number of each cation and anion is six.
- Column-I**
[Distance in terms of edge length of cube (a)]
(P) 0.866 a
(Q) 0.707 a
(R) 0.433 a
(S) a
Column-II
(I) Shortest distance between cation & anion in CsCl structure.
(II) Shortest distance between two cation in CaF_2 structure.
(III) Shortest distance between carbon atoms in diamond.
(IV) shortest distance between next nearest cations in rock salt structure.

Code :

P	Q	R	S
(1) IV	III	I	II
(2) I	II	III	IV
(3) III	II	I	IV
(4) I	II	IV	III

ALLEN

15. The no. of atoms per unit cell in B.C.C. & F.C.C. is respectively :
 (1) 8, 10 (2) 2, 4 (3) 1, 2 (4) 1, 3
16. How many unit cells are present in a cube-shaped ideal crystal of NaCl of mass 1.00g ?
 (1) 1.28×10^{21} unit cells (2) 1.71×10^{21} unit cells
 (3) 2.57×10^{21} unit cells (4) 5.14×10^{21} unit cells
17. What type of crystal defect is indicated in the diagram below ?
 $\begin{array}{cccccc} \text{Na}^+ & \text{Cl}^- & \text{Na}^+ & \text{Cl}^- & \text{Na}^+ & \text{Cl}^- \\ \text{Cl}^- & & \text{Cl}^- & \text{Na}^+ & & \text{Na}^+ \\ \text{Na}^+ & \text{Cl}^- & & \text{Cl}^- & \text{Na}^+ & \text{Cl}^- \\ \text{Cl}^- & \text{Na}^+ & \text{Cl}^- & \text{Na}^+ & & \text{Na}^+ \end{array}$
 (1) Frenkel defect
 (2) Schottky defect
 (3) Interstitial defect
 (4) Frenkel and Schottky defects
18. An ionic compound has a unit cell consisting of A ions at the corners of a cube and B ions on the centres of the faces of the cube. The empirical formula of this compound would be -
 (1) A_2B (2) AB (3) A_3B (4) AB_3
19. Lattice energy of an ionic compound depends upon-
 (1) Size of the ion only
 (2) Charge on the ion only
 (3) Charge on the ion and size of the ion
 (4) Packing of ions only
20. Total volume of atoms present in a face-centred cubic unit cell of a metal is (r is atomic radius) :
 (1) $\frac{24}{3}\pi r^3$ (2) $\frac{12}{3}\pi r^3$ (3) $\frac{16}{3}\pi r^3$ (4) $\frac{20}{3}\pi r^3$
21. In a compound, atoms of element Y form ccp lattice and those of element X occupy $\frac{2}{3}$ rd of tetrahedral voids. The formula of the compound will be -
 (1) X_4Y_3 (2) X_2Y_3 (3) X_2Y (4) X_3Y_4
22. The edge length of a face centred cubic cell of an ionic substance is 508 pm. If the radius of the cation is 110 pm, the radius of the anion is :-
 (1) 144 pm (2) 288 pm
 (3) 398 pm (4) 618 pm
23. Percentages of free space in cubic close packed structure and in body centred packed structure are respectively :-
 (1) 48% and 26% (2) 30% and 26%
 (3) 26% and 32% (4) 32% and 48%
24. The radius of a calcium ion is 94 pm and of the oxide ion is 146 pm. The possible crystal structure of calcium oxide will be :-
 (1) Octahedral (2) Tetrahedral
 (3) Pyramidal (4) Trigonal
25. The coordination number of a metal crystallising in a hcp structure is
 (1) 12 (2) 4 (3) 8 (4) 6
26. In a solid "AB" having NaCl structure "A" atoms occupy the corners of the cubic unit cell. If all the face-centred atoms along one of the axes are removed, then the resultant stoichiometry of the solid is
 (1) AB_2 (2) A_2B (3) A_4B_3 (4) A_3B_4
27. A substance A_xB_y crystallises in a FCC lattice in which atoms "A" occupy each corner of the cube and atoms "B" occupy the centres of each face of the cube. Identify the correct composition of the substance A_xB_y .
 (1) AB_3
 (2) A_4B_3
 (3) A_3B
 (4) composition cannot be specified
28. Which of the following FCC structure contains cations in alternate tetrahedral voids ?
 (1) NaCl (2) ZnS (3) Na_2O (4) CaF_2
29. The correct statement(s) regarding defects in solid is (are)
 (1) Frenkel defect is usually favoured by a very small difference in the sizes of cation and anion.
 (2) Frenkel defect is a dislocation defect
 (3) Trapping of an electron in the lattice leads to the formation of F-center.
 (4) Schottky defects have no effect on the physical properties of solids.
30. The packing efficiency of the two-dimensional square unit cell shown below is

 (1) 39.27% (2) 68.02%
 (3) 74.05% (4) 78.54%

ANSWER KEY

Exercise-I

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	3	3	2	4	3	3	1	1	4	3
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	4	2	4	2	2	3	2	4	3	3
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	1	1	3	1	1	4	1	2	2,3	4

PREVIOUS YEARS' QUESTIONS

EXERCISE-II

1. A metal crystallises into two cubic phases, FCC and BCC whose unit cell lengths are 3.5 and 3.0 Å respectively. Calculate the ratio of densities of FCC and BCC. [JEE-1999]
2. An element crystallises in FCC lattice having edge length 400 pm. Calculate the maximum diameter which can be placed in interstitial sites without disturbing the structure. [JEE 2005]
3. The edge length of unit cell of a metal having atomic weight 75 g/mol is 5 Å which crystallizes in cubic lattice. If the density is 2 g/cc then find the radius of metal atom. ($N_A = 6 \times 10^{23}$). Give the answer in pm. [JEE 2006]

Paragraph for Question No.4 to 6

In hexagonal systems of crystals, a frequently encountered arrangement of atoms is described as a hexagonal prism. Here, the top and bottom of the cell are regular hexagons and three atoms are sandwiched in between them. A space-filling model of this structure, called hexagonal close-packed (HCP), is constituted of a sphere on a flat surface surrounded in the same plane by six identical spheres as closely as possible. Three spheres are then placed over the first layer so that they touch each other and represent the second layer. Each one of these three spheres touches three spheres of the bottom layer. Finally, the second layer is covered with a third layer that is identical to the bottom layer in relative position. Assume radius of every sphere to be 'r'.

4. The number of atoms in this HCP unit cells is [JEE 2008]
(1) 4 (2) 6 (3) 12 (4) 17
5. The volume of this HCP unit cell is [JEE 2008]
(1) $24\sqrt{2}r^3$ (2) $16\sqrt{2}r^3$
(3) $12\sqrt{2}r^3$ (4) $\frac{64}{3\sqrt{3}}r^3$
6. The empty space in this HCP unit cell is [JEE 2008]
(1) 74% (2) 47.6 %
(3) 32% (4) 26%
7. Copper crystallises in fcc with a unit cell edge length of 361pm. What is the radius of copper atom? [AIEEE-2011]
(1) 181pm (2) 128pm
(3) 157pm (4) 108pm

8. Ammonium chloride crystallizes in a body centred cubic lattice with edge length of unit cell of 390 pm. If the size of chloride ion is 180 pm, the size of ammonium ion would be : [JEE-Main (online)-12]

- (1) 158 pm (2) 174 pm
(3) 142 pm (4) 126 pm

9. A solid has 'bcc' structure. If the distance of nearest approach between two atoms is 1.73 Å, the edge length of the cell is :- [JEE-Main (online)-12]

- (1) 314.20 pm (2) 216 pm
(3) 200 pm (4) 1.41 pm

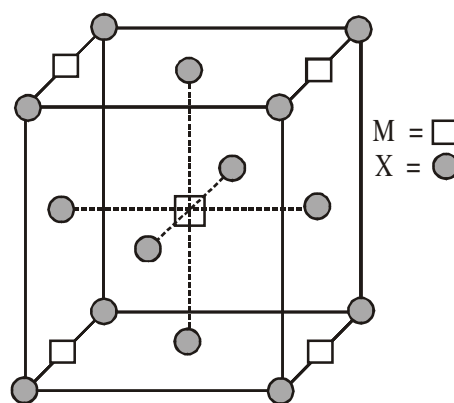
10. Among the following the incorrect statement is :- [JEE-Main (online)-12]

- (1) Density of crystals remains unaffected due to Frenkel defect
(2) In BCC unit cell the void space is 32%
(3) Electrical conductivity of semiconductors and metals increases with increase in temperature
(4) Density of crystals decreases due to Schottky defect

11. Lithium forms body centred cubic structure. The length of the side of its unit cell is 351 pm. Atomic radius of the lithium will be :- [JEE-Main (offline)-12]

- (1) 152 pm (2) 75 pm
(3) 300 pm (4) 240 pm

12. A compound M_pX_q has cubic close packing (ccp) arrangement of X. Its unit cell structure is shown below. The empirical formula of the compound is : [JEE-2012]



- (1) MX (2) MX_2
(3) M_2X (4) M_5X_{14}

13. In a face centred cubic lattice, atoms of A form the corner points and atoms of B form the face centred points. If two atoms of A are missing from the corner points, the formula of the ionic compound is

[JEE-Main (online)-13]

- (1) AB_2 (2) AB_3 (3) AB_4 (4) A_2B_5

14. Which one of the following statements about packing in solids is **incorrect** ?

[JEE-Main (online)-13]

- (1) Void space in ccp mode of packing is 26%
 (2) Coordination number in hcp mode of packing is 12
 (3) Void space in hcp mode of packing is 32%
 (4) Coordination number in bcc mode of packing is 8

15. An element having an atomic radius of 0.14 nm crystallizes in an fcc unit cell. What is the length of a side of the cell ?

[JEE-Main (online)-13]

- (1) 0.96 nm (2) 0.4 nm
 (3) 0.24 nm (4) 0.56 nm

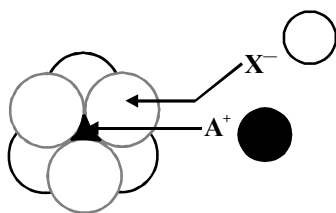
16. Experimentally it was found that a metal oxide has formula $M_{0.98}O$. Metal M, is present as M^{2+} and M^{3+} in its oxide. Fraction of the metal which exists as M^{3+} would be :-

[JEE-Main (offline)-13]

- (1) 7.01% (2) 4.08%
 (3) 6.05% (4) 5.08

17. The arrangement of X^- ions around A^+ ion in solid AX is given in the figure (not drawn to scale). If the radius of X^- is 250 pm, the radius of A^+ is -

[JEE-2013]



- (1) 104 pm (2) 125 pm
 (3) 183 pm (4) 57 pm

18. The total number of octahedral void(s) per atom present in a cubic close packed structure is :-

[JEE-Main (online)-14]

- (1) 1 (2) 2 (3) 3 (4) 4

19. In a monoclinic unit cell, the relation of sides and angles are respectively

[JEE-Main (online)-14]

- (1) $a \neq b \neq c$ and $\alpha \neq \beta \neq \gamma \neq 90^\circ$
 (2) $a \neq b \neq c$ and $\beta = \gamma = 90^\circ \neq \alpha$
 (3) $a = b \neq c$ and $\alpha = \beta = \gamma = 90^\circ$
 (4) $a \neq b \neq c$ and $\alpha = \beta = \gamma = 90^\circ$

20. The appearance of colour in solid alkali metal halides is generally due to :

[JEE-Main (online)-14]

- (1) Frenkel defect
 (2) F-centres
 (3) Schottky defect
 (4) Interstitial position

21. In a face centred cubic lattice atoms A are at the corner points and atoms B at the face centred points. If atom B is missing from one of the face centred points, the formula of the ionic compound is :

[AIEEE-2011, JEE-Main (online)-14]

- (1) AB_2 (2) A_2B_3
 (3) A_5B_2 (4) A_2B_5

22. CsCl crystallises in body centred cubic lattice. if 'a' is its edge length then which of the following expression is correct :

[JEE-Main (offline)-14]

- (1) $r_{Cs^+} + r_{Cl^-} = \frac{\sqrt{3}}{2}a$ (2) $r_{Cs^+} + r_{Cl^-} = \sqrt{3}a$
 (3) $r_{Cs^+} + r_{Cl^-} = 3a$ (4) $r_{Cs^+} + r_{Cl^-} = \frac{3a}{2}$

23. Sodium metal crystallizes in a body centred cubic lattice with a unit cell edge of 4.29 Å. The radius of sodium atom is approximately :-

[JEE-Main (offline)-15]

- (1) 5.72 Å (2) 0.93 Å
 (3) 1.86 Å (4) 3.022 Å

24. If the unit cell of a mineral has cubic close packed (ccp) array of oxygen atoms with m fraction of octahedral holes occupied by aluminium ions and n fraction of tetrahedral holes occupied by magnesium ions m and n respectively, are -

[JEE-2015]

- (1) $\frac{1}{2}, \frac{1}{8}$ (2) $1, \frac{1}{4}$
 (3) $\frac{1}{2}, \frac{1}{2}$ (4) $\frac{1}{4}, \frac{1}{8}$

25. Which of the following compounds is metallic and ferromagnetic ?

[JEE-Main (offline)-16]

- (1) MnO_2 (2) TiO_2
 (3) CrO_2 (4) VO_2

26. The **CORRECT** statement(s) for cubic close packed (ccp) three dimensional structure is (are)

[JEE-2016]

- (1) The number of the nearest neighbours of an atom present in the topmost layer is 12
- (2) The efficiency of atom packing is 74%
- (3) The number of octahedral and tetrahedral voids per atom are 1 and 2, respectively
- (4) The unit cell edge length is $2\sqrt{2}$ times the radius of the atom

27. A metal crystallises in a face centred cubic structure. If the edge length of its unit cell is 'a', the closest approach between two atoms in metallic crystal will be :-

[JEE-Main (offline)-17]

- (1) $2a$
- (2) $2\sqrt{2}a$
- (3) $\sqrt{2}a$
- (4) $\frac{a}{\sqrt{2}}$

28. Which type of defect has the presence of cations in the interstitial sites -

[JEE-Main (offline)-18]

- (1) Vacancy defect
- (2) Frenkel defect
- (3) Metal deficiency defect
- (4) Schottky defect

29. All of the following share the same crystal structure except :-

[JEE-Main (online)-18]

- (1) RbCl
- (2) CsCl
- (3) LiCl
- (4) NaCl

30. Which of the following arrangements shows the schematic alignment of magnetic moments of antiferromagnetic substance ?

[JEE-Main (online)-18]

- (1) $\uparrow \downarrow \downarrow \downarrow \downarrow \uparrow$
- (2) $\uparrow \uparrow \uparrow \uparrow \uparrow \uparrow$
- (3) $\uparrow \uparrow \downarrow \uparrow \uparrow \downarrow$
- (4) $\uparrow \downarrow \uparrow \downarrow \uparrow \downarrow$

PREVIOUS YEARS QUESTIONS				ANSWER KEY			Exercise-II			
Que.	1	2	3	4	5	6	7	8	9	10
Ans.	1,2,5,9	117.1 PM	216.59 PM	2	1	4	2	1	3	3
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	1	2	3	3	2	2	1	1	2	2
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	4	1	3	1	1	2,3,4	4	2	2	4

CHEMICAL EQUILIBRIUM

EXERCISE-I

- In a chemical equilibrium, the rate constant for the backward reaction is 2×10^{-4} and the equilibrium constant is 1.5. The rate constant for the forward reaction is:-
 (1) 2×10^{-3} (2) 5×10^{-4}
 (3) 3×10^{-4} (4) 9.0×10^{-4}
- For the reaction

$$\text{CuSO}_4 \cdot 5\text{H}_2\text{O}_{(s)} \rightleftharpoons \text{CuSO}_4 \cdot 3\text{H}_2\text{O}_{(s)} + 2\text{H}_2\text{O}_{(g)}$$
 Which one is correct representation :-
 (1) $K_p = (P_{\text{H}_2\text{O}})^2$ (2) $K_c = [\text{H}_2\text{O}]^2$
 (3) $K_p = K_c(RT)^2$ (4) All
- $\log \frac{K_p}{K_c} + \log RT = 0$ is true relationship for the following reaction:-
 (1) $\text{PCl}_5 \rightleftharpoons \text{PCl}_3 + \text{Cl}_2$
 (2) $2\text{SO}_2 + \text{O}_2 \rightleftharpoons 2\text{SO}_3$
 (3) $\text{N}_2 + 3\text{H}_2 \rightleftharpoons 2\text{NH}_3$
 (4) (2) and (3) both
- For any reversible reaction if concentration of reactants increases then effect on equilibrium constant:-
 (1) Depends on amount of concentration
 (2) Unchange
 (3) Decrease
 (4) Increase
- Sulfide ion in alkaline solution reacts with solid sulfur to form polysulfide ions having formulas S_2^{2-} , S_3^{2-} , S_4^{2-} and so on. The equilibrium constant for the formation of S_2^{2-} is 12 (K_1) & for the formation of S_3^{2-} is 132 (K_2), both from S and S^{2-} . What is the equilibrium constant for the formation of S_3^{2-} from S_2^{2-} and S?
 (1) 11 (2) 12
 (3) 132 (4) None of these
- What should be the value of K_c for the reaction

$$2\text{SO}_{2(g)} + \text{O}_{2(g)} \rightleftharpoons 2\text{SO}_{3(g)}$$
 if the amount are $\text{SO}_3 = 48$ g, $\text{SO}_2 = 12.8$ g and $\text{O}_2 = 9.6$ g at equilibrium and the volume of the container is one litre?
 (1) 64 (2) 30 (3) 42 (4) 8.5
- For the reaction $\text{A} + 2\text{B} \rightleftharpoons 2\text{C} + \text{D}$, initial concentration of A is a and that of B is 1.5 times that of A. Concentration of A and D are same at equilibrium. What should be the concentration of B at equilibrium ?
 (1) $\frac{a}{4}$ (2) $\frac{a}{2}$
 (3) $\frac{3a}{4}$ (4) All of the above.
- The degree of dissociation of SO_3 is α at equilibrium pressure p^0 .

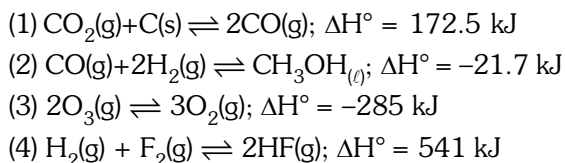
$$K_p \text{ for } 2\text{SO}_3(g) \rightleftharpoons 2\text{SO}_2(g) + \text{O}_2(g)$$
 (1) $\frac{p^0 \alpha^3}{2(1-\alpha)^3}$ (2) $\frac{p^0 \alpha^3}{(2+\alpha)(1-\alpha)^2}$
 (3) $\frac{p^0 \alpha^2}{2(1-\alpha)^2}$ (4) None of these
- The equilibrium constant for the reaction

$$\text{A}(g) + 2\text{B}(g) \rightleftharpoons \text{C}(g)$$
 is $0.25 \text{ dm}^6 \text{ mol}^{-2}$. In a volume of 5 dm^3 , what amount of A must be mixed with 4 mol of B to yield 1 mol of C at equilibrium.
 (1) 3 moles (2) 24 moles
 (3) 26 moles (4) None of these
- The equilibrium constant K_c for the reaction,

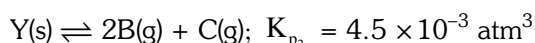
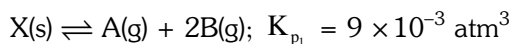
$$\text{A}(g) + 2\text{B}(g) \rightleftharpoons 3\text{C}(g)$$
 is 2×10^{-3}
 What would be the equilibrium partial pressure of gas C if initial pressure of gas A & B are 1 & 2 atm respectively.
 (1) 0.0625 atm (2) 0.1875 atm
 (3) 0.21 atm (4) None of these
- At 675 K, $\text{H}_2(g)$ and $\text{CO}_2(g)$ react to form $\text{CO}(g)$ and $\text{H}_2\text{O}(g)$, K_p for the reaction is 0.16.
 If a mixture of 0.25 mole of $\text{H}_2(g)$ and 0.25 mol of CO_2 is heated at 675 K, mole % of $\text{CO}(g)$ in equilibrium mixture is :
 (1) 7.14 (2) 14.28
 (3) 28.57 (4) 33.33

- 12.** The equilibrium constant K_p (in atm) for the reaction is 9 at 7 atm and 300 K.
- $$A_2(g) \rightleftharpoons B_2(g) + C_2(g)$$
- Calculate the average molar mass (in gm/mol) of an equilibrium mixture.
- Given :** Molar mass of A_2 , B_2 and C_2 are 70, 49 & 21 gm/mol respectively.
- (1) 50 (2) 45
(3) 40 (4) 37.5
- 13.** The equilibrium $2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$ shifts forward if :-
- (1) A catalyst is used.
(2) An adsorbent is used to remove SO_3 as soon as it is formed.
(3) Small amounts of reactants are removed.
(4) None of these
- 14.** Change in volume of the system does not alter the number of moles in which of the following equilibrium
- (1) $N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$
(2) $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$
(3) $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$
(4) $SO_2Cl_2(g) \rightleftharpoons SO_2(g) + Cl_2(g)$
- 15.** Densities of diamond and graphite are 3.5 and 2.3 gm/mL.
- $$C(\text{diamond}) \rightleftharpoons C(\text{graphite})$$
- $$\Delta_f H = -1.9 \text{ kJ/mole}$$
- favourable conditions for formation of diamond are
- (1) high pressure and low temperature
(2) low pressure and high temperature
(3) high pressure and high temperature
(4) low pressure and low temperature
- 16.** The equilibrium $SO_2Cl_2(g) \rightleftharpoons SO_2(g) + Cl_2(g)$ is attained at 25°C in a closed rigid container and an inert gas, helium is introduced. Which of the following statements is/are correct.
- (1) concentrations of SO_2 , Cl_2 and SO_2Cl_2 do not change
(2) more chlorine is formed
(3) concentration of SO_2 is reduced
(4) more SO_2Cl_2 is formed
- 17.** When $NaNO_3$ is heated in a closed vessel, oxygen is liberated and $NaNO_2$ is left. At equilibrium
- (1) addition of $NaNO_2$ favours reverse reaction
(2) addition of $NaNO_3$ favours forward reaction
(3) increasing temperature favours forward reaction
(4) increasing pressure favours forward reaction
- 18.** Consider the equilibrium $HgO(s) + 4I^-(aq) + H_2O(l) \rightleftharpoons HgI_4^{2-}(aq) + 2OH^-(aq)$, which changes will decrease the equilibrium concentration of HgI_4^{2-}
- (1) Addition of 0.1 M HI (aq)
(2) Addition of $HgO(s)$
(3) Removal of $H_2O(l)$
(4) Addition of KOH (aq)
- 19.** The equilibrium constant for the reaction $N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$ is 4×10^{-4} at 200 K. In presence of a catalyst, equilibrium is attained ten times faster. Therefore, the equilibrium constant in presence of the catalyst at 200 K is :-
- (1) 40×10^{-4}
(2) 4×10^{-4}
(3) 4×10^{-3}
(4) difficult to compute without more data
- 20.** In the reaction $X(g) + Y(g) \rightleftharpoons 2Z(g)$, 2 mol X, 1 mol Y and 1 mole Z are placed in a 10 litre vessel and allowed to reach equilibrium. If final concentration of Z is 0.2 M, then K_c for the given reaction is :-
- (1) 1.60 (2) $\frac{80}{3}$
(3) $\frac{16}{3}$ (4) $\frac{100}{3}$
- 21.** 9.2 grams of $N_2O_4(g)$ is taken in a closed one litre vessel and heated till the following equilibrium is reached $N_2O_4(g) \rightleftharpoons 2NO_2(g)$
- At equilibrium, 50% $N_2O_4(g)$ is dissociated. What is the equilibrium constant (in mol litre^{-1}) (molecular weight of $N_2O_4 = 92$) :-
- (1) 0.1
(2) 0.4
(3) 0.2
(4) 2

22. For which of the following reaction is product formation favoured by low pressure and low temperature?



23. Two solid compounds X and Y dissociates at a certain temperature as follows



The total pressure of gases over a mixture of X and Y is :-

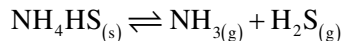
- (1) 4.5 atm (2) 0.45 atm
 (3) 0.6 atm (4) None of these

24. For the following reaction :

$\text{A} + 2\text{B} \rightleftharpoons \text{C} + 3\text{D}$; if the partial pressure of all the four substances A, B, C and D at equilibrium are 0.20, 0.10, 0.30 and 0.50 respectively, then calculate the equilibrium constant :-

- (1) 18.75 (2) 5.3
 (3) 11.25 (4) None of these

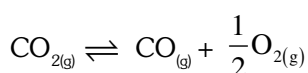
25. For the dissociation reaction :



if $K_p = 25 \text{ atm}^2$; then the equilibrium pressure of the system is :-

- (1) 25 atm (2) 5 atm
 (3) 10 atm (4) 15 atm

26. Consider the equilibrium



The equilibrium constant K is given by (when $\alpha < 1$)

- (1) $K = \frac{\alpha^{3/2}}{\sqrt{2}}$ (2) $K = \frac{\alpha^3}{2}$
 (3) $K = \frac{\alpha^{3/2}}{2}$ (4) $K = \frac{\alpha^{3/2}}{\sqrt{3}}$

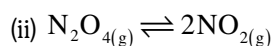
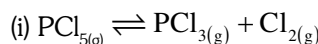
27. For the reaction, $\text{N}_2\text{O}_{4(\text{g})} \rightleftharpoons 2\text{NO}_{2(\text{g})}$, If percentage dissociation of N_2O_4 are 25%, 50%, 75% and 100% then the sequence of observed vapour densities d_1, d_2, d_3 and d_4 at these conditions will be :-

- (1) $d_1 > d_2 > d_3 > d_4$ (2) $d_4 > d_3 > d_2 > d_1$
 (3) $d_1 = d_2 = d_3 = d_4$ (4) None of these

28. For the reaction $\text{NH}_4\text{HS}(\text{s}) \rightleftharpoons \text{NH}_3(\text{g}) + \text{H}_2\text{S}(\text{g})$ in a closed flask, the equilibrium pressure is P atm. The standard free energy change (ΔG°) of the reaction would be :-

- (1) $-RT \ln p$
 (2) $-2.303 RT(\log P - \log 2)$
 (3) $-2RT (\ln P - \ln 2)$
 (4) $-2RT \ln P$

29. Consider the reactions :-



The addition of inert gas at constant volume :

- (1) will increase the dissociation of PCl_5 as well as N_2O_4 .
 (2) will reduce the dissociation of PCl_5 as well as N_2O_4 .
 (3) will increase the dissociation of PCl_5 while reduce the dissociation of N_2O_4 .
 (4) will not disturb the equilibrium of the reactions

30. When sulphur in the form of gaseous S_8 is heated at 900K, the initial pressure of 1 atm is decreased by 30% at equilibrium. This is because of conversion of some $\text{S}_{8(\text{g})}$ to $\text{S}_{2(\text{g})}$. Find the value of equilibrium constant for this reaction :-

- (1) 1.96 (2) 4.48
 (3) 5.56 (4) 2.96

ANSWER KEY

Exercise-I

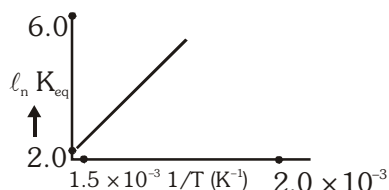
Que.	1	2	3	4	5	6	7	8	9	10
Ans.	3	4	2	2	1	2	2	2	3	2
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	2	3	2	1	3	1	3	4	2	3
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	3	3	2	1	3	1	1	3	4	4

PREVIOUS YEARS' QUESTIONS

EXERCISE-II

- When two reactants A and B are mixed to give products C and D, the reaction quotient Q, at the initial stages of the reaction : **[JEE 2000]**
 - (1) is zero
 - (2) decrease with time
 - (3) independent of time
 - (4) increases with time
- For the reversible reaction : $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$ at 500°C . The value of K_p is 1.44×10^{-5} , when partial pressure is measured in atmospheres. The corresponding value of K_c with concentration in mol L^{-1} is : **[JEE 2000]**
 - (1) $1.44 \times 10^{-5} / (0.082 \times 500)^2$
 - (2) $1.44 \times 10^{-5} / (8.314 \times 773)^2$
 - (3) $1.44 \times 10^{-5} / (0.082 \times 500)^2$
 - (4) $1.44 \times 10^{-5} / (0.082 \times 773)^{-2}$
- At constant temperature, the equilibrium constant (K_p) for the decomposition reaction, $\text{N}_2\text{O}_4 \rightleftharpoons 2\text{NO}_2$ is expressed by $K_p = 4x^2P/(1-x^2)$ where P is pressure, x is extent of decomposition. Which of the following statement is true ? **[JEE 2001]**
 - (1) K_p increases with increase of P
 - (2) K_p increases with increase of x
 - (3) K_p increases with decrease of x
 - (4) K_p remains constant with change in P or x
- One of the following equilibrium is not affected by change in volume of the flask - **[AIEEE-2002]**
 - (1) $\text{PCl}_5(\text{g}) \rightleftharpoons \text{PCl}_3(\text{g}) + \text{Cl}_2(\text{g})$
 - (2) $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$
 - (3) $\text{N}_2(\text{g}) + \text{O}_2 \rightleftharpoons 2\text{NO}(\text{g})$
 - (4) $\text{SO}_2\text{Cl}_2(\text{g}) \rightleftharpoons \text{SO}_2(\text{g}) + \text{Cl}_2(\text{g})$
- Consider the following equilibrium in a closed container : $\text{N}_2\text{O}_4(\text{g}) \rightleftharpoons 2\text{NO}_2(\text{g})$.
At a fixed temperature, the volume of the reaction container is halved. For this change, which of the following statements holds true regarding the equilibrium constant (K_p) and degree of dissociation (α) : **[JEE 2002]**
 - (1) Neither K_p nor α changes
 - (2) Both K_p and α change
 - (3) K_p changes, but α does not change
 - (4) K_p does not change, but α changes
- For the reaction equilibrium, $\text{N}_2\text{O}_4(\text{g}) \rightleftharpoons 2\text{NO}_2(\text{g})$ the concentration of N_2O_4 and NO_2 at equilibrium are 4.8×10^{-2} and $1.2 \times 10^{-2} \text{ mol L}^{-1}$ respectively. The value of K_c for the reaction is- **[AIEEE-2003]**
 - (1) $3 \times 10^{-3} \text{ mol L}^{-1}$
 - (2) $3 \times 10^3 \text{ mol L}^{-1}$
 - (3) $3.3 \times 10^2 \text{ mol L}^{-1}$
 - (4) $3 \times 10^{-1} \text{ mol L}^{-1}$
- What is the equilibrium expression for the reaction $\text{P}_{4(\text{s})} + 5\text{O}_{2(\text{g})} \rightleftharpoons \text{P}_4\text{O}_{10(\text{s})}$? **[AIEEE-2004]**
 - (1) $K_c = [\text{P}_4\text{O}_{10}] / [\text{P}_4] [\text{O}_2]^5$
 - (2) $K_c = [\text{P}_4\text{O}_{10}] / 5 [\text{P}_4] [\text{O}_2]$
 - (3) $K_c = [\text{O}_2]^5$
 - (4) $K_c = 1 / [\text{O}_2]^5$
- For the reaction $\text{CO}(\text{g}) + \text{Cl}_{2(\text{g})} \rightleftharpoons \text{COCl}_{2(\text{g})}$ the $\frac{K_p}{K_c}$ is equal to - **[AIEEE-2004]**
 - (1) $\frac{1}{RT}$
 - (2) RT
 - (3) $\sqrt{RT}$
 - (4) 1.0
- The equilibrium constant for the reaction $\text{N}_{2(\text{g})} + \text{O}_{2(\text{g})} \rightleftharpoons 2\text{NO}(\text{g})$ at temperature T is 4×10^{-4} . The value of K_c for the reaction $\text{NO}(\text{g}) \rightleftharpoons \frac{1}{2}\text{N}_{2(\text{g})} + \frac{1}{2}\text{O}_{2(\text{g})}$ **[AIEEE-2004]**
 - (1) 2.5×10^2
 - (2) 50
 - (3) 4×10^{-4}
 - (4) 0.02
- For the reaction $2\text{NO}_{2(\text{g})} \rightleftharpoons 2\text{NO}(\text{g}) + \text{O}_{2(\text{g})}$, ($K_c = 1.8 \times 10^{-6}$ at 184°C) ($R = 0.831 \text{ kJ}(\text{mol.K})$) When K_p and K_c are compared at 184°C it is found that **[AIEEE-2005]**
 - (1) K_p is less than K_c
 - (2) K_p is greater than K_c
 - (3) Whether K_p is greater than, less than or equal to K_c depends upon the total gas pressure
 - (4) $K_p = K_c$
- The exothermic formation of ClF_3 is represented by the equation **[AIEEE-2005]** $\text{Cl}_{2(\text{g})} + 3\text{F}_{2(\text{g})} \rightleftharpoons 2\text{ClF}_{3(\text{g})}$; $\Delta H_r = -329 \text{ kJ}$
Which of the following will increase the quantity of ClF_3 in an equilibrium mixture of Cl_2 , F_2 and ClF_3 ?
 - (1) Removing Cl_2
 - (2) Increasing the temperature
 - (3) Adding F_2
 - (4) Increasing the volume of the container

12. A schematic plot of $\ln K_{eq}$ versus inverse of temperature for a reaction is shown below. The reaction must be [AIEEE-2005]

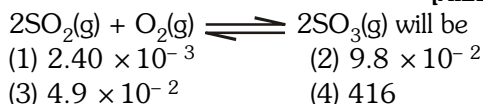


- (1) endothermic
(2) exothermic
(3) highly spontaneous at ordinary temperature
(4) one with negligible enthalpy change
13. Phosphorus pentachloride dissociates as follows, in a closed reaction vessel, [AIEEE-2006]



If total pressure at equilibrium of the reaction mixture is P and degree of dissociation of PCl_5 is x , the partial pressure of PCl_3 will be-

- (1) $\left(\frac{2x}{1-x}\right)P$ (2) $\left(\frac{x}{x-1}\right)P$
(3) $\left(\frac{x}{1-x}\right)P$ (4) $\left(\frac{x}{x+1}\right)P$
14. The equilibrium constant for the reaction $\text{SO}_3(\text{g}) \rightleftharpoons \text{SO}_2(\text{g}) + \frac{1}{2} \text{O}_2(\text{g})$ is $K_C = 4.9 \times 10^{-2}$. The value of K_C for the reaction [AIEEE-2006]



15. If $\text{Ag}^+ + \text{NH}_3 \rightleftharpoons [\text{Ag}(\text{NH}_3)]^+$; $K_1 = 1.6 \times 10^3$ and $[\text{Ag}(\text{NH}_3)]^+ + \text{NH}_3 \rightleftharpoons [\text{Ag}(\text{NH}_3)_2]^+$; $K_2 = 6.8 \times 10^3$. The formation constant of $[\text{Ag}(\text{NH}_3)_2]^+$ is : [JEE 2006]
- (1) 6.08×10^{-6} (2) 6.8×10^{-6}
(3) 1.6×10^3 (4) 1.088×10^7

16. The equilibrium constants K_{p1} and K_{p2} for the reaction $\text{X} \rightleftharpoons 2\text{Y}$ and $\text{Z} \rightleftharpoons \text{P} + \text{Q}$, respectively are in the ratio of 1 : 9. If the degree of dissociation of X and Z be equal then the ratio of total pressure at these equilibria is [AIEEE-2008]

- (1) 1 : 36 (2) 1 : 1
(3) 1 : 3 (4) 1 : 9

17. A vessel at 1000 K contains CO_2 with a pressure of 0.5 atm. Some of the CO_2 is converted into CO on the addition of graphite. If the total pressure at equilibrium is 0.8 atm, the value of K is :- [AIEEE-2011]

- (1) 0.3 atm (2) 0.18 atm
(3) 1.8 atm (4) 3 atm

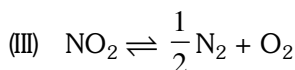
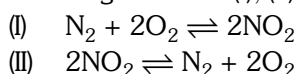
18. The equilibrium constant (K_C) for the reaction $\text{N}_2(\text{g}) + \text{O}_2(\text{g}) \longrightarrow 2\text{NO}(\text{g})$ at temperature T is 4×10^{-4} . The value of K_C for the reaction $\text{NO}(\text{g}) \longrightarrow \frac{1}{2} \text{N}_2(\text{g}) + \frac{1}{2} \text{O}_2(\text{g})$ at the same temperature is :- [AIEEE-2012]

- (1) 50.0 (2) 0.02
(3) 2.5×10^2 (4) 4×10^{-4}

19. One mole of $\text{O}_2(\text{g})$ and two moles of $\text{SO}_2(\text{g})$ were heated in a closed vessel of one litre capacity at 1098 K. At equilibrium 1.6 moles of $\text{SO}_3(\text{g})$ were found. The equilibrium constant K_C of the reaction would be :- [JEE-MAINS(online)-12]

- (1) 60 (2) 80
(3) 30 (4) 40

20. K_1 , K_2 and K_3 are the equilibrium constants of the following reactions (I), (II) and (III), respectively



The correct relation from the following is :

[JEE-MAINS(online)-12]

- (1) $K_1 = \sqrt{K_2} = K_3$ (2) $K_1 = \frac{1}{K_2} = \frac{1}{K_3}$
(3) $K_1 = \frac{1}{K_2} = K_3$ (4) $K_1 = \frac{1}{K_2} = \frac{1}{(K_3)^2}$

21. The value of K_p for the equilibrium reaction $\text{N}_2\text{O}_4(\text{g}) \rightleftharpoons 2\text{NO}_2(\text{g})$ is 2. The percentage dissociation of $\text{N}_2\text{O}_4(\text{g})$ at a pressure of 0.5 atm is [JEE-MAINS(online)-12]

- (1) 71 (2) 50
(3) 88 (4) 25

22. 8 mol of $\text{AB}_3(\text{g})$ are introduced into a 1.0 dm^3 vessel. If it dissociates as $2\text{AB}_3(\text{g}) \rightleftharpoons \text{A}_2(\text{g}) + 3\text{B}_2(\text{g})$

At equilibrium, 2mol of A_2 are found to be present. The equilibrium constant of this reaction is :-

[JEE-MAINS(online)-12]

- (1) 36 (2) 3
(3) 27 (4) 2

23. The thermal dissociation equilibrium of $\text{CaCO}_3(\text{s})$ is studied under different conditions.



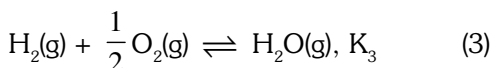
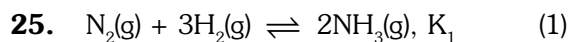
For this equilibrium, the correct statement(s) is(are)

- (1) ΔH is dependent on T
(2) K is independent of the initial amount of CaCO_3
(3) K is dependent on the pressure of CO_2 at a given T
(4) ΔH is independent of the catalyst, if any

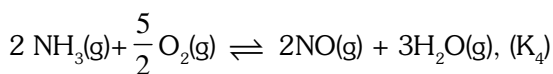
24. In reaction $A + 2B \rightleftharpoons 2C + D$, initial concentration of B was 1.5 times of $|A|$, but at equilibrium the concentrations of A and B became equal. The equilibrium constant for the reaction is :

[JEE-MAINS(online)-13]

- (1) 4 (2) 6
(3) 12 (4) 8



The equation for the equilibrium constant of the reaction



in terms of K_1 , K_2 and K_3 is :

[JEE-MAINS(online)-13]

- (1) $\frac{K_1 K_3^2}{K_2}$ (2) $\frac{K_2 K_3^3}{K_1}$ (3) $\frac{K_1 K_2}{K_3}$ (4) $K_1 K_2 K_3$

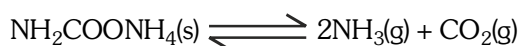
26. For the reaction $SO_{2(g)} + \frac{1}{2}O_{2(g)} \rightleftharpoons SO_{3(g)}$, if

$K_p = K_C (RT)^x$ where the symbols have usual meaning then the value of x is : (Assuming ideality)

[JEE-MAINS-14]

- (1) $\frac{1}{2}$ (2) 1 (3) -1 (4) $-\frac{1}{2}$

27. For the decomposition of the compound, represented as



the $K_p = 2.9 \times 10^{-5} \text{ atm}^3$.

If the reaction is started with 1 mol of the compound, the total pressure at equilibrium would be

[JEE-MAINS(online)-14]

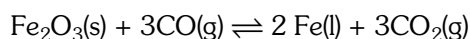
- (1) $38.8 \times 10^{-2} \text{ atm}$ (2) $1.94 \times 10^{-2} \text{ atm}$
(3) $5.82 \times 10^{-2} \text{ atm}$ (4) $7.66 \times 10^{-2} \text{ atm}$

28. The equilibrium constants at 298 K for a reaction $A + B \rightleftharpoons C + D$ is 100. If the initial concentration of all the four species were 1 M each, then equilibrium concentration of D (in mol L^{-1}) will be

[JEE-MAINS-16]

- (1) 1.182 (2) 0.182
(3) 0.818 (4) 1.818

29. The following reaction occurs in the Blast Furnace where iron ore is reduced to iron metal:



[JEE-MAINS(online)-17]

Using the Le Chatelier's principle, predict which one of the following will **not** disturb the equilibrium?

- (1) Removal of CO_2
(2) Addition of Fe_2O_3
(3) Addition of CO_2
(4) Removal of CO

30. For a reaction taking place in a container in equilibrium with its surroundings, the effect of temperature on its equilibrium constant K in terms of change in entropy is described by [JEE 2017]

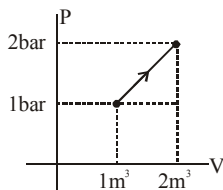
- (1) With increase in temperature, the value of K for exothermic reaction decreases because the entropy change of the system is positive
(2) With increase in temperature, the value of K for endothermic reaction increases because unfavourable change in entropy of the surroundings decreases
(3) With increase in temperature, the value of K for exothermic reaction decreases because favourable change in entropy of the surroundings decreases
(4) With increase in temperature, the value of K for endothermic reaction increases because the entropy change of the system negative

PREVIOUS YEARS QUESTIONS				ANSWER KEY			Exercise-II			
Que.	1	2	3	4	5	6	7	8	9	10
Ans.	4	4	4	3	4	1	4	1	2	2
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	3	2	4	4	4	1	3	1	2	4
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	1	3	1,2,4	1	2	4	3	4	2	2,3

THERMODYNAMICS & THERMOCHEMISTRY

EXERCISE-I

1. What is ΔU for the process described by figure. Heat supplied during the process $q = 200$ kJ.



- (1) +50 kJ
(2) -50 kJ
(3) -150 kJ
(4) + 150 kJ

2. What is the change in internal energy when a gas is compressed from 377 ml of 177 ml under a constant pressure of 1520 torr, while at the same time being cooled by removing 124 J heat ?

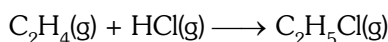
[Take : (1L atm) = 100 J]

- (1) -24 J (2) - 84 J
(3) - 164 J (4) - 248 J

3. For a reaction, $2X(s) + 2Y(s) \rightarrow 2C(l) + D(g)$
The q_p at 27°C is -28 KCal. mol^{-1} . The q_v is -----
----- K. Cal. mol^{-1} :-

- (1) - 27.4 (2) + 27.4
(3) - 28.6 (4) 28.6

4. Ethyl chloride ($\text{C}_2\text{H}_5\text{Cl}$), is prepared by reaction of ethylene with hydrogen chloride :



$$\Delta H = -72.3 \text{ kJ/mol}$$

What is the value of ΔU (in kJ), if 70 g of ethylene and 73 g of HCl are allowed to react at 300 K.

- (1) -69.8 (2) -180.75
(3) -174.5 (4) -139.6

5. One mole of an ideal monoatomic gas expanded irreversibly in two stage expansion.

State-1 (8.0 bar, 4.0 litre, 300 K)

State-2 (2.0 bar, 16litre, 300 K)

State-3 (1.0 bar, 32 litre, 300 K)

Total heat absorbed by the gas in the process is

- (1) 116J (2) 40 J
(3) 4000 J (4) None of these

6. When a gas is compressed adiabatically and reversibly, the final temperature is-

- (1) Higher than the initial temperature
(2) Lower than the initial temperature
(3) The same as initial temperature
(4) Dependent upon the rate of compression

7. A gas $\left(C_{v,m} = \frac{5}{2}R\right)$ behaving ideally was allowed to expand reversibly and adiabatically from 1 litre to 32 litre. Its initial temperature was 327°C . The molar enthalpy change (in J/mole) for the process is :-

- (1) -1125 R (2) - 575 R
(3) -1575 R (4) None of these

8. Two moles of an ideal gas $(C_v = \frac{5}{2}R)$ was compressed adiabatically against constant pressure of 2 atm. which was initially at 350 K and 1 atm pressure. The work involve in the process is equal to ?

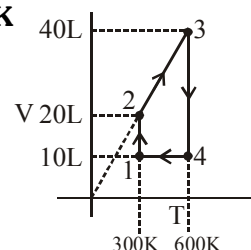
- (1) 250 R (2) 300 R
(3) 400 R (4) 500 R

9. What is the net work done when 1 mole of monoatomic ideal gas undergoes in a process described by 1, 2, 3, 4 in given V-T graph

Use : $R = 2\text{cal/mole K}$

$\ln 2 = 0.7$

- (1) -600 cal
(2) - 660 cal
(3) + 660 cal
(4) + 600 cal



10. The heat capacity of liquid water is 75.6 J/mol.K , while the enthalpy of fusion of ice is 6.0 kJ/mol . What is the smallest number of ice cubes at 0°C , each containing 9.0 g of water, needed to cool 500 g of liquid water from 20°C to 0°C ?

- (1) 1 (2) 7 (3) 14 (4) None of these

11. Change in entropy is negative for :-

- (1) Bromine (l) $\longrightarrow$ Bromine (g)
(2) $\text{C}(\text{s}) + \text{H}_2\text{O}(\text{g}) \longrightarrow \text{CO}(\text{g}) + \text{H}_2(\text{g})$
(3) $(3) \text{N}_2(\text{g}, 10 \text{ atm}, 298 \text{ K}) \longrightarrow \text{N}_2(\text{g}, 1 \text{ atm}, 298\text{K})$
(4) $\text{Fe}(\text{at } 400 \text{ K}) \longrightarrow \text{Fe}(\text{at } 300 \text{ K})$

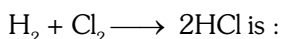
12. 5 mole of an ideal gas expand reversibly from a volume of 8 dm^3 to 80 dm^3 at a temperature of 27°C . The change in entropy is :-

- (1) 41.57 JK^{-1} (2) -95.73 JK^{-1}
(3) 95.73 JK^{-1} (4) -41.57 JK^{-1}

13. If $\Delta H_{\text{vapourisation}}$ of substance X(l) (molar mass : 30 g/mol) is 300 J/g at its boiling point 300 K, then molar entropy change for reversible condensation process is :-

(1) 30 J/mol.K (2) -300 J/mol.K
(3) -30 J/mol.K (4) None of these

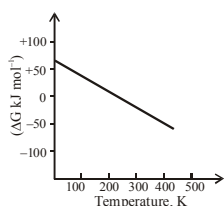
14. If S° for H_2 , Cl_2 and HCl are 0.13, 0.22 and 0.19 $\text{KJ K}^{-1} \text{mol}^{-1}$ respectively. The total change in standard entropy for the reaction,



(1) 30 $\text{JK}^{-1} \text{mol}^{-1}$ (2) 40 $\text{JK}^{-1} \text{mol}^{-1}$
(3) 60 $\text{JK}^{-1} \text{mol}^{-1}$ (4) 20 $\text{JK}^{-1} \text{mol}^{-1}$

15. What can be concluded about the values of ΔH and ΔS from this graph?

(1) $\Delta H > 0$, $\Delta S > 0$
(2) $\Delta H > 0$, $\Delta S < 0$
(3) $\Delta H < 0$, $\Delta S > 0$
(4) $\Delta H < 0$, $\Delta S < 0$



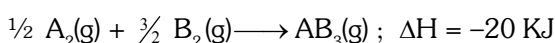
16. What is the free energy change (ΔG) when 1.0 mole of water at 100°C and 1 atm pressure is converted into steam at 100°C and 1 atm pressure ?

(1) 80 cal (2) 540 cal
(3) 620 cal (4) zero

17. For a reaction at 25°C enthalpy change (ΔH) and entropy change (ΔS) are $-11.7 \times 10^3 \text{ J mol}^{-1}$ and $-105 \text{ J mol}^{-1} \text{ K}^{-1}$ respectively. The reaction is :

(1) Spontaneous (2) Non spontaneous
(3) At equilibrium (4) Can't say anything

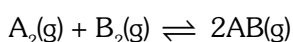
18. For hypothetical reversible reaction,



if standard entropies of A_2 , B_2 and AB_3 are 60, 40 and 50 $\text{JK}^{-1} \text{mole}^{-1}$ respectively. The above reaction will be in equilibrium at the temperature :-

(1) 400 K (2) 500 K
(3) 250 K (4) 200 K

19. For the hypothetical reaction ,



$\Delta_r G$ and $\Delta_r S$ are 20 kJ/mole and $-20 \text{ JK}^{-1} \text{mol}^{-1}$ respectively at 200 K.

If $\Delta_r C_p$ is $20 \text{ JK}^{-1} \text{mol}^{-1}$ then $\Delta_r H$ at 400 K is :-

(1) 20 kJ/mole (2) 7.98 kJ/mole
(3) 28 kJ/mole (4) None of these

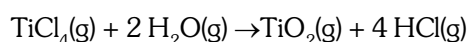
20. The value of ΔG_f° of gaseous mercury is 31 KJ/mole. At what total external pressure mercury start boiling at 25°C . [$R = 8.3 \text{ J/K mole}$]

(1) $10^{-5.44}$ (2) $10^{-12.5}$
(3) $10^{-6.52}$ (4) $10^{-3.12}$

21. Which of the following equations represents a reaction that provides the heat of formation of CH_3Cl ?

(1) $\text{C}(\text{s}) + \text{HCl}(\text{g}) + \text{H}_2(\text{g}) \rightarrow \text{CH}_3\text{Cl}(\text{g})$
(2) $\text{C}(\text{s}) + 3/2 \text{H}_2(\text{g}) + 1/2 \text{Cl}_2(\text{g}) \rightarrow \text{CH}_3\text{Cl}(\text{g})$
(3) $\text{C}(\text{s}) + 3 \text{H}(\text{g}) + \text{Cl}(\text{g}) \rightarrow \text{CH}_3\text{Cl}(\text{g})$
(4) $\text{CH}_4(\text{g}) + \text{Cl}_2(\text{g}) \rightarrow \text{CH}_3\text{Cl}(\text{g}) + \text{HCl}(\text{g})$

22. Use the given standard enthalpies of formation to determine the heat of reaction of the following reaction:



$$\Delta H_f^\circ \text{TiCl}_4(\text{g}) = -763.2 \text{ kJ/mole}$$

$$\Delta H_f^\circ \text{TiO}_2(\text{g}) = -944.7 \text{ kJ/mole}$$

$$\Delta H_f^\circ \text{H}_2\text{O}(\text{g}) = -241.8 \text{ kJ/mole}$$

$$\Delta H_f^\circ \text{HCl}(\text{g}) = -92.3 \text{ kJ/mole}$$

(1) -278.1 (2) +369.2
(3) +67.1 (4) -67.1

23. $2\text{NO}_2(\text{g}) \rightarrow \text{N}_2\text{O}_4(\text{g})$

$$\Delta U_f^\circ [\text{N}_2\text{O}_4(\text{g})] = 2 \text{ kcal/mole}$$

and $\Delta U_{\text{reaction}}^\circ = -16 \text{ kcal/mol}$ then calculate $\Delta H_{\text{formation}}^\circ$ of NO_2 at 727°C

(1) 9 kcal/mol (2) 4.5 kcal/mol
(3) 8 kcal/mol (4) 10 kcal/mol

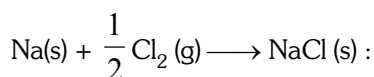
24. The reaction $\text{CH}_4(\text{g}) + \text{Cl}_2(\text{g}) \longrightarrow \text{CH}_3\text{Cl}(\text{g}) + \text{HCl}(\text{g})$ has $\Delta H = -25 \text{ kCal}$.

Bond	Bond Enthalpy kCal
$\epsilon_{\text{C}-\text{Cl}}$	84
$\epsilon_{\text{H}-\text{Cl}}$	103
$\epsilon_{\text{C}-\text{H}}$	x
$\epsilon_{\text{Cl}-\text{Cl}}$	y
$x : y = 9 : 5$	

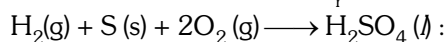
From the given data, what is the bond enthalpy of $\text{Cl}-\text{Cl}$ bond

(1) 70 kCal (2) 80 kCal
(3) 67.75 kCal (4) 57.75 kCal

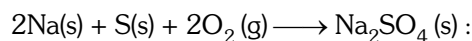
25. The enthalpy changes of the following reactions at 27°C are



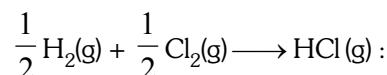
$$\Delta_f H = -411 \text{ kJ/mol}$$



$$\Delta_f H = -811 \text{ kJ/mol}$$

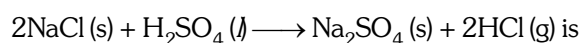


$$\Delta_f H = -1382 \text{ kJ/mol}$$



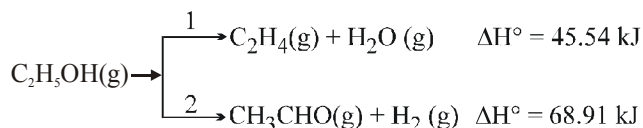
$$\Delta_f H = -92 \text{ kJ/mol};$$

from these data, the heat change of reaction at constant volume (in kJ/mol) at 27°C for the process ($R = 8.3 \text{ J/K-mol}$)



- (1) 67 (2) 62.02
(3) 71.98 (4) None

26. Ethanol can undergoes decomposition to form two sets of products



if the molar ratio of C_2H_4 to CH_3CHO is 8 : 1 in a set of product gases, then the enthalpy involved in the decomposition of 1 mole of ethano/is :

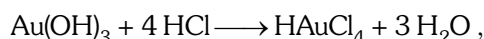
- (1) 65.98 kJ (2) 48.137 kJ
(3) 48.46 kJ (4) 57.22 kJ

27. ΔH_f° of water is $-285.8 \text{ kJ mol}^{-1}$. If enthalpy of neutralisation of monoacid strong base is $-57.3 \text{ kJ mol}^{-1}$, ΔH_f° of OH^- ion will be
(1) $-228.5 \text{ kJ mol}^{-1}$ (2) $228.5 \text{ kJ mol}^{-1}$
(3) $114.25 \text{ kJ mol}^{-1}$ (4) $-114.25 \text{ kJ mol}^{-1}$

28. The molar heat capacities at constant pressure (assume constant with respect to temperature) of A, B and C are in ratio of **1.5 : 3.0 : 2.0**. If enthalpy change for the exothermic reaction $\text{A} + 2\text{B} \longrightarrow 3\text{C}$ at 300 K is -10 kJ/mol & $C_{p,m}$ (2) is 300 J/mol then enthalpy change at 310 K is

- (1) -8.5 kJ/mol (2) 8.5 kJ/mol
(3) -11.5 kJ/mol (4) none of these

29. Reactions involving gold have been of particular interest to a chemist. Consider the following reactions,



$$\Delta H = -28 \text{ kCal}$$

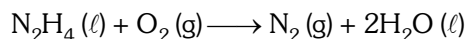


$$\Delta H = -36.8 \text{ kCal}$$

In an experiment there was an absorption of 0.44 kCal when one mole of HAuBr_4 was mixed with 4 moles of HCl . What is the percentage conversion of HAuBr_4 into HAuCl_4 ?

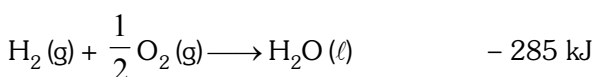
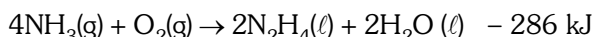
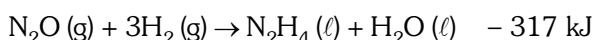
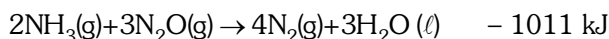
- (1) 0.5 % (2) 0.6 %
(3) 5 % (4) 50 %

30. Hydrazine, a component of rocket fuel, undergoes combustion to yield N_2 and H_2O .



What is the enthalpy change of combustion of N_2H_4 (kJ/mole)

Given Reaction **$\Delta H/\text{kJ}$**



- (1) -620.5 (2) -622.75
(3) 1167.5 (4) $+622.75$

ANSWER KEY

Exercise-I

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	1	2	3	4	3	1	3	4	3	3
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	4	3	3	1	1	4	2	2	1	1
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	2	4	3	4	2	2	1	3	3	1

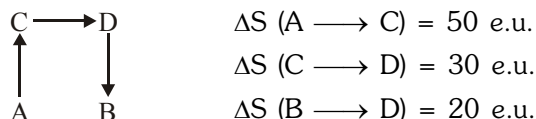
PREVIOUS YEARS' QUESTIONS

EXERCISE-II

THERMODYNAMICS

1. A process $A \longrightarrow B$ is difficult to occur directly instead it takes place in three successive steps.

[JEE 2006]



Where e.u. is entropy unit.

Then the entropy change for the process

$\Delta S (A \longrightarrow B)$ is :-

- (1) + 100 e.u. (3) - 60 e.u.
 (2) - 100 e.u. (4) + 60 e.u.
2. Assuming that water vapour is an ideal gas, the internal energy change (ΔU) when 1 mol of water is vapourised at 1 bar pressure and 100°C , (Given : Molar enthalpy of vapourisation of water at 1 bar and $373 \text{ K} = 41 \text{ kJ mol}^{-1}$ and $R = 8.3 \text{ J mol}^{-1} \text{ K}^{-1}$ will be) :- [AIEEE-2007]
- (1) $4.100 \text{ kJ mol}^{-1}$ (2) $3.7904 \text{ kJ mol}^{-1}$
 (3) $37.904 \text{ kJ mol}^{-1}$ (4) $41.00 \text{ kJ mol}^{-1}$
3. In conversion of lime-stone to lime,
 $\text{CaCO}_3(\text{s}) \longrightarrow \text{CaO}(\text{s}) + \text{CO}_2(\text{g})$
 the values of ΔH° and ΔS° are $+179.1 \text{ kJ mol}^{-1}$ and 160.2 J/K respectively at 298 K and 1 bar. Assuming that ΔH° and ΔS° do not change with temperature, temperature above which conversion of limestone to lime will be spontaneous is :- [AIEEE-2007]
- (1) 1008 K (2) 1200 K
 (3) 845 K (4) 1118 K
4. For the process $\text{H}_2\text{O}(\text{l}) (1 \text{ bar}, 373 \text{ K}) \longrightarrow \text{H}_2\text{O}(\text{g}) (1 \text{ bar}, 373 \text{ K})$, the correct set of thermodynamic parameters is :- [JEE 2007]
- (1) $\Delta G = 0, \Delta S = +ve$
 (2) $\Delta G = 0, \Delta S = -ve$
 (3) $\Delta G = +ve, \Delta S = 0$
 (4) $\Delta G = -ve, \Delta S = +ve$
5. Among the following, the state function(s) is (are) [JEE 2009]
- (1) Internal energy
 (2) Irreversible expansion work
 (3) Reversible expansion work
 (4) Molar enthalpy

6. For a particular reversible reaction at temperature T , ΔH and ΔS were found to be both +ve. If T_e is the temperature at equilibrium, then reaction would be spontaneous when :- [AIEEE-2010]

- (1) $T = T_e$
 (2) $T_e > T$
 (3) $T > T_e$
 (4) T_e is 5 times T

7. The value of enthalpy change (ΔH) for the reaction
 $\text{C}_2\text{H}_5\text{OH}(\text{l}) + 3\text{O}_{2(\text{g})} \rightarrow 2\text{CO}_{2(\text{g})} + 3\text{H}_2\text{O}(\text{l})$
 at 27°C is $-1366.5 \text{ kJ mol}^{-1}$. The value of internal energy change for the above reaction at this temperature will be :- [AIEEE-2011]

- (1) -1371.5 kJ
 (2) -1369.0 kJ
 (3) -1364.0 kJ
 (4) -1361.5 kJ

8. The entropy change involved in the isothermal reversible expansion of 2 moles of an ideal gas from a volume of 10 dm^3 to a volume of 100 dm^3 at 27°C is :- [AIEEE-2011]

- (1) $32.3 \text{ J mol}^{-1} \text{ K}^{-1}$
 (2) $42.3 \text{ J mol}^{-1} \text{ K}^{-1}$
 (3) $38.3 \text{ J mol}^{-1} \text{ K}^{-1}$
 (4) $35.8 \text{ J mol}^{-1} \text{ K}^{-1}$

9. The incorrect expression among the following is :-

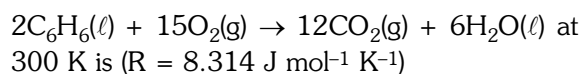
- (1) $K = e^{-\Delta G^\circ/RT}$ [AIEEE-2012]
 (2) $\frac{\Delta G_{\text{system}}}{\Delta S_{\text{total}}} = -T$

- (3) In isothermal process, $W_{\text{reversible}} = -nRT \ln \frac{V_f}{V_i}$

- (4) $\ln K = \frac{\Delta H^\circ - T\Delta S^\circ}{RT}$

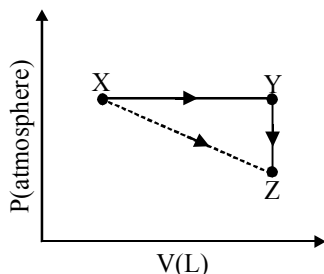
10. The difference between the reaction enthalpy change ($\Delta_r H$) and reaction internal energy change ($\Delta_r U$) for the reaction :

[JEE-MAINS(online)-2012]



- (1) 0 J mol^{-1}
 (2) 2490 J mol^{-1}
 (3) -2490 J mol^{-1}
 (4) -7482 J mol^{-1}

11. For an ideal gas, consider only P-V work in going from an initial state X to the final state Z. The final state Z can be reached by either of the two paths shown in the figure. Which of the following choice(s) is (are) correct ? [take ΔS as change in entropy and w as work done] [JEE 2012]



- (1) $\Delta S_{X \rightarrow Z} = \Delta S_{X \rightarrow Y} + \Delta S_{Y \rightarrow Z}$
 (2) $W_{X \rightarrow Z} = W_{X \rightarrow Y} + W_{Y \rightarrow Z}$
 (3) $W_{X \rightarrow Y \rightarrow Z} = W_{X \rightarrow Y}$
 (4) $\Delta S_{X \rightarrow Y \rightarrow Z} = \Delta S_{X \rightarrow Z}$
12. Which of the following statements/relationships is not correct in thermodynamic changes ? [JEE-MAINS(online)-2014]

- (1) $q = -nRT \ln \frac{V_2}{V_1}$ (isothermal reversible expansion of an ideal gas)
 (2) For a system at constant volume, heat involved merely changes to internal energy.
 (3) $w = -nRT \ln \frac{V_2}{V_1}$ (isothermal reversible expansion of an ideal gas)
 (4) $\Delta U = 0$ (isothermal reversible expansion of a gas)

13. The molar heat capacity (C_p) of CD_2O is 10 cal at 1000 K. The change in entropy associated with cooling of 32 g of CD_2O vapour from 1000 K to 100 K at constant pressure will be [JEE-MAINS(online)-2014]

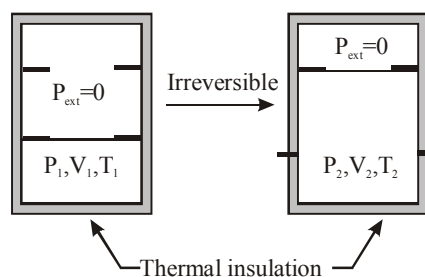
- (D = deuterium, at. mass = 2u)
 (1) $-23.03 \text{ cal deg}^{-1}$
 (2) $2.303 \text{ cal deg}^{-1}$
 (3) $23.03 \text{ cal deg}^{-1}$
 (4) $-2.303 \text{ cal deg}^{-1}$

14. The entropy (S°) of the following substances are [JEE-MAINS(online)-2014]

$CH_4(g)$ $186.2 \text{ J K}^{-1} \text{ mol}^{-1}$
 $O_2(g)$ $205.0 \text{ J K}^{-1} \text{ mol}^{-1}$
 $CO_2(g)$ $213.6 \text{ J K}^{-1} \text{ mol}^{-1}$
 $H_2O(l)$ $69.9 \text{ J K}^{-1} \text{ mol}^{-1}$

The entropy change (ΔS°) for the reaction $CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(l)$ is:-

- (1) $-312.5 \text{ JK}^{-1} \text{ mol}^{-1}$
 (2) $-37.6 \text{ JK}^{-1} \text{ mol}^{-1}$
 (3) $-108.1 \text{ JK}^{-1} \text{ mol}^{-1}$
 (4) $-242.8 \text{ JK}^{-1} \text{ mol}^{-1}$
15. An ideal gas in thermally insulated vessel at internal pressure = P_1 , volume = V_1 and absolute temperature = T_1 expands irreversibly against zero external pressure, as shown in the diagram. The final internal pressure, volume and absolute temperature of the gas are P_2 , V_2 and T_2 , respectively. For this expansion, [JEE 2014]



- (1) $q = 0$
 (2) $T_2 = T_1$
 (3) $P_2 V_2 = P_1 V_1$
 (4) $P_2 V_2^\gamma = P_1 V_1^\gamma$
16. ΔU is equal to [JEE-MAINS(offline)-2017]
 (1) Isochoric work
 (2) Isobaric work
 (3) Adiabatic work
 (4) Isothermal work
17. The enthalpy change on freezing of 1 mol of water at 5°C to ice at -5°C is : [JEE-MAINS(online)-2017]

(Given $\Delta_{\text{fus}}H = 6 \text{ kJ mol}^{-1}$ at 0°C ,
 $C_p(H_2O, l) = 75.3 \text{ J mol}^{-1} \text{ K}^{-1}$,
 $C_p(H_2O, s) = 36.8 \text{ J mol}^{-1} \text{ K}^{-1}$)

- (1) 6.56 kJ mol^{-1}
 (2) 5.81 kJ mol^{-1}
 (3) 6.00 kJ mol^{-1}
 (4) 5.44 kJ mol^{-1}
18. An ideal gas undergoes isothermal expansion at constant pressure. During the process:- [JEE-MAINS(online)-2017]
 (1) enthalpy remains constant but entropy increases.
 (2) enthalpy increases but entropy decreases.
 (3) Both enthalpy and entropy remain constant.
 (4) enthalpy decreases but entropy increases.

THERMOCHEMISTRY

19. Which of the following is not an endothermic reaction? [JEE 1999]

(1) Combustion of methane
(2) Decomposition of water
(3) Dehydrogenation of ethene to acetylene
(4) Conversion of graphite to diamond

20. ΔH_f° for $\text{CO}_2(\text{g})$, $\text{CO}(\text{g})$ and $\text{H}_2\text{O}(\text{g})$ are -393.5 , -110.5 and $-241.8 \text{ kJ mol}^{-1}$ respectively. The standard enthalpy change (in kJ) for the reaction $\text{CO}_2(\text{g}) + \text{H}_2(\text{g}) \longrightarrow \text{CO}(\text{g}) + \text{H}_2\text{O}(\text{g})$ is [JEE 2000]

(1) 524.1 (2) 41.2
(3) -262.5 (4) -41.2

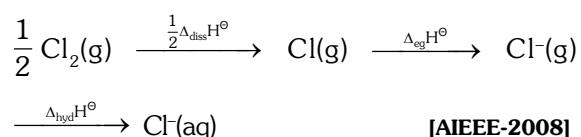
21. The enthalpy changes for the following processes are listed below : [AIEEE-2006]

$\text{Cl}_2(\text{g}) = 2\text{Cl}(\text{g}), \quad 242.3 \text{ kJ mol}^{-1}$
 $\text{I}_2(\text{g}) = 2\text{I}(\text{g}) \quad 151.0 \text{ kJ mol}^{-1}$
 $\text{ICl}(\text{g}) = \text{I}(\text{g}) + \text{Cl}(\text{g}), \quad 211.3 \text{ kJ mol}^{-1}$
 $\text{I}_2(\text{s}) = \text{I}_2(\text{g}), \quad 62.76 \text{ kJ mol}^{-1}$

Given that the standard states for iodine and chlorine are $\text{I}_2(\text{s})$ and $\text{Cl}_2(\text{g})$, the standard enthalpy of formation for $\text{ICl}(\text{g})$ is :-

(1) $-16.8 \text{ kJ mol}^{-1}$
(2) $+16.8 \text{ kJ mol}^{-1}$
(3) $+244.8 \text{ kJ mol}^{-1}$
(4) $-14.6 \text{ kJ mol}^{-1}$

22. Oxidising power of chlorine in aqueous solution can be determined by the parameters indicated below:



The energy involved in the conversion of $\frac{1}{2} \text{Cl}_2(\text{g})$ to $\text{Cl}^-(\text{aq})$

(using the data $\Delta_{\text{diss}} H_{\text{Cl}_2}^\circ = 240 \text{ kJ mol}^{-1}$,

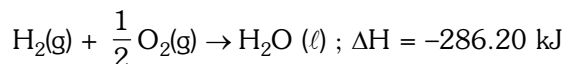
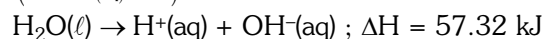
$$\Delta_{\text{eg}} H_{\text{Cl}}^\circ = -349 \text{ kJ mol}^{-1},$$

$$\Delta_{\text{hyd}} H_{\text{Cl}^-}^\circ = -381 \text{ kJ mol}^{-1})$$

will be:-

(1) -610 kJ mol^{-1}
(2) -850 kJ mol^{-1}
(3) $+120 \text{ kJ mol}^{-1}$
(4) $+152 \text{ kJ mol}^{-1}$

23. On the basis of the following thermochemical data ($\Delta G_f^\circ H_{(\text{aq})}^+ = 0$)

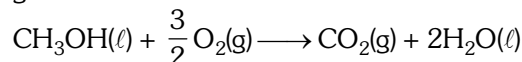


The value of enthalpy of formation of OH^- ion at 25°C is :-

[AIEEE-2009]

(1) $+228.88 \text{ kJ}$ (2) -343.52 kJ
(3) -22.88 kJ (4) -228.88 kJ

24. In a fuel cell methanol is used as fuel and oxygen gas is used as an oxidizer. The reaction is



At 298 K standard Gibbs energies of formation for $\text{CH}_3\text{OH}(\ell)$, $\text{H}_2\text{O}(\ell)$ and $\text{CO}_2(\text{g})$ are -166.2 , -237.2 and $-394.4 \text{ kJ mol}^{-1}$ respectively. If standard enthalpy of combustion of methanol is -726 kJ mol^{-1} , efficiency of the fuel cell will be

[AIEEE-2009]

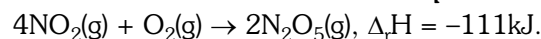
(1) 90% (2) 97%
(3) 80% (4) 87%

25. The standard enthalpy of formation of NH_3 is $-46.0 \text{ kJ mol}^{-1}$. If the enthalpy of formation of H_2 from its atoms is -436 kJ mol^{-1} and that of N_2 is -712 kJ mol^{-1} , the average bond enthalpy of N-H bond in NH_3 is :-

[AIEEE-2010]

(1) $-1102 \text{ kJ mol}^{-1}$ (2) -964 kJ mol^{-1}
(3) $+352 \text{ kJ mol}^{-1}$ (4) $+1056 \text{ kJ mol}^{-1}$

26. Consider the reaction : [AIEEE-2011]



If $\text{N}_2\text{O}_5(\text{s})$ is formed instead of $\text{N}_2\text{O}_5(\text{g})$ in the above reaction, the $\Delta_r H$ value will be :-

(Given, ΔH of sublimation for N_2O_5 is 54 kJ mol^{-1})

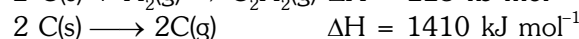
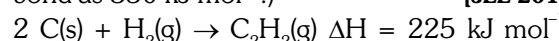
(1) -165 kJ (2) $+54 \text{ kJ}$
(3) $+219 \text{ kJ}$ (4) -219 kJ

27. The enthalpy of neutralisation of NH_4OH with HCl is $-51.46 \text{ kJ mol}^{-1}$ and the enthalpy of neutralisation of NaOH with HCl is $-55.90 \text{ kJ mol}^{-1}$. The enthalpy of ionisation of NH_4OH is:

[JEE-MAINS (online) 2012]

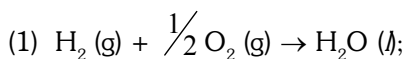
(1) $+107.36 \text{ kJ mol}^{-1}$
(2) $-4.44 \text{ kJ mol}^{-1}$
(3) $-107.36 \text{ kJ mol}^{-1}$
(4) $+4.44 \text{ kJ mol}^{-1}$

28. Using the data provided, calculate the multiple bond energy (kJ mol^{-1}) of a $\text{C} \equiv \text{C}$ bond in C_2H_2 . That energy is (take the bond energy of a C-H bond as 350 kJ mol^{-1}). [JEE 2012]

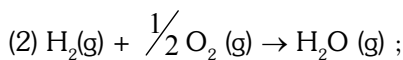


(1) 1165 (2) 837
(3) 865 (4) 815

29. Given : [JEE-MAINS (online) 2013]



$$\Delta H^\circ_{298\text{K}} = -285.9 \text{ kJ mol}^{-1}$$



$$\Delta H^\circ_{298\text{K}} = -241.8 \text{ kJ mol}^{-1}$$

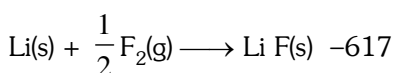
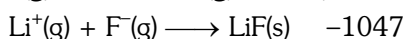
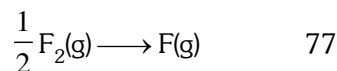
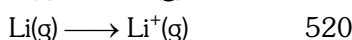
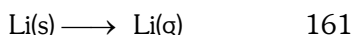
The molar enthalpy of vapourisation of water will be :-

$$(1) 241.8 \text{ kJ mol}^{-1} \quad (2) 527.7 \text{ kJ mol}^{-1}$$

$$(3) 44.1 \text{ kJ mol}^{-1} \quad (4) 22.0 \text{ kJ mol}^{-1}$$

30. Given [JEE-MAINS (online) 2013]

Reaction Energy Change (in kJ)



Based on data provided, the value of electron gain enthalpy of fluorine would be :-

$$(1) -300 \text{ kJ mol}^{-1} \quad (2) -328 \text{ kJ mol}^{-1}$$

$$(3) -350 \text{ kJ mol}^{-1} \quad (4) -228 \text{ kJ mol}^{-1}$$

31. For complete combustion of ethanol,
 $\text{C}_2\text{H}_5\text{OH}(\text{l}) + 3\text{O}_2(\text{g}) \rightarrow 2\text{CO}_2(\text{g}) + 3\text{H}_2\text{O}(\text{l})$,
 the amount of heat produced as measured in bomb calorimeter, is $1364.47 \text{ kJ mol}^{-1}$ at 25°C . Assuming ideality the Enthalpy of combustion, $\Delta_c H$, for the reaction will be ($R = 8.314 \text{ kJ mol}^{-1}$) :-

[JEE-MAINS(offline)2014]

$$(1) -1460.50 \text{ kJ mol}^{-1}$$

$$(2) -1350.50 \text{ kJ mol}^{-1}$$

$$(3) -1366.95 \text{ kJ mol}^{-1}$$

$$(4) -1361.95 \text{ kJ mol}^{-1}$$

32. The standard enthalpy of formation ($\Delta_f H^\circ_{298}$) for methane, CH_4 is $-74.9 \text{ kJ mol}^{-1}$. In order to calculate the average energy given out in the formation of a C-H bond from this it is necessary to know which one of the following?

[JEE-MAINS(online) 2014]

(1) the dissociation energy of the hydrogen molecule, H_2 .

(2) the dissociation energy of H_2 and enthalpy of sublimation of carbon (graphite).

(3) the first four ionisation energies of carbon and electron affinity of hydrogen.

(4) the first four ionisation energies of carbon.

33. The heats of combustion of carbon and carbon monoxide are -393.5 and $-285.5 \text{ kJ mol}^{-1}$, respectively. The heat of formation (in kJ) of carbon monoxide per mole is :- [JEE-MAINS(offline)2016]

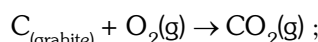
$$(1) -110.5$$

$$(2) 110.5$$

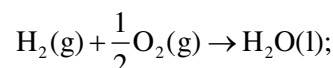
$$(3) 676.5$$

$$(4) -676.5$$

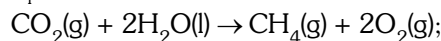
34. Given [JEE-MAINS(offline)2017]



$$\Delta_f H^\circ = -393.5 \text{ kJ mol}^{-1}$$

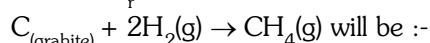


$$\Delta_f H^\circ = -285.8 \text{ kJ mol}^{-1}$$



$$\Delta_r H^\circ = +890.3 \text{ kJ mol}^{-1}$$

Based on the above thermochemical equations, the value of $\Delta_f H^\circ$ at 298 K for the reaction



$$(1) +74.8 \text{ kJ mol}^{-1} \quad (2) +144.0 \text{ kJ mol}^{-1}$$

$$(3) -74.8 \text{ kJ mol}^{-1} \quad (4) -144.0 \text{ kJ mol}^{-1}$$

35. The enthalpy change on freezing of 1 mol of water at 5°C to ice at -5°C is :

(Given $\Delta_{\text{fus}} H = 6 \text{ kJ mol}^{-1}$ at 0°C ,

$$C_p(\text{H}_2\text{O}, \text{l}) = 75.3 \text{ J mol}^{-1} \text{ K}^{-1},$$

$$C_p(\text{H}_2\text{O}, \text{s}) = 36.8 \text{ J mol}^{-1} \text{ K}^{-1})$$

[JEE-MAINS(online)2017]

$$(1) 6.56 \text{ kJ mol}^{-1}$$

$$(2) 5.81 \text{ kJ mol}^{-1}$$

$$(3) 6.00 \text{ kJ mol}^{-1}$$

$$(4) 5.44 \text{ kJ mol}^{-1}$$

PREVIOUS YEARS QUESTIONS				ANSWER KEY			Exercise-II			
Que.	1	2	3	4	5	6	7	8	9	10
Ans.	4	3	4	1	1,4	3	3	3	4	4
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	1,3	1	1	4	1,2,3	3	1	1	1	2
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	1	1	4	2	3	4	4	4	3	2
Que.	31	32	33	34	35					
Ans.	3	2	1	3	1					

REDOX REACTION

EXERCISE-I

- Which reaction does not represent autoredox or disproportionation :-
 (1) $\text{Cl}_2 + \text{OH}^- \longrightarrow \text{Cl}^- + \text{ClO}_3^- + \text{H}_2\text{O}$
 (2) $2\text{H}_2\text{O}_2 \longrightarrow \text{H}_2\text{O} + \text{O}_2$
 (3) $2\text{Cu}^+ \longrightarrow \text{Cu}^{+2} + \text{Cu}$
 (4) $(\text{NH}_4)_2\text{Cr}_2\text{O}_7 \longrightarrow \text{N}_2 + \text{Cr}_2\text{O}_3 + 4\text{H}_2\text{O}$
- Which of the following is not a redox reaction?
 (1) $\text{BaO}_2 + \text{H}_2\text{SO}_4 \rightarrow \text{BaSO}_4 + \text{H}_2\text{O}_2$
 (2) $2\text{BaO} + \text{O}_2 \rightarrow 2\text{BaO}_2$
 (3) $2\text{KClO}_3 \rightarrow 2\text{KCl} + 3\text{O}_2$
 (4) $\text{SO}_2 + 2\text{H}_2\text{S} \rightarrow 2\text{H}_2\text{O} + 3\text{S}$
- H_2O_2 acts as a reducing agent in :
 (1) $\text{FeCl}_2 + \text{HCl} + \text{H}_2\text{O}_2 \longrightarrow \text{FeCl}_3 + \text{H}_2\text{O}$
 (2) $\text{Cl}_2 + \text{H}_2\text{O}_2 \longrightarrow \text{HCl} + \text{O}_2$
 (3) $\text{HI} + \text{H}_2\text{O}_2 \longrightarrow \text{I}_2 + \text{H}_2\text{O}$
 (4) $\text{H}_2\text{SO}_3 + \text{H}_2\text{O}_2 \longrightarrow \text{H}_2\text{SO}_4 + \text{H}_2\text{O}$
- Match List-I (Compounds) with List-II (Oxidation states of nitrogen) and select answer using the codes given below the lists :-

List-I

- (a) NaN_3
 (b) N_2H_2
 (c) NO
 (d) N_2O_5

List-II

1. +5
 2. +2
 3. -1/3
 4. -1

Code : (a) (b) (c) (d)

- (1) 3 4 2 1
 (2) 4 3 2 1
 (3) 3 4 1 2
 (4) 4 3 1 2

- In the reaction
 $x\text{HI} + y\text{HNO}_3 \longrightarrow \text{NO} + \text{I}_2 + \text{H}_2\text{O}$
 (1) $x = 3, y = 2$ (2) $x = 2, y = 3$
 (3) $x = 6, y = 2$ (4) $x = 6, y = 1$
- The number of electrons to balance the following equation :-
 $\text{NO}_3^- + 4\text{H}^+ + e^- \rightarrow 2\text{H}_2\text{O} + \text{NO}$ is
 (1) 5 (2) 4
 (3) 3 (4) 2
- Number of moles of electrons taken up when 1 mole of NO_3^- ions is reduced to 1 mole of NH_2OH is
 (1) 2 (2) 4 (3) 5 (4) 6

- For the redox reaction,
 $\text{MnO}_4^- + \text{C}_2\text{O}_4^{2-} + \text{H}^+ \rightarrow \text{Mn}^{2+} + \text{CO}_2 + \text{H}_2\text{O}$
 the correct coefficients of the reactants for the balanced reaction are :

MnO_4^-	$\text{C}_2\text{O}_4^{2-}$	H^+
(1) 2	5	16
(2) 16	5	2
(3) 5	16	2
(4) 2	16	5
- In a redox reaction, the equivalent weight of HNO_2 is found to be 23.5. The reaction products might contain
 (1) NO_2 (2) NO
 (3) NH_3 (4) HNO_3
- When KBrO_3 ion reacts with Br^- ion in acid solution Br_2 is liberated. The equivalent weight of KBrO_3 in this reaction is
 (1) $\text{M}/8$ (2) $\text{M}/3$
 (3) $\text{M}/5$ (4) $\text{M}/6$
- In the reaction $\text{CrO}_5 + \text{H}_2\text{SO}_4 \rightarrow \text{Cr}_2(\text{SO}_4)_3 + \text{H}_2\text{O} + \text{O}_2$ one mole of CrO_5 will liberate how many moles of O_2 :-
 (1) $5/2$ (2) $5/4$
 (3) $9/2$ (4) $7/2$
- A solution of KMnO_4 is reduced to MnO_2 . The normality of solution is 0.6. The molarity is:
 (1) 1.8M (2) 0.6M
 (3) 0.1M (4) 0.2M
- 0.52 g of a dibasic acid required 100 mL of 0.2 N NaOH for complete neutralization. The equivalent weight of acid is
 (1) 26 (2) 52
 (3) 104 (4) 156
- The number of moles of KMnO_4 that will be required to react with 2 mol of ferrous oxalate is

(1) $\frac{6}{5}$	(2) $\frac{2}{5}$	(3) $\frac{4}{5}$	(4) 1
-------------------	-------------------	-------------------	-------
- The mass of oxalic acid crystals ($\text{H}_2\text{C}_2\text{O}_4 \cdot 2\text{H}_2\text{O}$) required to prepare 50 mL of a 0.2 N solution is :-
 (1) 4.5 g (2) 6.3 g
 (3) 0.63 g (4) 0.45 g

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				ANSWER KEY			Exercise-I			
Que.	1	2	3	4	5	6	7	8	9	10
Ans.	4	1	2	1	3	3	4	1	4	3
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	4	4	1	1	3	3	3	3	3	2
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	4	4	1	2	4	2	3	3	1	3

PREVIOUS YEARS' QUESTIONS

EXERCISE-II

1. The oxidation states of the most electronegative element in the products of the reaction of BaO_2 with dilute H_2SO_4 . [JEE 1991]
 (1) 0 and -1 (2) -1 and -2
 (3) -2 and 0 (4) -2 and +2
 2. For the redox reaction, [JEE 1992]
 $\text{MnO}_4^- + \text{C}_2\text{O}_4^{2-} + \text{H}^+ \rightarrow \text{Mn}^{2+} + \text{CO}_2 + \text{H}_2\text{O}$
 the correct coefficients of the reactants for the balanced reaction are :

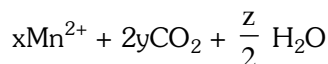
MnO_4^-	$\text{C}_2\text{O}_4^{2-}$	H^+
(1) 2	5	16
(2) 16	5	2
(3) 5	16	2
(4) 2	16	5
 3. A 5.0 cm^3 solution of H_2O_2 liberates 0.508 g of iodine from an acidified KI solution. Calculate the strength of H_2O_2 solution in terms of volume strength at STP. [JEE' 1995]
 (1) 2.48 (2) 3.48 (3) 4.48 (4) 5.48
 4. The number of mole of KMnO_4 that will need to react completely with one mole ferrous oxalate in acidic solution is : [JEE 1997]
 (1) 2/5 (2) 3/5 (3) 4/5 (4) 1
 5. The equivalent mass of MnSO_4 is half its molecular mass when it is converted to : [JEE 1998]
 (1) Mn_2O_3 (2) MnO_2
 (3) MnO_4^- (4) MnO_4^{2-}
 6. How many millilitre of 0.5 M H_2SO_4 are needed to dissolve 0.5 g of copper II carbonate ? [JEE 1999]
 (1) 7.097 (2) 8.097 (3) 10 (4) 12
 7. Among the following species in which oxidation state of the element is +6 : [JEE 2000]
 (1) MnO_4^- (2) $\text{Cr}(\text{CN})_6^{3-}$
 (3) NiF_6^{2-} (4) CrO_2Cl_2
 8. Oxidation number of iron in $\text{Na}_2[\text{Fe}(\text{CN})_5\text{NO}^\oplus]$ is: [JEE 2001]
 (1) +2 (2) +3
 (3) +8/3 (4) none of these
 9. An aqueous solution of 6.3 g of oxalic acid dihydrate is made upto 250 mL. The volume of 0.1 N NaOH required to completely neutralise 10 mL of this solution is : [JEE 2001]
 (1) 40 mL (2) 20 mL
 (3) 10 mL (4) 4 mL
 10. MnO_4^- is good oxidising agent in different medium changing to - [AIEEE-02]

MnO_4^-	$\longrightarrow$	Mn^{2+}
	$\longrightarrow$	MnO_4^{2-}
	$\longrightarrow$	MnO_2
	$\longrightarrow$	Mn_2O_3

 Changes in oxidation number respectively are -
 (1) 1, 3, 4, 5 (2) 5, 4, 3, 2
 (3) 5, 1, 3, 4 (4) 2, 6, 4, 3
 11. Oxidation number of Cl in CaOCl_2 (bleaching powder is) [AIEEE-02]
 (1) Zero, since it contains Cl_2
 (2) -1, since it contains Cl^-
 (3) +1, since it contains ClO^-
 (4) +1 and -1 since it contains ClO^- and Cl^-
 12. Which of the following is a redox [AIEEE-02]
 (1) $2\text{NaAg}(\text{CN})_2 + \text{Zn} \longrightarrow \text{Na}_2\text{Zn}(\text{CN})_4 + 2\text{Ag}$
 (2) $\text{BaO}_2 + \text{H}_2\text{SO}_4 \longrightarrow \text{BaSO}_4 + \text{H}_2\text{O}_2$
 (3) $\text{N}_2\text{O}_5 + \text{H}_2\text{O} \longrightarrow 2\text{HNO}_3$
 (4) $\text{AgNO}_3 + \text{KI} \longrightarrow \text{AgI} + \text{KNO}_3$
 13. In the coordination compound, $\text{K}_4[\text{Ni}(\text{CN})_6]$, the oxidation state of nickel is [AIEEE-03]
 (1) +1 (2) +2
 (3) -1 (4) 0
 14. The oxidation state of Cr in $[\text{Cr}(\text{NH}_3)_4\text{Cl}_2]^+$ is - [AIEEE-05]
 (1) +2 (2) +3 (3) 0 (4) +1
 15. The oxidation state of chromium in the final product formed by the reaction between KI and acidified potassium dichromate solution is - [AIEEE-05]
 (1) +6 (2) +4 (3) +3 (4) +2
 16. Reduction of the metal centre in aqueous permanganate ion involves - [JEE-2011]
 (1) 3 electrons in neutral medium
 (2) 5 electrons in neutral medium
 (3) 3 electrons in alkaline medium
 (4) 5 electrons in acidic medium
 17. Which ordering of compounds is according to the decreasing order of the oxidation state of nitrogen- [JEE- 2012]
 (1) HNO_3 , NO, NH_4Cl , N_2
 (2) HNO_3 , NO, N_2 , NH_4Cl
 (3) HNO_3 , NH_4Cl , NO, N_2
 (4) NO, HNO_3 , NH_4Cl , N_2
 18. 25 mL of household bleach solution was mixed with 30 mL of 0.50 M KI and 10 mL of 4 N acetic acid. In the titration of the liberated iodine, 48 mL of 0.25 N $\text{Na}_2\text{S}_2\text{O}_3$ was used to reach the end point. The molarity of the household bleach solution is [JEE- 2012]
 (1) 0.48 M (2) 0.96 M
 (3) 0.24 M (4) 0.024 M
 19. Given :
 $\text{X Na}_2\text{HASO}_3 + \text{Y NaBrO}_3 + \text{ZHCl} \rightarrow$
 $\text{NaBr} + \text{H}_3\text{AsO}_4 + \text{NaCl}$
 The values of X, Y and Z in the above redox reaction are respectively : [JEE(Main-online)-2013]
 (1) 2, 1, 3 (2) 3, 1, 6
 (3) 2, 1, 2 (4) 3, 1, 4

20. Consider the following reaction:

[JEE(Main)-2013]

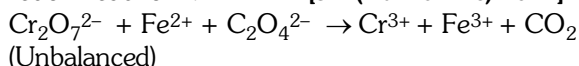


The values of x, y and z in the reaction are respectively :-

(1) 5, 2 and 16 (2) 2, 5 and 8

(3) 2, 5 and 16 (4) 5, 2 and 8

21. How many electrons are involved in the following redox reaction ? [JEE(Main-online)-2014]

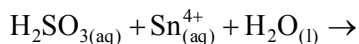


(1) 3 (2) 4

(3) 5 (4) 6

22. Consider the reaction

[JEE(Main-online)-2014]



Which of the following statements is correct?

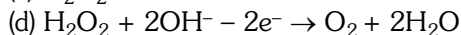
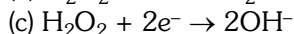
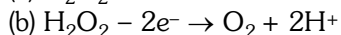
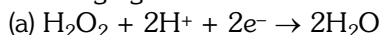
(1) H_2SO_3 is the reducing agent because it undergoes oxidation

(2) H_2SO_3 is the reducing agent because it undergoes reduction

(3) Sn^{4+} is the reducing agent because it undergoes oxidation

(4) Sn^{4+} is the oxidizing agent because it undergoes oxidation

23. In which of the following reaction H_2O_2 acts as a reducing agent ? [JEE(Main)-2014]



(1) (a), (c) (2) (b), (d) (3) (a), (b) (4) (c), (d)

24. The molecular formula of a commercial resin used for exchanging ions in water softening is $\text{C}_8\text{H}_7\text{SO}_3\text{Na}$ (Mol. wt 206). What would be the maximum uptake of Ca^{2+} ions by the resin when expressed in mole per gram resin ? [JEE(Main)-2015]

(1) $\frac{2}{309}$ (2) $\frac{1}{412}$ (3) $\frac{1}{103}$ (4) $\frac{1}{206}$

25. The volume of 0.1N dibasic acid sufficient to neutralize 1 g of a base that furnishes 0.04 mole of OH^- in aqueous solution is :

[JEE(Main)-OnLine-2016]

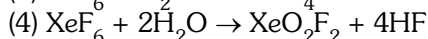
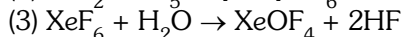
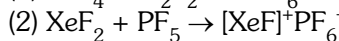
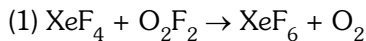
(1) 400 mL

(2) 200 mL

(3) 600 mL

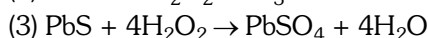
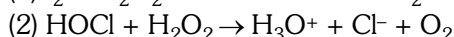
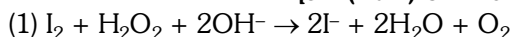
(4) 800 mL

26. Which of the following reactions is an example of a redox reaction ? [JEE(Main)-2017]

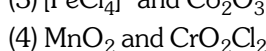
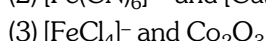
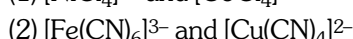
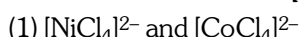


27. In which of the following reaction, hydrogen peroxide acts as an oxidizing agent ?

[JEE(Main)-OnLine-2017]



28. The pair of compounds having metal in their highest oxidation state is : [JEE(Main)-OnLine-2017]



29. In KO_2 , the nature of oxygen species and the oxidation state of oxygen atom are, respectively

[JEE(Main)-OnLine-2018]

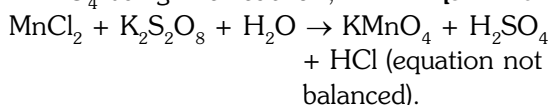
(1) Superoxide and $-1/2$

(2) Oxide and -2

(3) Peroxide and $-1/2$

(4) Superoxide and -1

30. To measure the quantity of MnCl_2 dissolved in an aqueous solution, it was completely converted to KMnO_4 using the reaction, [JEE- 2018]



Few drops of concentrated HCl were added to this solution and gently warmed. Further, oxalic acid (225 g) was added in portions till the colour of the permanganate ion disappeared. The quantity of MnCl_2 (in mg) present in the initial solution is _____. (Atomic weights in g mol^{-1} : $\text{Mn} = 55$, $\text{Cl} = 35.5$)

PREVIOUS YEARS QUESTIONS				ANSWER KEY				Exercise-II			
Que.	1	2	3	4	5	6	7	8	9	10	
Ans.	2	1	3	2	2	2	4	1	1	3	
Que.	11	12	13	14	15	16	17	18	19	20	
Ans.	4	1	2	2	3	1,2,4	2	3	2	3	
Que.	21	22	23	24	25	26	27	28	29	30	
Ans.	4	1	2	2	1	1	3	2	1	126	

IONIC EQUILIBRIUM & ACID BASE

EXERCISE-I

- The conjugate acid of NH_2^- is
 (1) NH_3 (2) NH_2OH
 (3) NH_4^+ (4) N_2H_4
 - Which of the following is not a Bronsted acid:-
 (1) CH_3NH_4^+ (2) CH_3COO^-
 (3) H_2O (4) HSO_4^-
 - In the reaction
 $\text{HNO}_3 + \text{H}_2\text{O} \rightleftharpoons \text{H}_3\text{O}^+ + \text{NO}_3^-$, the conjugate base of HNO_3 is :-
 (1) H_2O (2) H_3O^+
 (3) NO_3^- (4) H_3O^+ and NO_3^-
 - Out of the following, amphiprotic species in aqueous medium are
 I: HPO_3^{2-} II OH^-
 III H_2PO_4^- IV HCO_3^-
 (1) I, III, IV (2) I and III
 (3) III and IV (4) All
 - When ammonia is added to water, it decreases the concentration of which of the following ion
 (1) OH^- (2) H_3O^+
 (3) NH_4^+ (4) NH_4^+ & OH^-
 - Which of the following pair is Lewis acid & Lewis base & Product of these is also Lewis base
 (1) BF_3 , NH_3 (2) SiCl_4 , 2Cl^-
 (3) CH_3^+ , $^-\text{OC}_2\text{H}_5$ (4) All of these
 - Ionic product of water will increase, if :-
 (1) Pressure is decreased
 (2) H^+ is added
 (3) OH^- is increased
 (4) Temperature is increased
 - At 60°C , pure water has $[\text{H}_3\text{O}^+] = 10^{-6.7} \text{ mol/lit.}$ what is the value of K_w at 60°C :-
 (1) 10^{-6} (2) 10^{-12} (3) 10^{-67} (4) $10^{-13.4}$
 - The pH of solution is increased from 3 to 6. Its H^+ ion conc. will be :-
 (1) Reduced to half
 (2) Doubled
 (3) Reduced by 1000 times
 (4) Increased by 1000 times
 - Degree of dissociation of 0.1 N CH_3COOH is :- (Dissociation constant = 1×10^{-5})
 (1) 10^{-5} (2) 10^{-4} (3) 10^{-3} (4) 10^{-2}
 - The pH of a 0.02 M ammonia solution which is 5% ionised will be :-
 (1) 2 (2) 11 (3) 5 (4) 7
 - The pH of an aqueous solution of 1.0 M solution of a weak monoprotic acid which is 1% ionised is
 (1) 1 (2) 2 (3) 3 (4) 11
 - The concentration of $[\text{H}^+]$ and concentration of $[\text{OH}^-]$ of a 0.1 M aqueous solution of 2% ionised weak acid is [ionic product of water = 1×10^{-14}]
 (1) $0.02 \times 10^{-3} \text{ M}$ and $5 \times 10^{-11} \text{ M}$
 (2) $1 \times 10^{-3} \text{ M}$ and $3 \times 10^{-11} \text{ M}$
 (3) $2 \times 10^{-3} \text{ M}$ and $5 \times 10^{-12} \text{ M}$
 (4) $3 \times 10^{-2} \text{ M}$ and $4 \times 10^{-13} \text{ M}$
 - What is the quantity of NaOH present in 250 cc of the solution, so that it gives a pH = 13 :-
 (1) 10^{-13} g (2) 10^{-1} g (3) 1.0 g (4) 4.0 g
 - An aqueous solution of HCl is 10^{-9} M HCl. The pH of the solution should be:-
 (1) 9 (2) Between 6 and 7
 (3) 7 (4) Unpredictable
 - Which one of the following has highest pH:-
 (1) Distilled water
 (2) 1 M NH_3
 (3) 1 M NaOH
 (4) Water saturated with chlorine
 - 8 gm NaOH and 4.9 gm H_2SO_4 are present in one litre of the solution. What is its pH
 (1) 1 (2) 13 (3) 12 (4) 2
 - 10 ml of $\frac{\text{M}}{200} \text{H}_2\text{SO}_4$ is mixed with 40 ml of $\frac{\text{M}}{200} \text{H}_2\text{SO}_4$. The pH of the resulting solution is
 (1) 1 (2) 2
 (3) 2.3 (4) none of these
 - Which of the following solution will have pH close to 1.0?
 (1) 100 ml of M/100 HCl + 100 ml of M/10 NaOH
 (2) 55 ml of M/10 HCl + 45 ml of M/10 NaOH
 (3) 10 ml of M/10 HCl + 90 ml of M/10 NaOH
 (4) 75 ml of M/5 HCl + 25 ml of M/5 NaOH
 - A solution with pH 2.0 is more acidic than the one with pH 6.0 by a factor of:
 (1) 3 (2) 4 (3) 3000 (4) 10000

- 21.** The first and second dissociation constants of an acid H_2A are 1.0×10^{-5} and 5.0×10^{-10} respectively. The overall dissociation constant of the acid will be:
 (1) 5.0×10^{-5} (2) 5.0×10^{15}
 (3) 5.0×10^{-15} (4) 0.2×10^5
- 22.** pH of an aqueous solution of NaCl at $85^\circ C$ should be
 (1) 7 (2) > 7 (3) < 7 (4) 0
- 23.** The degree of hydrolysis of a salt of weak acid and weak base in its 0.1 M solution is found to be 50%. If the molarity of the solution is 0.2 M, the percentage hydrolysis of the salt should be
 (1) 100% (2) 50%
 (3) 25% (4) none of these
- 24.** If 40 ml of 0.2 M KOH is added to 160 ml of 0.1 M $HCOOH$ [$K_a = 2 \times 10^{-4}$], the pOH of the resulting solution is
 (1) 3.4 (2) 3.7
 (3) 7 (4) 10.3
- 25.** The pK_a of a weak acid (HA) is 4.5. The pOH of an aqueous buffered solution of HA in which 50% of the acid is ionized is :
 (1) 4.5 (2) 2.5
 (3) 9.5 (4) 7.0
- 26.** To a 50 ml. of 0.05M formic acid, how much volume of 0.10M sodium formate must be added to get a buffer solution of $pH = 4.0$? (pK_a of the acid is 3.7) ($\log 2 = 0.3$)
 (1) 40 ml. (2) 4 ml.
 (3) 50 ml. (4) 100 ml.
- 27.** Which can act as buffer :-
 (1) $NH_4OH + NaOH$
 (2) $HCOOH + HCl$
 (3) 40 ml. of 0.1 M NaCN + 20 ml. of 0.1 M HCl
 (4) All of them
- 28.** Calculate the pH of a buffer prepared by mixing 600 cc of 0.6 M NH_3 and 400 cc of 0.5 M NH_4Cl . K_b for $NH_3 = 1.8 \times 10^{-5}$, ($\log 1.8 = 0.26$)
 (1) 11.3 (2) 9.0 (3) 9.52 (4) 5
- 29.** When 0.02 moles of NaOH are added to a litre of buffer solution, its pH changes from 5.75 to 5.80. What is its buffer capacity :-
 (1) 0.4 (2) 0.05 (3) - 0.05 (4) 2.5
- 30.** Calculate the pH range in which an acid indicator with $K_{acid}(\text{indicator}) = 1.0 \times 10^{-5}$ changes colour when the concentration of the indicator is $1 \times 10^{-3} M$.
 (1) 5 ± 1 (2) 11 ± 1 (3) 3 ± 1 (4) 8 ± 1

				ANSWER KEY						Exercise-I	
Que.	1	2	3	4	5	6	7	8	9	10	
Ans.	1	2	3	3	2	3	4	4	3	4	
Que.	11	12	13	14	15	16	17	18	19	20	
Ans.	2	2	3	3	2	3	2	2	4	4	
Que.	21	22	23	24	25	26	27	28	29	30	
Ans.	3	3	2	4	3	3	3	3	1	1	

PREVIOUS YEARS' QUESTIONS

EXERCISE-II

- 1.** The pH of 0.1 M solution of the following salts increases in the order [JEE 1999]
 - (1) NaCl < NH₄Cl < NaCN < HCl
 - (2) HCl < NH₄Cl < NaCl < NaCN
 - (3) NaCN < NH₄Cl < NaCl < HCl
 - (4) HCl < NaCl < NaCN < NH₄Cl

 - 2.** The solubility of Mg(OH)₂ is x mole/lit. then its solubility product is - [AIIEEE-2002]
 - (1) x³
 - (2) 5x³
 - (3) 4x³
 - (4) 2x²

 - 3.** A solution which is 10⁻³ M each in Mn²⁺, Fe²⁺, Zn²⁺ and Hg²⁺ is treated with 10⁻¹⁶M sulphide ion. If K_{sp}, MnS, FeS, ZnS and HgS are 10⁻¹⁵, 10⁻²³, 10⁻²⁰ and 10⁻⁵⁴ respectively, which one will precipitate first ? [JEE 2003]
 - (1) FeS
 - (2) MnS
 - (3) HgS
 - (4) ZnS

 - 4.** The solubility in water of a sparingly soluble salt AB₂ is 1.0 × 10⁻⁵ mol L⁻¹. Its solubility product will be [AIIEEE-2003]
 - (1) 1 × 10⁻¹⁵
 - (2) 1 × 10⁻¹⁰
 - (3) 4 × 10⁻¹⁵
 - (4) 4 × 10⁻¹⁰

 - 5.** The molar solubility in mol L⁻¹ of a sparingly soluble salt MX₄ is 's'. The corresponding solubility product is K_{SP}. 's' is given in terms of K_{SP} by relation : [AIIEEE-2004]
 - (1) s = (K_{SP} / 128)^{1/4}
 - (2) s = (128K_{SP})^{1/4}
 - (3) s = (256K_{SP})^{1/5}
 - (4) s = (K_{SP}/256)^{1/5}

 - 6.** The solubility product of a salt having general formula MX₂, in water is : 4 × 10⁻¹². The concentration of M²⁺ ions in the aqueous solution of the salt is - [AIIEEE-2005]
 - (1) 1.0 × 10⁻⁴ M
 - (2) 2.0 × 10⁻⁶ M
 - (3) 4.0 × 10⁻¹⁰ M
 - (4) 1.6 × 10⁻⁴ M

 - 7.** Hydrogen ion concentration in mol/L in a solution of pH = 5.4 will be - [AIIEEE-2005]
 - (1) 3.88 × 10⁶
 - (2) 3.98 × 10⁸
 - (3) 3.98 × 10⁻⁶
 - (4) 3.68 × 10⁻⁶

 - 8.** In a saturated solution of the sparingly soluble strong electrolyte AgIO₃ (molecular mass = 283) the equilibrium which sets in is - [AIIEEE-2007]

$$\text{AgIO}_3 \rightleftharpoons \text{Ag}_{(\text{aq})}^+ + \text{IO}_{3(\text{aq})}^-$$

If the solubility product constant K_{sp} of AgIO₃ at a given temperature is 1.0 × 10⁻⁸, what is the mass of AgIO₃ contained in 100 ml of its saturated solution ?

 - (1) 28.3 × 10⁻² g
 - (2) 2.83 × 10⁻³ g
 - (3) 1.0 × 10⁻⁷ g
 - (4) 1.0 × 10⁻⁴ g

 - 9.** The pK_a of a weak acid, HA, is 4.80. The pK_b of a weak base, BOH, is 4.78. The pH of an aqueous solution of the corresponding salt. BA, will be - [AIIEEE-2008]
 - (1) 9.58
 - (2) 4.79
 - (3) 7.01
 - (4) 9.22

 - 10.** Solid Ba(NO₃)₂ is gradually dissolved in a 1.0 × 10⁻⁴ M Na₂CO₃ solution. At what concentration of Ba²⁺ will a precipitate begin to form? (K_{SP} for Ba CO₃ = 5.1 × 10⁻⁹) [AIIEEE-2009]
 - (1) 8.1 × 10⁻⁸ M
 - (2) 8.1 × 10⁻⁷ M
 - (3) 4.1 × 10⁻⁵ M
 - (4) 5.1 × 10⁻⁵ M

 - 11.** Solubility product of silver bromide is 5.0 × 10⁻¹³. The quantity of potassium bromide (molar mass taken as 120 g mol⁻¹) to be added to 1 litre of 0.05 M solution of silver nitrate to start the precipitation of AgBr is :- [AIIEEE-2010]
 - (1) 5.0 × 10⁻⁸ g
 - (2) 1.2 × 10⁻¹⁰ g
 - (3) 1.2 × 10⁻⁹ g
 - (4) 6.2 × 10⁻⁵ g

 - 12.** In aqueous solution the ionization constants for carbonic acid are

$$K_1 = 4.2 \times 10^{-7} \text{ and } K_2 = 4.8 \times 10^{-11}$$

Select the correct statement for a saturated 0.034 M solution of the carbonic acid :- [AIIEEE-2010]

 - (1) The concentration of H⁺ is double that of CO₃²⁻
 - (2) The concentration of CO₃²⁻ is 0.034 M
 - (3) The concentration of CO₃²⁻ is greater than that of HCO₃⁻
 - (4) The concentrations of H⁺ and HCO₃⁻ are approximately equal

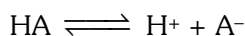
13. At 25° C, the solubility product of Mg(OH)_2 is 1.0×10^{-11} . At which pH, will Mg^{2+} ions start precipitating in the form of Mg(OH)_2 from a solution of 0.001 M Mg^{2+} ions ? [AIEEE-2010]

- (1) 8 (2) 9 (3) 10 (4) 11

14. The K_{sp} for Cr(OH)_3 is 1.6×10^{-30} . The molar solubility of this compound in water is :- [AIEEE-2011]

- (1) $\sqrt[2]{1.6 \times 10^{-30}}$ (2) $\sqrt[4]{1.6 \times 10^{-30}}$
(3) $\sqrt[4]{1.6 \times 10^{-30} / 27}$ (4) $1.6 \times 10^{-30} / 27$

15. An acid HA ionises as



The pH of 1.0 M solution is 5. Its dissociation constant would be :- [AIEEE-2011]

- (1) 1×10^{-10} (2) 5
(3) 5×10^{-8} (4) 1×10^{-5}

16. If K_{sp} of CaF_2 at 25°C is 1.7×10^{-10} , the combination amongst the following which gives a precipitate of CaF_2 is :- [JEE-MAIN(online)-2012]

- (1) 1×10^{-2} M Ca^{2+} and 1×10^{-5} M F^-
(2) 1×10^{-4} M Ca^{2+} and 1×10^{-4} M F^-
(3) 1×10^{-3} M Ca^{2+} and 1×10^{-5} M F^-
(4) 1×10^{-2} M Ca^{2+} and 1×10^{-3} M F^-

17. The pH of a 0.1 molar solution of the acid HQ is 3. The value of the ionization constant, K_a of this acid is :- [AIEEE-2012]

- (1) 1×10^{-7} (2) 3×10^{-7} (3) 1×10^{-3} (4) 1×10^{-5}

18. How many litres of water must be added to 1 litre of an aqueous solution of HCl with a pH of 1 to create an aqueous solution with pH of 2 ? [AIEEE-2013]

- (1) 0.1 L (2) 0.9 L (3) 2.0 L (4) 9.0 L

19. Solid $\text{Ba(NO}_3)_2$ is gradually dissolved in a 1.0×10^{-4} M Na_2CO_3 solution. At which concentration of Ba^{2+} , precipitate of BaCO_3 begins to form ? (K_{sp} for $\text{BaCO}_3 = 5.1 \times 10^{-9}$) [JEE-MAIN(Online)-2013]

- (1) 5.1×10^{-5} M (2) 8.1×10^{-7} M
(3) 4.1×10^{-5} M (4) 7.1×10^{-8} M

20. NaOH is a strong base. What will be pH of 5.0×10^{-2} M NaOH solution ? ($\log 2 = 0.3$) [JEE-MAIN(Online)-2013]

- (1) 13.70 (2) 13.00 (3) 14.00 (4) 12.70

21. Which one of the following arrangements represents the correct order of solubilities of sparingly soluble salts Hg_2Cl_2 , $\text{Cr}_2(\text{SO}_4)_3$, BaSO_4 and CrCl_3 respectively ? [JEE-MAIN(Online)-2013]

- (1) $\left(\frac{K_{sp}}{4}\right)^{\frac{1}{3}}$, $\left(\frac{K_{sp}}{108}\right)^{\frac{1}{5}}$, $(K_{sp})^{\frac{1}{2}}$, $\left(\frac{K_{sp}}{27}\right)^{\frac{1}{4}}$
(2) $(K_{sp})^{\frac{1}{2}}$, $\left(\frac{K_{sp}}{4}\right)^{\frac{1}{3}}$, $\left(\frac{K_{sp}}{27}\right)^{\frac{1}{4}}$, $\left(\frac{K_{sp}}{108}\right)^{\frac{1}{5}}$
(3) $(K_{sp})^{\frac{1}{2}}$, $\left(\frac{K_{sp}}{108}\right)^{\frac{1}{5}}$, $\left(\frac{K_{sp}}{27}\right)^{\frac{1}{4}}$, $\left(\frac{K_{sp}}{4}\right)^{\frac{1}{3}}$
(4) $\left(\frac{K_{sp}}{108}\right)^{\frac{1}{5}}$, $\left(\frac{K_{sp}}{27}\right)^{\frac{1}{4}}$, $(K_{sp})^{\frac{1}{2}}$, $\left(\frac{K_{sp}}{4}\right)^{\frac{1}{3}}$

22. What would be the pH of a solution obtained by mixing 5 g of acetic acid and 7.5 g of sodium acetate and making the volume equal to 500 mL? [JEE-MAIN(Online)-2013]

($K_a = 1.75 \times 10^{-5}$, $\text{p}K_a = 4.76$)

- (1) $4.76 < \text{pH} < 5.0$
(2) $\text{pH} < 4.70$
(3) pH of solution will be equal to pH of acetic acid
(4) $\text{pH} = 4.70$

23. Zirconium phosphate $[\text{Zr}_3(\text{PO}_4)_4]$ dissociates into three zirconium cations of charge +4 and four phosphate anions of charge -3. If molar solubility of zirconium phosphate is denoted by S and its solubility product by K_{sp} then which of the following relationship between S and K_{sp} is correct ? [JEE-MAIN(Online)-2014]

- (1) $S = \{K_{sp}/144\}^{1/7}$
(2) $S = \{K_{sp}/(6912)\}^{1/7}$
(3) $S = (K_{sp}/6912)^{1/7}$
(4) $S = \{K_{sp}/6912\}^{1/7}$

24. $\text{p}K_a$ of a weak acid (HA) and $\text{p}K_b$ of a weak base (BOH) are 3.2 and 3.4, respectively. The pH of their salt (AB) solution is [JEE-MAIN(Offline)-2017]

- (1) 7.2 (2) 6.9 (3) 7.0 (4) 1.0

MOLE CONCEPT & GASEOUS STATE

EXERCISE-I

- Which of the following contain largest number of carbon atoms ?
 (1) 15 gm ethane, C_2H_6
 (2) 40.2 gm sodium oxalate, $Na_2C_2O_4$
 (3) 72 gm glucose, $C_6H_{12}O_6$
 (4) 35 gm pentene, C_5H_{10}
- The percentage by mole of NO_2 in a mixture of $NO_2(g)$ and $NO(g)$ having average molecular mass 34 is :
 (1) 25% (2) 20% (3) 40% (4) 75%
- Volume of O_2 obtained at 2 atm & 546K, by the complete decomposition of 8.5 g $NaNO_3$ is
 $2NaNO_3 \rightarrow 2NaNO_2 + O_2$
 (1) 2.24 lit (2) 1.12 lit (3) 0.84 lit (4) 0.56 lit
- A metal carbonate decomposes according to following reaction

$$M_2CO_3(s) \longrightarrow M_2O(s) + CO_2(g)$$
 Percentage loss in mass on complete decomposition of $M_2CO_3(s)$
 (Atomic mass of M = 102)
 (1) $\frac{100}{3}\%$ (2) $\frac{50}{3}\%$ (3) $\frac{25}{3}\%$ (4) 15%
- 25.4 gm of iodine and 14.2 gm of chlorine are made to react completely to yield a mixture of ICl and ICl_3 . Ratio of moles of ICl & ICl_3 formed is (Atomic mass : I = 127, Cl = 35.5)
 (1) 1 : 1 (2) 1 : 2 (3) 1 : 3 (4) 2 : 3
- 100 ml of PH_3 decomposes according to given reaction.

$$PH_3(g) \longrightarrow P(s) + 3/2 H_2(g)$$
 The change in volume of the gas is :-
 (1) 50 ml increase (2) 500 ml decrease
 (3) 900 ml decrease (4) 150 ml increase
- 44 g of a sample of organic compound on complete combustion gives 88 g CO_2 and 36 g of H_2O . The molecular formula of the compound may be :-
 (1) C_4H_6 (2) C_2H_6O (3) C_2H_4O (4) C_3H_6O
- A metal oxide has the formula X_2O_3 . It can be reduced by hydrogen to give free metal and water. 0.156 gm of metal oxide requires 6 mg of hydrogen for complete reduction. The atomic mass of metal in amu is :-
 (1) 15.6 (2) 156 (3) 108 (4) 54
- Two oxides of a metal contain 22.22% and 30% oxygen by mass respectively. If the formula of the first oxide is MO , then the formula of the second oxide is:-
 (1) MO_2 (2) M_2O_3 (3) M_2O (4) M_2O_5
- In the reaction $4A + 2B + 3C \longrightarrow A_4B_2C_3$, what will be the number of moles of product formed, starting from 1 mol of A, 0.6 mol of B and 0.72 mol of C :-
 (1) 2.32 (2) 0.24 (3) 0.3 (4) 0.25
- Two isotopes of an element Q are Q^{97} (23.4% abundance) and Q^{94} (76.6% abundance). Q^{97} is 8.082 times heavier than C^{12} and Q^{94} is 7.833 times heavier than C^{12} . What is the average atomic weight of the element Q ?
 (1) 94.702 (2) 78.913 (3) 96.298 (4) 94.695
- 12 g of Mg was burnt in a closed vessel containing 32 g oxygen. Remaining unreacted gm-molecules of O_2 will be :-
 (1) 0.5 (2) 0.75 (3) 1 (4) 0.25
- The total number of electrons present in 9 mL of water (density of water is 1 gmL^{-1}) :-
 (1) 3.01×10^{23} (2) 3.01×10^{22}
 (3) 3.01×10^{24} (4) 3.01×10^{25}
- From 200 mg of CO_2 , 10^{21} molecules are removed. How many moles of CO_2 are left ?
 (1) 126.9×10^{-1} (2) 2.88×10^{-3}
 (3) 7.31×10^{-3} (4) 4.4×10^{-3}
- On heating 10 g of $CaCO_3$, 5.6 g CaO is formed. Moles of CO_2 obtained in this reaction will be :-
 (1) 2.2 (2) 4.4 (3) 0.1 (4) 0.2
- 40 gm of a carbonate of an **alkali metal** or **alkaline earth metal** containing some inert impurities was made to react with excess HCl solution. The liberated CO_2 occupied 12.315 lit. at 1 atm & 300 K. The correct option is
 (1) Mass of impurity is 1 gm and metal is Be
 (2) Mass of impurity is 3 gm and metal is Li
 (3) Mass of impurity is 5 gm and metal is Be
 (4) Mass of impurity is 2 gm and metal is Mg
- 1 mole of H_2SO_4 will not exactly neutralise :
 (1) 2 mole of ammonia (2) 1 mole of $Ba(OH)_2$
 (3) 0.5 mole of $Ca(OH)_2$ (4) 2 mole of KOH
- 12 g of Mg was burnt in a closed vessel containing 32 g oxygen. Which of the following is incorrect.
 (1) 2 gm of Mg will be left unburnt.
 (2) 0.75 gm-molecule of O_2 will be left unreacted.
 (3) 20 gm of MgO will be formed.
 (4) The mixture at the end will weight 44 g.

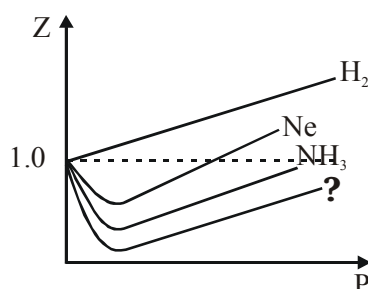
19. 50 gm of CaCO_3 is allowed to react with 68.6 gm of H_3PO_4 then select the incorrect option-
 $3\text{CaCO}_3 + 2\text{H}_3\text{PO}_4 \rightarrow \text{Ca}_3(\text{PO}_4)_2 + 3\text{H}_2\text{O} + 3\text{CO}_2$
 (1) 51.67 gm salt is formed
 (2) Amount of unreacted reagent = 35.93 gm
 (3) $n_{\text{CO}_2} = 0.5$ moles
 (4) 0.7 mole CO_2 is evolved
20. Select the correct statement(s) for $(\text{NH}_4)_3\text{PO}_4$.
 (1) Ratio of number of oxygen atoms to number of hydrogen atoms is 1 : 3
 (2) Ratio of number of cations to number of anions is 3 : 2
 (3) Ratio of number of gm-atoms of nitrogen to gm-atoms of oxygen is 3 : 2
 (4) Total number of atoms in one mole of $(\text{NH}_4)_3\text{PO}_4$ is 20.
21. At STP, the order of root mean square speed of molecules H_2 , N_2 , O_2 and HBr is :
 (1) $\text{H}_2 > \text{N}_2 > \text{O}_2 > \text{HBr}$ (2) $\text{HBr} > \text{O}_2 > \text{N}_2 > \text{H}_2$
 (3) $\text{HBr} > \text{H}_2 > \text{O}_2 > \text{N}_2$ (4) $\text{N}_2 > \text{O}_2 > \text{H}_2 > \text{HBr}$
22. Most probable speed, average speed and RMS speed are related as :
 (1) 1 : 1.128 : 1.224 (2) 1 : 1.128 : 1.424
 (3) 1 : 2.128 : 1.224 (4) 1 : 1.428 : 1.442
23. If the average velocity of N_2 molecules is 0.3 m/sec. at 27°C , then the velocity of 0.6 m/sec will take place at:
 (1) 273 K (2) 927 K (3) 1000 K (4) 1200 K
24. The total KE of an ideal monoatomic gas at 27°C is
 (1) 900 cal (2) 1800 cal (3) 300 cal (4) None
25. The correct expression for the van der Waals' equation of state is :
 (1) $\left(P + \frac{a}{n^2V^2}\right)(V - nb) = nRT$
 (2) $\left(P + \frac{an^2}{V^2}\right)(V - nb) = \Delta nRT$
 (3) $\left(P + \frac{an^2}{V^2}\right)(V - b) = nRT$
 (4) $\left(P + \frac{an^2}{V^2}\right)(V - nb) = nRT$

- 26.** At relatively high pressure, van der Waals' equation reduces to :

$$(1) PV_m = RT \qquad (2) PV_m = RT + \frac{a}{V_m}$$

$$(3) \text{ PV}_m = RT + Pb \quad (4) \text{ PV}_m = RT - \frac{a}{V_m^2}$$

- 27.** Observe the following Z vs P graph.



The missing gas in the above graph can be :

- (1) He (2) Ar
(3) C₅H₁₂ (4) All are correct
- 28.** The values of van der Waals' constant '**a**' for the gases O₂, N₂, NH₃ and CH₄ are 1.360, 1.390, 4.170 and 2.253 L atm mol⁻² respectively. The gas which can most easily be liquefied is :
- (1) O₂ (2) N₂ (3) NH₃ (4) CH₄
- 29.** A gas can be liquefied by :
- (1) Cooling (2) Compressing
(3) Both (1) and (2) (4) None
- 30.** Which set of conditions represents easiest way to liquefy a gas :
- (1) Low temperature and high pressure
(2) High temperature and low pressure
(3) Low temperature and low pressure
(4) High temperature and high pressure

				ANSWER KEY			Exercise-1			
Que.	1	2	3	4	5	6	7	8	9	10
Ans.	4	1	2	2	1	1	3	4	2	2
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	4	2	3	2	3	2	3	1	4	1
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	1	1	4	1	4	3	3	3	3	1

PREVIOUS YEARS' QUESTIONS

EXERCISE-II

- How many moles of e^- weight one Kg :
[JEE '2002 (Scr), 1]
(1) 6.023×10^{23}
(2) $\frac{1}{9.108} \times 10^{31}$
(3) $\frac{6.023}{9.108} \times 10^{54}$
(4) $\frac{1}{9.108 \times 6.023} \times 10^8$
- As the temperature is raised from 20°C to 40°C , the average kinetic energy of neon atoms changes by factor of which of the following ? [AIEEE-2004]
(1) $1/2$ (2) $\sqrt{(313/293)}$
(3) $313/293$ (4) 2
- In Van der Waals equation of state of the gas law, the constant 'b' is a measure of : [AIEEE-04]
(1) intermolecular repulsions
(2) intermolecular attractions
(3) volume occupied by the molecules
(4) intermolecular collisions per unit volume
- Calculate the amount of Calcium oxide required when it reacts with 852 g of P_4O_{10} . [JEE 2005]
 $6\text{CaO} + \text{P}_4\text{O}_{10} \longrightarrow 2\text{Ca}_3(\text{PO}_4)_2$ [Ca = 40, P = 31]
(1) 2000 g (2) 3008 g
(3) 1008 g (4) 500 g
- Given that the abundances of isotopes ^{54}Fe , ^{56}Fe and ^{57}Fe are 5%, 90% and 5%, respectively, the atomic mass of Fe is : [JEE 2009]
(1) 55.85 (2) 55.95
(3) 55.75 (4) 56.05
- A 5.2 molal aqueous solution of methyl alcohol, CH_3OH , is supplied. What is the mole fraction of methyl alcohol in the solution ? [AIEEE-2011]
(1) 0.086 (2) 0.050
(3) 0.100 (4) 0.190
- 'a' and 'b' are Van der Waals constants for gases. Chlorine is more easily liquefied than ethane because : [AIEEE-2011]
(1) a for $\text{Cl}_2 < \text{a for C}_2\text{H}_6$ but b for $\text{Cl}_2 > \text{b for C}_2\text{H}_6$
(2) a for $\text{Cl}_2 > \text{a for C}_2\text{H}_6$ but b for $\text{Cl}_2 < \text{b for C}_2\text{H}_6$
(3) a and b for $\text{Cl}_2 > \text{a and b for C}_2\text{H}_6$
(4) a and b for $\text{Cl}_2 < \text{a and b for C}_2\text{H}_6$

- The compressibility factor for a real gas at high pressure is :- [AIEEE-2012]
(1) $1 - \frac{\text{Pb}}{\text{RT}}$ (2) $1 + \frac{\text{RT}}{\text{Pb}}$ (3) 1 (4) $1 + \frac{\text{Pb}}{\text{RT}}$
- The concentrated sulphuric acid that is peddled commercially is 95% H_2SO_4 by weight. If the density of this commercial acid is 1.834 g cm^{-3} , the molarity of this solution is :- [JEE(Main)-2012]
(1) 17.8 M (2) 15.7 M (3) 10.5 M (4) 12.0 M
- The density of a solution prepared by dissolving 120 g of urea (mol. mass = 60 u) in 1000 g of water is 1.15 g/mL . The molarity of this solution is [JEE(Main)-2012]
(1) 2.05 M (2) 0.50 M (3) 1.78 M (4) 1.02 M
- A transition metal M forms a volatile chloride which has a vapour density of 94.8. If it contains 74.75% of chlorine the formula of the metal chloride will be [AIEEE 2012 (Online)]
(1) MCl_2 (2) MCl_4 (3) MCl_5 (4) MCl_3
- The ratio of number of oxygen atoms (O) in 16.0 g ozone (O_3), 28.0 g carbon monoxide (CO) and 16.0 g oxygen (O_2) is :- (Atomic mass : C = 12, O = 16 and Avogadro's constant $N_A = 6.0 \times 10^{23} \text{ mol}^{-1}$) [AIEEE 2012 (Online)]
(1) 3 : 1 : 1 (2) 1 : 1 : 2
(3) 3 : 1 : 2 (4) 1 : 1 : 1
- The relationship among most probable velocity, average velocity and root mean square velocity is respectively :- [JEE(Main-online)-2012]
(1) $\sqrt{2} : \sqrt{8/\pi} : \sqrt{3}$ (2) $\sqrt{2} : \sqrt{3} : \sqrt{8/\pi}$
(3) $\sqrt{3} : \sqrt{8/\pi} : \sqrt{2}$ (4) $\sqrt{8/\pi} : \sqrt{3} : \sqrt{2}$
- A gaseous hydrocarbon gives upon combustion 0.72 g of water and 3.08 g of CO_2 . The empirical formula of the hydrocarbon is [JEE(Main)-2013]
(1) C_2H_4 (2) C_3H_4 (3) C_6H_5 (4) C_7H_8
- For gaseous state, if most probable speed is denoted by C, average speed by $\bar{C}$ and mean square speed by C , then for a large number of molecules the ratios of these speeds are :- [JEE(Main-offline)-2013]
(1) $C : \bar{C} : C = 1.225 : 1.128 : 1$
(2) $C : \bar{C} : C = 1.128 : 1.225 : 1$
(3) $C : \bar{C} : C = 1 : 1.128 : 1.225$
(4) $C : \bar{C} : C = 1 : 1.225 : 1.128$

- 16.** 10 mL of 2(M) NaOH solution is added to 200 mL of 0.5 (M) of NaOH solution. What is the final concentration ? **[JEE(Main-online)-2013]**
 (1) 0.57 M (2) 5.7 M
 (3) 11.4 M (4) 1.14 M

17. The density of 3M solution of sodium chloride is 1.252 g mL^{-1} . The molality of the solution will be (molar mass, NaCl = 58.5 g mol^{-1}) **[JEE(Main-online)-2013]**
 (1) 2.18 m (2) 3.00 m
 (3) 2.60 m (4) 2.79 m

18. The ratio of masses of oxygen and nitrogen in a particular gaseous mixture is 1 : 4. The ratio of number of their molecule is : **[JEE(Main)-2014]**
 (1) 1 : 8 (2) 3 : 16
 (3) 1 : 4 (4) 7 : 32

19. If Z is the compressibility factor, Van der Waals equation **at low** pressure can be written as : **[JEE-MAINS-2014]**
 (1) $Z = 1 - \frac{Pb}{RT}$ (2) $Z = 1 + \frac{Pb}{RT}$
 (3) $Z = 1 + \frac{RT}{Pb}$ (4) $Z = 1 - \frac{a}{V_m RT}$

20. The amount of BaSO_4 formed upon mixing 100 mL of 20.8% BaCl_2 solution with 50 mL of 9.8% H_2SO_4 solution will be : **[JEE(Main-online)-2014]**
 (Ba = 137, Cl = 35.5, S=32, H = 1 and O = 16)
 (1) 33.2 g (2) 11.65 g
 (3) 23.3 g (4) 30.6 g

21. For the estimation of nitrogen, 1.4 g of an organic compound was digested by Kjeldahl method and the evolved ammonia was absorbed in 60 mL of $\frac{M}{10}$ sulphuric acid. The unreacted acid required 20 mL of $\frac{M}{10}$ sodium hydroxide for complete neutralization. The percentage of nitrogen in the compound is : **[JEE(Main-online)-2014]**
 (1) 3% (2) 5%
 (3) 6% (4) 10%

22. The temperature at which oxygen molecules have the same root mean square speed as helium atoms have at 300 K is : **[JEE(Main-online)-2014]**
 (Atomic masses : He = 4 u, O = 16 u)
 (1) 1200 K (2) 600 K
 (3) 300 K (4) 2400 K

23. In Carius method of estimation of halogens, 250 mg of an organic compound gave 141 mg of AgBr. The percentage of bromine in the compound is : (Atomic mass Ag = 108; Br = 80) **[JEE(Main)-2015]**
 (1) 48 (2) 60
 (3) 24 (4) 36

24. When does a gas deviate the most from its ideal behaviour ? **[JEE-MAINS(ONLINE)-2015]**
 (1) At high pressure and low temperature
 (2) At high pressure and high temperature
 (3) At low pressure and low temperature
 (4) At low pressure and high temperature

25. The most abundant elements by mass in the body of a healthy human adult are :
 Oxygen (61.4%) ; Carbon (22.9%), Hydrogen (10.0%) ; and Nitrogen (2.6%). The weight which a 75 kg person would gain if all ^1H atoms are replaced by ^2H atoms is **[JEE(Main)-2017]**
 (1) 15 kg (2) 37.5 kg
 (3) 7.5 kg (4) 10 kg

26. 1 gram of a carbonate (M_2CO_3) on treatment with excess HCl produces 0.01186 mole of CO_2 . The molar mass of M_2CO_3 in g mol^{-1} is :- **[JEE(Main)-2017]**
 (1) 1186 (2) 84.3
 (3) 118.6 (4) 11.86

Previous Years Questions				Answer Key			Exercise-II			
Que.	1	2	3	4	5	6	7	8	9	10
Ans.	4	3	3	3	2	1	2	4	1	1
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	2	4	1	4	3	1	4	4	4	2
Que.	21	22	23	24	25	26				
Ans.	4	4	3	1	3	2				